What Does an Observation Identify? A Synthesis with Supporting Notes

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Scope

The synthesis states, with examples, which properties of a specified model are fixed by a specified family of observations. The supporting notes that follow it give the detailed arguments for individual examples: source, gain and detector ambiguity, graph and scale observations, predictive dimension and history-dependent readout, and constraints on phase contrast, purity and paired-input data for conditional operations. Each note is restricted to its stated state space, observation menu and calibration premises.

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From Prephysical Hypotheses to Physical Models

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Abstract

What would justify interpreting a relational structure as a physical one? We study the part of this question that admits an operational answer: which properties are shared by every model consistent with specified preparations, interventions and measurements? Finite operator algebras illustrate when invariant rank-one choices exist. Spectral filters show how calibrated port measurements can recover a graph, while a static projector or a finite list of moments can leave its generating dynamics ambiguous. Fermionic relative entropy and a scalar-field response illustrate how a physical model and its scale enter such comparisons. For a finite-dimensional instrument, we determine the greatest possible weight of its operation in a specified local operator subspace from all basis-input first-output marginals. For every finite discarded dimension of at least two, joint purification gives the attainable ceiling and zero remains possible. Reading the second output label can give a positive lower bound. Matched coherent preparations and output projections can determine the weight even when all basis outputs leave it ambiguous. We construct both endpoints for each observation class and give an all-attempt sequential test for the calibrated four-input model. The results specify what each observation determines and which calibration would make the inference applicable to a particular physical system.

1 What does an observation identify?

A prephysical proposal may specify relations, compositions or symmetries before it specifies a clock, a position observable or a set of physical preparations. The problem is to give those symbols an observable interpretation. A mathematical map from relations into operators is insufficient on its own: the same experimental records may be compatible with several operator assignments, tensor factorizations or dynamical laws. The question can be made precise without assuming that an underlying history is deterministic.

Let Θ be the class of descriptions under consideration, $\mathcal{E}$ a specified set of experiments, and P_θ^e the outcome law for experiment e. Include source, detector, gain and loss parameters in θ whenever the evidence allows them to vary. Define $\theta \sim \theta'$ when $P_\theta^e = P_{\theta'}^e$ for every $e \in \mathcal{E}$. A proposed property h is identified by this experiment family exactly when it is constant on each equivalence class. Indeed, constancy defines a function of the observed laws; conversely any function of those laws takes the same value on equivalent descriptions. For actual data the relevant object is the range of h over the jointly compatible descriptions, with sampling and calibration uncertainty included. A second experiment can refine this range only if its outcomes distinguish descriptions with different values of the property.

This criterion connects the mathematical and ontological questions. If a description assigns the same observations to two inequivalent referents, those observations cannot decide between them. Independently specifying a preferred port, tensor factor, unit of time, or detector can change the answer. Such a specification is itself a physical premise. We use examples in which the compatible class, or a sharp property of it, can be computed. We start with the selection of operators, then ask what a calibrated spectrum reveals about a generator. The same issue reappears when the measuring operation is itself unknown.

From Prephysical Hypotheses to Physical Models

Structure, observation and physical identification

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Proposed relational structures
Relations and symmetries specify a mathematical description. Its observable meaning and physical units require a justified assignment.

Representation and apparatus
Specify states, operators, preparations, accessible effects and units. Retain uncertain detector, source and clock calibrations in the model class.

Which properties are shared by every description with the same outcome laws?

Specified observations

Conditional inferences and their limits

Algebra and symmetry
Known finite operators, their commutant, and spin fluctuations in a specified state.

Commutant multiplicities control invariant rank-one choices. Vanishing averaged commutators alone do not identify independent subsystems.

Labelled spectral probes
Matched populations, known ports, zero-mode probabilities and unconditional first scale moments.

Nonzero zero-mode contrast on every labelled pair recovers the graph generator in the given units. Imaginary coherences can conceal it.

States, spectra and dynamics
Equal-time fermionic covariances, positive spectral moments and calibrated dynamical measurements.

Distinct dispersions can share an occupied projector. Moments constrain support only in the stated measure class. Wilson entropy depends on the mass-scaling regime.

Phase and purity
A known match population and a common phase detector with nonzero overlap on both probed directions.

A saturating harmonic fixes a rank-one block at positive weight. A detector at an overlap endpoint can leave the coherence undetermined.

Four-input quantum instrument
All four success rates and the specified binary records, with one common first-qubit detector and the stated product structure.

For the stated example, calibration narrows the compatible local Choi-weight interval from $[0, 0.9907077\ldots]$ to $[0, 37/38]$. This does not identify a unique operation.

What would test the physical interpretation?

Independent controls and observations reserved for testing
Anchor preparations, port bases, units and readout effects. Vary source populations, losses and detector orientation. Fix eligibility before private randomization and retain failed preparations. Include sampling and calibration uncertainty jointly.

Compatible descriptions
Report the surviving class or property interval. Seek an additional observation that separates members of that class.

Outside the admissible observation set
The test challenges the joint physical model and calibration assumptions at its stated error level.

Meaning of the arrows. The dashed arrow marks an open physical correspondence. Solid horizontal arrows connect specified observations to conditional mathematical results. The lower branches are possible conclusions of a prospective test, not reported experimental outcomes. The five examples illustrate observation problems; they do not exhaust the program or constitute a derivation of a unique physical ontology.

Sources. K. Justus, *Identification from Restricted Measurements*; *Operational Identification and Conditional Quantum Instruments*; *Finite Part of Wilson–Dirac Relative Entropy*; and *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* (2026), in the accompanying *Mathematical Supplement*.

Method and scope

For each example we specify a model class, the interventions an observer may perform, and what is recorded on one attempt. We then ask which values of a property survive all those observations. For example, two Hamiltonians may give the same occupied state at every momentum while assigning different excitation energies. A timed measurement distinguishes them only if the time unit and the measured coupling are independently known.

The graph and instrument examples illustrate different observation contracts. For the tested graph pair, the scale readout retains 36.08% of spectral total variation. The calibrated four-input example allows local weight in $[0, 37/38]$. Orthogonal binary outputs with perfect return fix $w = 1/2$ over the complete input basis. These are results for the specified examples and their stated assumptions.

For a collection of exact outcome laws $p = (p^e)_{e \in \mathcal{E}}$, the *identified set* for h is

$$\mathcal{J}_h(p) = \{h(\theta) : \theta \in \Theta, \quad P_\theta^e = p^e \text{ for all } e\}.$$

An empty set excludes the model class at those population values. A singleton identifies the property. To prove that a bound is *sharp*, we give an inequality for every compatible description and a description attaining the endpoint, or state explicitly when an endpoint is only a limit. These notions use the standard observational criterion for identification [3].

Actual measurements give counts and calibration uncertainties. If $\mathcal{C}$ jointly covers the true outcome laws and calibration parameters with stated probability, the union of the corresponding identified sets inherits that coverage under the model. The sequential test in Section 10 instead bounds the probability of ever rejecting a specified conditional model. Neither procedure assigns a probability to the competing physical descriptions. Figure 1 shows the role of each measurement, and the glossary gives the conventions for the quantities used below.

Proposition 1.1 (Stability of an identified property)**.** *For a finite list of experiments on finite outcome spaces, put* $d(\theta, \theta') = \max_e \mathrm{TV}(P_\theta^e, P_{\theta'}^e)$*. Suppose the description space is compact metric, all the outcome maps are continuous and* h *is a continuous real property. If the exact observations identify* h*, then* $\sup_{d(\theta,\theta') \le \delta} |h(\theta) - h(\theta')| \to 0$ *as* $\delta \downarrow 0$*.*

Proof. Otherwise pairs with observational distance tending to zero have a property difference bounded away from zero. A subsequence converges in the product space. Continuity makes the limiting descriptions observationally equivalent, while their property values are distinct. This contradicts identification. □

This result assumes a topology on the model class and implies no numerical rate. For a prephysical structure x and representation $b(x)$, the relevant observational map is $x \mapsto (P_{b(x)}^e)_e$. If b is itself undetermined, competing assignments must be included in the description space. Fixing a representation can shrink the equivalence class before any observation has been made.

2 Invariant structure and physical labels

Consider a finite-dimensional Hilbert space and specified self-adjoint operators $A_1, \dots, A_k$. Their algebra $\mathcal{A}$ is the unital $*$-algebra they generate. A data-preserving unitary commutes with every A_j. Ask whether these data select a complete unordered family of rank-one orthogonal projections.

Theorem 2.1 (Invariant selection)**.** *Write the finite-dimensional representation as*

$$\mathcal{A} = \bigoplus_{\alpha} (M_{n_\alpha} \otimes I_{m_\alpha}).$$

A complete resolution invariant under all data-preserving unitaries exists exactly when every multiplicity m_α equals one. When it exists, it is unique exactly when $\mathcal{A}$ is maximal abelian.

Proof. The commutant's unitary group is a product of connected unitary groups. It acts on an invariant finite set of projections by permutations. Every orbit is connected and hence a single point. Each projection therefore commutes with the entire commutant and belongs to the finite-dimensional bicommutant $\mathcal{A}$. A projection in a block of multiplicity m_α has rank divisible by that multiplicity. Such a block cannot be covered by rank-one projections when $m_\alpha > 1$. In multiplicity one any choice of orthonormal basis within each block gives an invariant resolution. A block of dimension exceeding one admits more than one such basis. If every block is one-dimensional, its minimal projections give the unique resolution. □

There is a calculable difference between exact selection and stable selection. Let $A_1 = Z$, $A_2 = \varepsilon X$ on a qubit, with X, Y, Z the Pauli matrices. For every $\varepsilon > 0$ they generate M_2. Yet the minimum eigenvalue of $\sum_j [A_j, [A_j, \cdot]]$ on the traceless Hermitian matrices is $4\varepsilon^2$ for $0 < \varepsilon \le 1$: the Pauli basis diagonalizes this map with eigenvalues 4, $4(1+\varepsilon^2)$ and $4\varepsilon^2$, respectively. An arbitrarily small additional coupling removes the exact commutant while leaving a weakly distinguished direction.

A similar calculation locates what a relational phase means. If one known operator has simple spectrum, rephasing its eigenvectors sends an off-diagonal entry B_{ij} of a second operator to $e^{i(\phi_j - \phi_i)} B_{ij}$. Its magnitude and every phase product on a closed cycle are invariant. Conversely, choose phases on a spanning tree of each support component to make the tree entries positive. The remaining edge phases are fixed by their fundamental cycles. With the diagonal entries and labels fixed, this identifies the pair up to the allowed rephasings. A position basis or a physical separation requires an additional identification of the vertices.

Fluctuation variables. Even a familiar physical representation can depend on which states and operations are observed. In a product of N spin-$1/2$ systems with single-site state $(I + mZ)/2$, $0 < m \le 1$, set $Q_N = \sum_j X_j / \sqrt{2mN}$ and $P_N = \sum_j Y_j / \sqrt{2mN}$. Then

$$[Q_N, P_N] = \frac{i}{mN} \sum_j Z_j, \qquad \mathbb{E}|[Q_N, P_N] - iI|^2 = \frac{1 - m^2}{Nm^2}.$$

The minimum eigenvalue of $\sum Z_j$ is $-N$, so the operator-norm discrepancy is $1 + 1/m$ at every N. For fixed u, v, putting $a = u/\sqrt{2mN}$, $b = v/\sqrt{2mN}$ gives

$$\mathbb{E}(e^{iuQ_N} e^{ivP_N}) = (\cos a \cos b - im \sin a \sin b)^N \longrightarrow \exp[-(u^2+v^2)/(4m) - iuv/2].$$

This follows by multiplying the one-site factors and expanding at the origin. In the tracial state $m = 0$ the appropriate $1/\sqrt{N}$ normalization instead gives $e^{-(u^2+v^2)/2}$ for the ordered characteristic function. The limiting commutator and representation are state-dependent. These spin examples are elementary instances of quantum fluctuation theory [7]. They begin with quantum spins. Identifying subsystems further requires the generated local algebras to commute as operators. An expected commutator can vanish even for two Pauli matrices generating all of M_2.

3 From labelled spectral data to a generator

A graph Laplacian provides a case in which stronger observations close an inverse problem. Let L be the real symmetric combinatorial Laplacian of a connected weighted graph on n labelled vertices. Set $P_0 = \mathbf{1}\mathbf{1}^T/n$ and $K = L^+$. Assume a source can address the indicated vertex modes. Consider the continuous positive operator-valued measure $\Pi(ds) = 4sL^2e^{-2sL}\,ds$ on positive scales, completed by a separate output P_0. This model uses graph spectral filters and the established resistance pseudoinverse [8, 10]. The unit of s is set by the generator and its calibration.

Theorem 3.1 (Pair reconstruction)**.** *For two normalized preparations supported on the same pair i, j, suppose their diagonal populations match. Let $\Delta\mu$ be their difference in unconditional first scale moments and Δp_0 their difference in zero-mode probabilities. If $\Delta p_0 \neq 0$, then*

$$K_{ij} = \frac{\Delta\mu}{n\Delta p_0}, \qquad R_{ij} = K_{ii} + K_{jj} - 2K_{ij}.$$

Observing all off-diagonal entries fixes K and L in the given labels and units.

Proof. At positive eigenvalue λ the density is $4s\lambda^2e^{-2s\lambda}$. Its zeroth and first moments are 1 and $1/\lambda$. It follows that $\int \Pi = I - P_0$ and $\int s\Pi = K$. The difference of the preparations has entries $z, \bar{z}$ off diagonal and zeros elsewhere. Its traces against K and P_0 are respectively $2\Re(z)K_{ij}$ and $2\Re(z)/n$. Division gives the result. Since $K\mathbf{1} = 0$, the off-diagonal entries determine each diagonal, and $K^+ = L$. □

Observing every pair does not ensure a usable contrast. The preparations $(|i\rangle \pm i|j\rangle)/\sqrt{2}$ have identical probabilities under every real symmetric spectral filter. On either the 4×4 rook graph or the Shrikhande graph, the weights at Laplacian eigenvalues $0, 4, 8$ are $(1, 6, 9)/16$ on every one of the 120 labelled pairs. Translation invariance fixes each projector diagonal to its rank divided by 16; the imaginary off-diagonal coherence cancels. The graphs have eight and zero four-cliques, respectively. Their full scale laws therefore coincide under these preparations despite their inequivalent adjacency. Real plus/minus preparations instead have $\Delta p_0 = 2/n$. This control, proved in the mathematical review, concerns different graphs and observations from the finite-moment comparison below [2].

The loss is quantitative on the labelled pair $(0, 0), (0, 2)$ for the two graphs just considered. Prepare $(|i\rangle + e^{i\phi}|j\rangle)/\sqrt{2}$ with $c = \cos\phi$. Measuring the Laplacian eigenvalue gives total variation $|c|/4$ between the two laws. The scale record, including its separate zero outcome, gives $(3 + 4\log 2)|c|/64$. For this pair, therefore, the scale filter retains about 36.1% of the one-record spectral distinction when $c \neq 0$. Both readouts agree at $c = 0$. The review derives the contraction and the signed-phase bounds for independent repetitions.

Population recovery leaves a separate precision question. For one edge of conductance g, the minus preparation has a gamma scale law of mean $1/(2g)$. In fixed units, g and $g+\sqrt{g}$ diverge in absolute difference while their laws for any fixed budget approach each other, even with adaptive preparations. Independent minus preparations nevertheless give an exact relative confidence interval. Both statements are proved in the mathematical review.

For a common finite measurement window $[a,b]$, the omitted first-moment operator is positive. With positive spectrum in $[\lambda_2,\lambda_{\max}]$ its norm is bounded by

$$B=\tfrac{4}{3}\lambda_{\max}^2a^3+\lambda_2^{-1}e^{-2\lambda_2 b}(1+2\lambda_2 b+2\lambda_2^2b^2).$$

The early integral uses $e^{-2s\lambda}\le 1$. The late integral is the decreasing upper Gamma tail evaluated at λ_2, times $1/\lambda_2$. Subtracting $BI/2$ bounds each omitted off-diagonal entry by $B/2$. A small coherence contrast still amplifies statistical error in the ratio. Means conditioned on a positive click require separate normalization for the zero mode and lost events.

Local precision and global ambiguity. For a known eigenmode and scale u, let $f_u=u\lambda e^{-u\lambda}$. A coherent probe with unknown real amplitude a produces homodyne means $\sqrt{2}av$ with independent variance $1/2$ for $v=(f_s,f_t,\sqrt{1-f_s^2-f_t^2})$. Normalization gives $v^Tv'=0$, so the amplitude and eigenvalue scores are orthogonal and the Fisher information for λ is $4a^2\|v'\|^2$. This is a local sensitivity. When $s=t$, xe^{-x} takes each value in $(0,e^{-1})$ twice, and the complete output vector agrees at both positive eigenvalues. More repetitions of that fixed experiment leave this ambiguity intact.

Which receiver optimizes local precision? With $x=s\lambda$, $y=t\lambda$, $f(u)=ue^{-u}$ and $z(u)=u(1-u)e^{-u}$, the full-output information is $(a^2/\lambda^2)C(x,y)$, where

$$C/4=z(x)^2+z(y)^2+\frac{[f(x)z(x)+f(y)z(y)]^2}{1-f(x)^2-f(y)^2}.$$

Set $g(u)=z(u)^2/(1-2f(u)^2)$. Cauchy–Schwarz bounds $C/4$ by $[z(x)^2+z(y)^2]/[1-f(x)^2-f(y)^2]\le 2\max g$. Equal scales at a maximizer attain this upper bound. Equality requires $z(x)/f(x)=z(y)/f(y)$, hence $x=y$. The derivative of g has the sign of $-(u-1)F(u)$, $F=u^2-3u+1+2u^3e^{-2u}$, and $F'=(2u-3)(1-2u^2e^{-2u})$. There is exactly one root in $(0,1)$ and one above $3/2$. On $(0,1)$, $g<(1/16)/(1-2/e^2)<2/23$; at $u=5/2$, $g>(225/16)e^{-5}>2/23$ using $e<11/4$. The upper root $r=2.5212815658\ldots$ is therefore the global maximizer, with $C_{\max}=8r^3e^{-2r}=0.8279358532\ldots$. The positive root below one of $f(u)=f(r)$ is $0.2637461879\ldots$. This calculation establishes local optimality over the specified two scales and full readout, together with a global ambiguity at that design.

A single labelled probe can likewise conceal the generator when only a few moments are available. A concrete example compares the pair vector $(e_0-e_1)/\sqrt{2}$ on a 16-vertex rook graph with a weighted graph. For $m_j=\langle K^j\rangle$, the rook edge gives $(m_0,\ldots,m_4)=(1,5/32,7/256,11/2048,19/16384)$. The rook edge has weights $1/4$ and $3/4$ at Laplacian eigenvalues 4 and 8, respectively, giving these inverse moments. The constructive graph certificate in [9] fixes three conductances of a four-vertex complete graph at $1/50$ and encloses a root for the other three using a contraction on a rational box. In the rational certificate the unknown weights are (w_{02},w_{12},w_{13}), near $(15.6899984583,3.9688719038,0.0235628564)$. The reproducibility packet gives the full rational center and interval verifier. For radius 10^{-34}, the exact Newton displacement is

below 4×10^{-63} and the contraction bound is below 5×10^{-27}. Thus the Newton map with derivative frozen at the center maps the box into itself contractively. Its fixed point matches m_1, m_2, m_3 and puts m_4 in $(0.006804564875115, 0.006804564875116)$. To obtain the same vertex count, connect each of twelve new vertices to the original four by conductance $\varepsilon = 3/500$ and subtract 3ε from every original edge. The weights remain positive. On vectors of sum zero supported on the original four vertices, the diagonal shift $12\varepsilon I$ cancels the subtraction $3\varepsilon L(K_4)$, and the cross block vanishes. Hence the full pair measure is preserved under the extension; the added vertices can, for example, be connected as a positive clique. The distinction at the fourth moment is specific to this probe and these graphs.

Orientation and finite size. On a bipartite hopping model let the real diagonal grading G satisfy $GhG = -h$ and let $B(t)$ be any real bond control obeying $GB(t)G = -B(t)$. The propagators satisfy $U_{\bar{h}+B}(t) = G\overline{U_{h+B}(t)}G$: complex-conjugating the evolution equation and applying the grading produces the same differential equation and initial condition. In the site basis their transition intensities agree, even when a closed-loop phase changes sign. A calibrated diagonal detuning has the opposite parity under this transformation and can give a distinguishing experiment.

A heat-trace slope can also change for reasons other than geometry. For $L_c = L + c(I - P_0)$, let $B(s) = \operatorname{Tr} e^{-sL} - 1$. Direct differentiation gives

$$-2s\partial_s \log \operatorname{Tr} e^{-sL_c} = \frac{e^{-sc}B(s)}{1 + e^{-sc}B(s)}[-2sB'(s)/B(s) + 2sc].$$

At fixed finite volume, increasing c at positive s suppresses all nonconstant modes. A thermodynamic limit taken first can suppress the stationary mode's density instead. Interpreting either slope as a spacetime dimension requires a specified diffusion process, scale and limit, as well as evidence for their physical meaning.

4 Energy, time and the observable channel

A spectral parameter acquires physical units through an independently timed or spatially resolved intervention. Even within a specified lattice theory, the static state need not determine the Hamiltonian. If $h(k) = d(k) \cdot \sigma$ is gapped, its negative-band projector is $(I - \hat{d}(k) \cdot \sigma)/2$. Replacing $h(k)$ by $a(k)h(k)$ for a positive real function leaves this projector fixed while changing the excitation energies. A finite-range version uses $h_m(k) = \sin k\, X + (1 + m - \cos k)Z$ with $m > 0$ and multiplier $1 + \varepsilon \cos k$, $|\varepsilon| < 1$. The occupied projector and every equal-time quasifree correlation on the same algebra agree. A calibrated interband oscillation has frequency proportional to $(1 + \varepsilon \cos k)\sqrt{m^2 + 4(m+1)\sin^2(k/2)}$. At $m = 1$ the ratios at $k = \pi/3, \pi$ relative to $k = 2\pi/3$ obey $\sqrt{21}R_{12} + 2\sqrt{7}R_{32} = 9$. This follows by substitution. The first ratio is strictly increasing in ε and identifies it within this one-parameter family [15].

There are also inequivalent finite-range dispersions that agree on every uniform spectral integral. Let $h_n(k) = (E_* + d \cos nk)Z$, $E_* > |d| > 0$. For integer $n \geq 1$, the map $k \mapsto nk$ preserves normalized uniform measure on the circle, so all such integrals of a function of the band energy agree. But a probe with density $(1 + \eta \cos k)/(2\pi)$, $0 < \eta \leq 1$, gives mean energy $E_* + \eta d/2$ for $n = 1$ and E_* for $n = 2$. The new datum breaks the equivalence through its calibrated momentum dependence.

A sharp limitation from Euclidean moments. Let a positive spectral measure be observed at integer multiples of known time spacing $a > 0$. Write $x = e^{-aE}$ and $m_j = \int_0^1 x^j \, d\rho(x)$. Assume $H_0 = (m_{i+j})_{0 \le i,j < n}$ is positive definite and put $H_1 = (m_{i+j+1})$. For a representing measure supported in $[0, v]$, positivity of $\int (v - x)|p(x)|^2 d\rho$ for every polynomial of degree less than n gives $H_1 \preceq v H_0$. Thus v is at least the largest generalized eigenvalue x_n of (H_1, H_0). The orthogonal polynomial of degree n has these generalized eigenvalues as its Gaussian quadrature nodes. Its positive quadrature measure matches degrees 0 through $2n - 1$ and has largest node x_n. One can see exactness by dividing any such polynomial by the degree-n orthogonal polynomial. The quotient is orthogonal to it, and the remainder is integrated by interpolation at the nodes. The weights are positive because the squared Lagrange polynomials integrate to positive numbers. This realizes the sharp lower bound on the largest accessible x, hence the sharp upper bound $-a^{-1} \log x_n$ on the lowest coupled energy for $x_n > 0$. For interior moment data a sufficiently small atom at $x = 1$ can be subtracted from the moment vector while leaving a representable residual. Adding the atom back yields zero lowest energy with the same observations. Boundary moment data need separate treatment. The quadrature/transfer-matrix connection also appears in Lanczos spectroscopy [11].

Positive moments with uncertainty. The quadrature result concerns equally spaced times. Another exact support bound uses a positive probability measure on energy with data $y = (\langle E \rangle, \langle E^2 \rangle, \langle E^3 \rangle, \langle e^{-hE} \rangle, \langle e^{-2hE} \rangle)$, $h > 0$. Suppose $y - y_0$ lies in a compact convex uncertainty set and a feasible measure has three positive masses at $0 < x_1 < x_2 < x_3$. Form

$$F(E) = E^3 + c_2 E^2 + c_1 E + c_0 + b_1 e^{-hE} + b_2 e^{-2hE}.$$

If F vanishes simply at x_1 and doubly at x_2, x_3, it is nonnegative on $[x_1, \infty)$. Indeed $F^{(4)}$ is an exponential times a linear combination of 1 and e^{hE}, with at most one zero unless identically zero. Rolle's theorem bounds the zero count of F by five, with multiplicity, and F is positive at large E. Suppose further that the three-atom data uniquely maximize the linear functional $y \mapsto (c_1, c_2, 1, b_1, b_2) \cdot y$ over the allowed data region. Then every feasible measure integrates F to at most zero. Any such measure supported above or at x_1 must therefore be concentrated on the three zeros, with exactly the same data. Its weights follow from normalization and the first two moments by a Vandermonde system. This proves that x_1 is the largest feasible lower edge and that the maximizing measure is unique under the stated condition. The exact, rectangular and ellipsoidal examples have different optimizers. A proposed transition in the sign of a correction remains a numerical candidate [16].

In a weak Ricker source experiment, a normalized leading response can take the form $F(z) = \int e^{-z\lambda} d\mu(\lambda)$ with $\lambda = \omega^2$ and positive spectral weight proportional to ω^4 times the coupled transition measure. Where $0 < F < \infty$, differentiation gives

$$-(\log F)' = \mathbb{E}_z \lambda, \qquad \frac{d}{dz} \mathbb{E}_z \lambda = -\operatorname{Var}_z(\lambda).$$

Consequently a two-scale logarithmic secant bounds the lowest *coupled* squared frequency from above. A selection rule may remove a lower excitation from μ. A physical clock and a model of the excited channel must already be available to interpret ω.

A field-theory comparison. For a scalar with radial amplitude ρ in a canonically normalized classical Abelian Higgs model, add a homogeneous, independently normalized

source $J\rho^2/2$. On the stable positive branch the potential is $\lambda(\rho^2-v^2)^2/4-J\rho^2/2$. Its minimum satisfies $\rho_J^2 = v^2+J/\lambda > 0$. The linearized squared radial and vector masses are $2\lambda\rho_J^2$ and $g^2\rho_J^2$, from the second derivative and the covariant kinetic term. At $J = 0$ these two poles alone leave a continuous parameter ambiguity: rescale v and adjust g, λ to keep both squared masses fixed. A calibrated pole residue of $\Phi^*\Phi$ in this tree-level convention is v^2, since its linear fluctuation is v times the canonical radial fluctuation. It fixes the remaining scale with the stated field and source normalization. Loop corrections, operator renormalization, or a different action change this inference. The example is within an already specified spacetime field theory [12].

A calibrated compact-group transfer law presents the same issue in a different setting. For the two-dimensional pure $SU(2)$ Wilson plaquette measure the normalized spin-j eigenvalue is $I_{2j+1}(\beta)/I_1(\beta)$. At fixed j the large-β expansion of the modified Bessel functions gives $\log\lambda_j = -2j(j+1)/\beta + O_j(\beta^{-2})$. With $\beta = 4/(g^2a^2)$ and $N = A/a^2$ this yields $\lambda_j^N \to \exp[-Ag^2j(j+1)/2]$. In the dimensionless transfer convention $a_{\gamma,n} = -\gamma\log[I_n(\gamma)/I_1(\gamma)]$ and $\ell_n = (n^2-1)/2$, the stronger fixed-time operator-norm limit on either the character space or the full regular representation follows by a separate argument [17]. The integral representation gives $0 < I_{n+1}/I_n < 1$, so $a_{\gamma,n}$ increases with n. At fixed n it tends to ℓ_n. For any cutoff M the supremum of $|e^{-sa_{\gamma,n}} - e^{-s\ell_n}|$ above M is bounded by $e^{-sa_{\gamma,M}} + e^{-s\ell_M}$. Take $\gamma \to \infty$ and then $M \to \infty$ for fixed $s > 0$; the same split works for $(1+a_{\gamma,n})^{-1}$. At moving s the tail bound changes. The proposed full-representation moving-scale trace-asymptotic coefficient in [17] remains an open target. This argument uses a known area, lattice spacing and dimension. A fixed-representation limit alone gives no uniform control over the entire character expansion. Continuum lattice convergence in two-dimensional Yang–Mills theory is a separate, established analysis [14].

In finite dimension the normalized Gibbs states of A and $A+V$ at inverse-temperature parameter $s \geq 0$ have half-trace distance at most $\tanh[s(\lambda_{\max}(V)-\lambda_{\min}(V))/4]$. A commuting two-level system attains this bound. The review gives the proof and distinguishes its domain from the trace-class perturbation result.

What would determine a metric? Even complete unparametrized light paths have a residual ambiguity. For smooth positive Ω, g and $\tilde{g} = \Omega^2 g$ have the same null cones. The change of connection on a null tangent k contributes $2(k\cdot d\log\Omega)k - g(k,k)\nabla\log\Omega$, proportional to k. Thus the two metrics have the same unparametrized null geodesics. On a timelike comparison curve, however, $d\tilde{\tau} = \Omega\, d\tau$ (units $c = 1$). An independently calibrated proper-time comparison can distinguish conformal factors that give different elapsed times on the selected curve. This standard result of Lorentzian geometry [13] begins with a differentiable manifold, a metric class and a causal interpretation. A diffusion scale on a graph does not supply them.

5 What an equal-time fermionic state can determine

The measurement problem is clearer when both the missing observable and the compatible set can be written down. In two number-conserving fermionic modes with gauge-invariant density matrix, let $q_{00}, q_{10}, q_{01}, q_{11}$ be the four occupation probabilities. The only off-diagonal entry allowed is a coherence z within the one-particle sector. Positivity gives $0 \leq |z|^2 \leq q_{10}q_{01}$. Every modulus in this interval gives a positive state and

$$\operatorname{Tr}\rho^2 = \sum_i q_i^2 + 2|z|^2.$$

Thus even noiseless number counting can leave a purity interval. The range can collapse at a boundary probability. This simple projection calculation will reappear for quantum channels.

An exact comparison can also test whether the two-mode state is quasifree, meaning that its correlations obey Wick factorization. Let $C_{ij} = \langle a_i^\dagger a_j \rangle$ and let ω_C be the gauge-invariant quasifree state with that covariance. Its double occupation is $C_{11}C_{22} - |C_{12}|^2$. Subtracting the observed q_{11} yields $W = q_{10}q_{01} - q_{00}q_{11} - |C_{12}|^2$. The two states have the same one-particle coherence and one-mode populations. Their difference has diagonal entries of magnitudes $|W|$ in all four occupation sectors and zero off-diagonal entries. Thus $\|\rho - \omega_C\|_1 = 4|W|$, and the trace distance is $2|W|$. Counting alone does not determine $|C_{12}|$ [18].

A full covariance carries substantially more information than its occupation probabilities. For a Wilson–Dirac chain take

$$h_m(k) = \sin k\, X + (1 + m - \cos k)Z, \qquad P_m(k) = \tfrac{1}{2}(I - h_m(k)/|h_m(k)|).$$

Compress the infinite-chain occupied projector to L consecutive two-orbital cells, giving $C_{L,m}$. Set $m = \mu/L$, $\mu > 0$ fixed. For two strictly interior fermionic covariances A, B, write $S(A) = -\operatorname{Tr}[A \log A + (I - A)\log(I - A)]$ and use the quasifree relative entropy

$$D(A\|B) = \operatorname{Tr}\{A(\log A - \log B) + (I - A)[\log(I - A) - \log(I - B)]\}.$$

Direct expansion gives

$$D(A\|B) = \operatorname{Tr}[(A - B)(\log(I - B) - \log B)] - [S(A) - S(B)].$$

This identity needs no simultaneous diagonalization of the two covariances.

Proposition 5.1 (Wilson comparison and its analytic inputs)**.** *In the common interval scalarization let $N = 2L$, A be the massive covariance, B the massless covariance, $\Delta = A - B$ and $J_{j-1,j} = J_{j,j-1} = j(N - j)/N$ with zero diagonal. Put $K_B = \log(I - B) - \log B$ and*

$$\eta(t) = \pi t\, {}_3F_2(1/4, 1/2, 3/4; 1, 3/2; 4t^2), \qquad E_N = K_B + N\eta(J/N).$$

The finite identity gives $D = P + Q + M - \Delta S$, where

$$P = -\pi \operatorname{Tr}(\Delta J), \quad Q = -N \operatorname{Tr}\{\Delta[\eta(J/N) - \pi J/N]\}, \quad M = \operatorname{Tr}(\Delta E_N), \quad \Delta S = S(A) - S(B).$$

The collected proof gives $D = (\mu^2/6)\log L + O_\mu(1)$ and states separately sufficient conditions for convergence of the finite part [19]; the leading-law proof appears in [20]. With the two-bond, nonlinear, modular and boundary-entropy estimates of [24, Secs. 2–7], the finite part is

$$D(C_{L,\mu/L}\|C_{L,0}) = \frac{\mu^2}{6}\log L + \mu^2\left(\frac{13}{36} - \frac{\pi}{12} - \frac{\log\mu}{6}\right) - \mathcal{S}(\mu) + O_\mu((\log L)^{-1}).$$

Proof. Insert $K_B = E_N - N\eta(J/N)$ in the covariance identity and split the linear term. The precise estimates used in the asymptotic implication are

$$P = \frac{\mu^2}{6}\log L + \mu^2\left(\frac{1}{6}\log\frac{8}{\mu} - \frac{1}{4}\right) + O_\mu(\log L/L), \qquad M = O_\mu(N^{-1/2}),$$
$$Q = \mu^2 c_\eta + O_\mu((\log N/N)^{1/3}), \qquad \Delta S = \mathcal{S}(\mu) + O_\mu((\log L)^{-1}),$$

with $c_\eta = 11/18 - \pi/12 - (\log 8)/6$. Here $\mathcal{S}(\mu)$ is the finite cross-boundary trace-path limit of the entropy difference. The constant follows by addition, and the largest of the four remainder orders is $O_\mu(1/\log L)$. For a concrete definition of the boundary term, let χ restrict to the unit interval and let P_μ^c have continuum symbol $[I-(uX+\mu Z)/\sqrt{u^2+\mu^2}]/2$. Set $Y_\mu = \chi P_\mu^c(I-\chi)$, form the off-diagonal Hermitian block matrix $\mathcal{H}(Y)$ with upper right block Y, and put $\mathcal{T} = \mathcal{H}(Y_\mu - Y_0)$, $A_t = \mathcal{H}(Y_0) + t\mathcal{T}$. Then $\mathcal{S}(\mu) = \frac{1}{2}\int_0^1 \mathrm{Tr}[h'(A_t)\mathcal{T}]\,dt$, with $h(x)$ the binary entropy at $(1+\sqrt{1-4x^2})/2$. The reference spectral interpolation, nonlinear Jacobi estimates and trace-class boundary theorem enter through the cited estimates. The modular bound uses Theorem 4 and Corollary 5 of [24], the nonlinear rate uses its Lemma 6 and Theorem 7, and the entropy limit uses its Section 7 and [21]. The two-bond expansion is in [19] and [22]. □

One can see precisely how an estimate on the reference spectrum controls the entropy error. For a positive majorant G suppose $\mathrm{Tr}\,G \le M$, $\mathrm{Tr}(G\,\mathbf{1}_{|J|>x}) \le a/x + b/x^2$, $\|E_N\| \le 4\sqrt{N}$, and the simultaneous spectral modes of J, E_N have discrepancy at most C/N on $|\lambda| \le N/4$. Splitting the trace at $N/4$ gives

$$\mathrm{Tr}(G|E_N|) \le CM/N + 16a/\sqrt{N} + 64b/N^{3/2}.$$

This proof does not need G to commute with J. The interior approximation and the tail majorant enter in different regions of the same estimate.

For the same half-filled finite matrices, the stronger estimate $\|E_N\| \le 7$ holds for every even $N \ge 2$ [23, Theorem 3]. With $N = 2L$ and fixed $\mu > 0$, Corollary 4 gives a full modular correction $O_\mu(\sqrt{\log N}/N)$, and Theorem 6 gives a nonlinear remainder $O_\mu(\sqrt{(\log N)/N})$. These statements apply at the actual finite eigenmodes; they do not assert a moving-edge interpolation expansion. The full relative-entropy remainder remains $O_\mu(1/\log L)$ in the stated argument because the boundary entropy estimate controls that order.

The leading coefficient and finite constant describe a specific equal-time restriction of specified states. There is no uniform assertion as $\mu \downarrow 0$. Even an exact value for this functional cannot resolve Hamiltonians with the same occupied projectors. The commuting-operator treatment of entanglement Hamiltonians in free chains gives related spectral context [25].

6 Identifying states when the detector also varies

If both state preparations and measurements are unknown, a similarity transformation can preserve complete experimental sequences. For the labelled directions $j = X, Y, Z$, take $\rho_{\pm j} = (I \pm a\sigma_j)/2$ and binary effects $E_{\pm j} = (I \pm b\sigma_j)/2$. Bloch scaling by $3/2$ carries $(a, b) = (1/2, 3/4)$ into $(3/4, 1/2)$ with the inverse scaling on effects. All one-step probabilities agree. With each accessible map transformed by the same similarity, internal factors cancel in every sequence probability whenever the two restricted models remain physical. The preparation purities are $5/8$ and $25/32$. This is a special case of gate-set gauge freedom [26].

Entangling capability by itself does not universally fix that scaling. In the CNOT example $\Phi = 0.37\,\mathrm{Ad}_{\mathrm{CNOT}} + 0.63\,\mathcal{D}$, with $\mathcal{D}$ the two-qubit depolarizing channel, let G multiply a Pauli string of weight k by $(3/2)^k$ and put $\widetilde{\Phi} = G\Phi G^{-1}$. Its unnormalized Choi matrix

$J = \frac{1}{4}\sum_P P^T \otimes \widetilde{\Phi}(P)$, summed over the sixteen Pauli strings including identity, has spectrum and multiplicities

$$(1/300, 4), \quad (23/240, 1), \quad (63/400, 6), \quad (187/600, 4), \quad (2039/1200, 1).$$

These follow by applying the CNOT Pauli permutation in the displayed matrix. The map is trace preserving, and the listed eigenvalues are positive. On input $|+\rangle|0\rangle$, the least eigenvalues after output partial transposition are $-11/400$ and $-3/25$ respectively. Both channels can therefore entangle a product input. A sharp extra hypothesis changes the problem: if the gate is a unitary conjugation, its scaled conjugate must have determinant of magnitude one. A unital TP completely positive map contracts the Hilbert–Schmidt norm by Schwarz's inequality. The determinant forces all singular values to equal one, so the unitary conjugation must commute with the squared Bloch scaling. It preserves the one-body Pauli space. The two commuting simple Lie ideals in this space must be preserved or exchanged, giving local unitaries, possibly followed by SWAP. Thus a genuinely entangling unitary excludes this particular similarity. Other gauges and other noise models have their own identifiability conditions.

Incomplete calibration. With trusted states ρ_i and a fixed effect $0 \preceq E \preceq I$, reference click probabilities constrain $\operatorname{Tr} E\rho_i$. The sharp possible target probability is its minimum or maximum over that convex set [27]. On a qubit, trusted computational-basis click rates p_0, p_1 fix the diagonal of E. Both E and $I - E$ must be positive, giving

$$|E_{01}| \le R = \min\{\sqrt{p_0 p_1}, \sqrt{(1-p_0)(1-p_1)}\}.$$

The click interval on $|+\rangle$ is $[(p_0+p_1)/2 - R, (p_0+p_1)/2 + R]$, attained by the two real endpoint coherences. Thus incomplete tomography can give exact predictions for some calibration outcomes and almost no prediction for others.

A reference with which to test a detector. For a scale input with density $f_c(u) = 4c^2 u e^{-2cu}$ suppose a common detector event has response $0 \le h(u) \le 1$. For couplings 1 and 2 the likelihood ratio $4e^{-2u}$ decreases. To maximize the second click rate at fixed first rate p, allocate acceptance to the lowest values of u; to minimize it, allocate acceptance to the highest. The maximizing rule follows by integrating the difference from a threshold rule against $f_2 - k f_1$, whose sign changes at the threshold. With F_c the incident CDF the exact interval is

$$[1 - F_2(F_1^{-1}(1-p)),\ F_2(F_1^{-1}(p))].$$

Randomizing between these rules realizes the interior. Concavity of the upper envelope and convexity of the lower preserve necessity under a stable unknown mode mixture when all eigenvalues scale by the same factor. Passing the test shows compatibility and cannot alone distinguish a changing source from a changing detector.

Phase interventions. Let P be a rank-one match projector, $Q = I - P$, $s = \operatorname{Tr} P\rho$, $h = \operatorname{Tr} EP$ and $U_\phi = P + e^{i\phi}Q$. With a common conditional state and effect, $q_\phi = \operatorname{Tr}(EU_\phi \rho U_\phi^*)$ contains just one harmonic. Put $a = sh$, $b = (q_0 + q_\pi)/2 - a$ and $c = [q_0 - q_\pi + i(q_{\pi/2} - q_{3\pi/2})]/4$. A quartet is feasible exactly if its two opposite-arm sums agree and

$$0 \le b \le 1 - s, \qquad |c|^2 \le \min\{shb,\ s(1-h)(1-s-b)\}.$$

Indeed an expansion gives $q_\phi = a + b + 2\Re(e^{-i\phi}c)$. Cauchy–Schwarz applied to the two projected components gives the first disk from E and the second from $I-E$. Conversely take the pure vector with components $\sqrt{s}, \sqrt{1-s}$ in two directions and an effect with diagonals $h, b/(1-s)$ and the appropriate off-diagonal of modulus $|c|/\sqrt{s(1-s)}$. The two determinant conditions give positivity of the effect and of its complement. At $s = 0, 1$ the phase is unobservable by support. Even a fully measured quartet is compatible with a pure-state realization, so its harmonics alone do not identify purity. A physical implementation must characterize the joint phase operation as well as the transfer of its readout.

7 Purity and the probability that an operation succeeds

A history-conditioned quantum operation describes a selected outcome of an instrument. Let $\mathcal{B}$ be a nonzero CP trace-nonincreasing map on two qubits, and $H = \mathcal{B}^\dagger(I)$ its success effect. Put $p_* = \operatorname{Tr} H/4$. With output-first vectorization, the trace-one Choi state C has input marginal $H^T/(4p_*)$. Trace preservation gives $I/4$. Conditioning on success generally changes that marginal. The branch conditional on input i is meaningful only if $s_i = \langle i|H|i\rangle > 0$. The success of a particular preparation must therefore be related to the mean success on $I_4/4$.

Theorem 7.1 (Purity interval from one input)**.** *Fix a pure basis input i, its successful joint output σ, and its Choi weight share $t = s_i/(4p_*) \in (0, 1]$. Write $P = \operatorname{Tr}\sigma^2$, $\ell = \lambda_{\max}(\sigma)$. In the class fixing only these data and permitting a common absolute scaling of a nonzero branch, the exact possible values of $\operatorname{Tr} C^2$ form the interval*

$$\left[t^2P + \frac{(1-t)^2}{12}, \quad t^2P + (1-t)^2 + 2t(1-t)\ell\right].$$

Proof. The input sector has dimension four and complementary dimension twelve. Write the corresponding positive block matrix as $C = \left(\begin{smallmatrix} t\sigma & B \\ B^* & D \end{smallmatrix}\right)$, $\operatorname{Tr} D = 1 - t$. Then $\operatorname{Tr} C^2 = t^2P + \operatorname{Tr} D^2 + 2\|B\|_F^2$. The minimum uses $B = 0$ and $D = (1-t)I/12$. Block positivity gives $B = (t\sigma)^{1/2}KD^{1/2}$ for a contraction K, also for singular blocks by restriction to their supports. Hence $\|B\|_F^2 \le t\ell(1-t)$ and $\operatorname{Tr} D^2 \le (1-t)^2$. To attain both bounds, coherently couple a top eigenvector of the input block with one complementary direction, with squared amplitudes $t\ell$ and $1-t$. Keep all other eigenvalues of $t\sigma$ in their original directions. Every positive trace-one matrix so obtained can be scaled to the Choi matrix of a trace-nonincreasing map. A continuous mixture between the endpoints yields every intermediate purity. □

For TP channels $t = 1/4$, giving numerical endpoints $(P+3/4)/16$ and $(9+6\ell+P)/16$. The TP constraint is stronger than $t = 1/4$, but both endpoints can still be attained. The lower witness prepares $I/4$ on each other basis input and dephases across inputs. For the upper witness, take a Schmidt purification of σ as the first isometry column. Complete it with three orthogonal system vectors perpendicular to the top system Schmidt vector, all tensored with the top environment vector. These columns form an isometry. The environment state on the maximally mixed input is the mixture of the first environment state of weight $1/4$ and its top ray of weight $3/4$. Its nonzero spectrum is that of the Choi state. Squaring this mixture yields the upper endpoint. A pure normalized branch can instead have a single nonunitary Kraus operator.

With two local readings on the single prepared input, let a be the larger eigenvalue of the first marginal, β a rank-one outcome probability on the second, $b = \min(\beta, 1-\beta)$ and $m = \min(1, a+b)$. A unitary fits those readings exactly if $1-a \le \beta \le a$: for a pure bipartite output the two marginal spectra agree, and rotating the second measurement basis attains that entire interval. In general block positivity bounds the largest joint-output eigenvalue by m. For $a + b < 1$, the state

$$(\sqrt{a}|00\rangle + \sqrt{b}|11\rangle)(\sqrt{a}\langle 00| + \sqrt{b}\langle 11|) + (1-a-b)|10\rangle\langle 10|$$

attains it. For $a + b \ge 1$ use the pure rotated Schmidt state. Hence the largest joint-output purity is $m^2 + (1-m)^2$. Dephasing both measured bases minimizes the joint-output purity for given diagonal marginals: minimize $(a-x)^2 + x^2 + (1-a-b+x)^2 + (b-x)^2$ over $\max(0, a+b-1) \le x \le \min(a, b)$. The unconstrained minimizer is $(a+b)/2 - 1/4$. These extremizers inserted into Theorem 7.1 give the sharp interval for local data alone. Additional measurements need not share their extremizers.

Multiple measurements of a fixed Choi state. The same geometry explains how additional tomography can improve a lower purity bound. Suppose calibrated data fix the populations d_j of mutually orthogonal sectors of ranks k_j covering the compatible support. If $C_* = \sum_j d_j P_j / k_j$ satisfies the remaining constraints, then

$$\operatorname{Tr} C^2 = \sum_j d_j^2 / k_j + \|C - C_*\|_F^2$$

because $\operatorname{Tr}(CC_*) = \operatorname{Tr}(C_*^2)$. Extra linear observables M_i can only restrict $C - C_*$ in the tangent space T of allowed Hermitian perturbations. Project M_i onto T, let G be their Hilbert–Schmidt Gram matrix and $u_i = \operatorname{Tr} M_i(C - C_*)$. Orthogonal least squares gives $\|C - C_*\|_F^2 \ge u^T G^+ u$. The least-norm perturbation attains equality whenever C_* plus that perturbation is positive.

For a concrete phase example take normalized Choi vectors $p = |I_4\rangle, u = |iZ \otimes I_2\rangle, v = |iX \otimes I_2\rangle$, with vectorization divided by two. Use the phase about p and detector ray $\sqrt{h}\, p + \sqrt{1-h}\, u$. Assume $0 < h < 1$ and $d = s + k > 0$. A complete curve $q_\phi = |\sqrt{sh} - e^{i\phi}\sqrt{k(1-h)}|^2$ with the separate match population s fixes the p, u block to a single ray of weight $d = s + k$: the curve fixes the two diagonal populations and the off-diagonal saturates their positivity bound. In detail, the mean equals $hs + (1-h)k$, and the modulus of the first Fourier coefficient equals $\sqrt{h(1-h)}|C_{pu}|$. The specified curve gives $|C_{pu}|^2 = sk$, hence a positive two-dimensional block of rank one. At $h = 0$ or 1 the harmonic vanishes for any coherence. The mathematical review exhibits channels of purities $3/8$ and $1/2$ with the same constant scan at either endpoint and the same basis-input outputs. A coherent input distinguishes them. Under the stated interior assumptions the orthogonal ray is in the kernel. An additional measurement of $|v\rangle\langle v|$ with rate f leaves complementary mass $1 - d - f$ in a space of rank 13. The sector formula gives $J = d^2 + f^2 + (1-d-f)^2/13$. The sectors can be filled by maximally entangled vectors, so their uniform filling satisfies the TP marginal. For first-qubit Pauli readouts on input 00, the observable operators are $4[(I+P)/2 \otimes I] \otimes |00\rangle\langle 00|$, $P = Z, X, Y$. Let $B_0 = I - |\theta\rangle\langle\theta|$, where $\theta = (\sqrt{k}\, p + \sqrt{s}\, u)/\sqrt{d}$ is the null ray. For each observable, first form $M_i' = B_0 M_i B_0 - \sum_j P_j \operatorname{Tr}(P_j M_i)/k_j$. Its partial trace $L_i = \operatorname{Tr}_{\text{out}} M_i'$ is traceless. The ray θ is maximally entangled, so for traceless L, $\operatorname{Tr}_{\text{out}}[B_0(I \otimes L)B_0] = (7/2)L$. Each sector projector has scalar input marginal. Thus $\widehat{M}_i = M_i' - (2/7)B_0(I \otimes L_i)B_0$ removes the partial-trace component without changing any fixed

population. Multiplication of the stated Pauli matrices gives $\mathrm{Tr}(M_i' M_j') - (2/7)\,\mathrm{Tr}(L_i L_j)$ for the projected Gram entries. Its diagonal is $(1230/91, 97/7, 97/7)$ and the cross terms vanish. The corresponding increments in the lower bound are $91u_Z^2/1230 + 7(u_X^2 + u_Y^2)/97$, with reference readings $(d + 6(1-d-f)/13, 1/2, 1/2)$. This bound is attained for small feasible interior perturbations. A boundary support must be recomputed. Its use after conditioning requires the same process to transfer to the product input.

8 All four preparations constrain a two-qubit operation

The single-input interval does not exploit the fact that other preparations must share the same operation. We now identify precisely what the four inputs constrain. Write them as $i = 00, 01, 10, 11$, in order $i = 0, 1, 2, 3$. The output factors will be called A and B. For a common branch with Kraus operators K_j, let

$$S = \sum_i s_i = 4p_* > 0, \quad \mathcal{L} = \{V \otimes I_2 : V \in M_2\}, \quad w = \frac{\sum_j \|\,\mathrm{proj}_{\mathcal{L}}\, K_j\|_F^2}{S}.$$

The orthogonal projection uses the Hilbert–Schmidt inner product. w is the population of a specified operator subspace in the normalized Choi state. Its definition uses a particular tensor decomposition. Let $\tau_i = \mathrm{Tr}_B\, \mathcal{B}(|i\rangle\langle i|)$ be the unnormalized first-output marginal, so $\mathrm{Tr}\,\tau_i = s_i$. For positive subnormalized operators define $F_*(\tau, \omega) = \|\sqrt{\tau}\sqrt{\omega}\|_1$ and

$$d(\tau, \omega) = \mathrm{Tr}\,\tau + \mathrm{Tr}\,\omega - 2F_*(\tau, \omega).$$

This convention measures the successful subdistributions. No extra outcome for the two failure populations enters d.

Theorem 8.1 (Local weight from four output marginals)**.** *Given any positive first-qubit matrices $\tau_0, \ldots, \tau_3$ of traces $0 \le s_i \le 1$ with $S > 0$, the maximum over common CP trace-nonincreasing branches realizing those marginals is*

$$\boxed{w_{\max} = 1 - \frac{d(\tau_0, \tau_1) + d(\tau_2, \tau_3)}{2S}.}$$

Thus a prescribed lower bound $w \ge w_0$ is compatible exactly when $w_0 \le w_{\max}$ in this observation class.

Proof. With respect to the second-bit input and output, write $K_j = \left(\begin{smallmatrix} A_j & B_j \\ C_j & D_j \end{smallmatrix}\right)$. Projection onto $\mathcal{L}$ replaces the diagonal blocks by their average and the other blocks by zero. Therefore

$$S(1-w) = \sum_j [\tfrac{1}{2}\|A_j - D_j\|_F^2 + \|B_j\|_F^2 + \|C_j\|_F^2]. \tag{1}$$

Let $u_i = \sum_j K_j |i\rangle \otimes |j\rangle$. Write X_B for the second-output bit flip, extended by the identity elsewhere. Pair the columns with equal first input bit. The squared distances of aligned columns sum to

$$\sum_{b=0}^{1} \|u_{2b} - X_B u_{2b+1}\|^2 = \sum_j [\|A_j - D_j\|_F^2 + \|C_j - B_j\|_F^2] \le 2S(1-w).$$

Alignment leaves each first marginal unchanged. By the purification characterization of root fidelity [28, Ch. 3], the real overlap in each pair is bounded by F_*. Hence each squared distance is at least its d.

For attainment, take purifications ξ_{2b}, ξ_{2b+1} of the two first marginal matrices in pair b with real overlap $F_*(\tau_{2b}, \tau_{2b+1})$. Use disjoint environment spaces for $b = 0, 1$. Attach output flag $|t\rangle_B$ to the column for input $2b+t$. Distinct flags or environment spaces make the four columns mutually orthogonal with squared norms s_i. They define a contraction because its Gram matrix is $\mathrm{diag}(s_i) \leq I$. Tracing out its environment gives a common CP trace-nonincreasing operation. The off-diagonal second-bit blocks vanish and both overlap bounds are attained. Zero-trace states correspond to zero columns. This proves the formula including its boundary cases. □

Corollary 8.2 (Complete interval of local weights)**.** *For the observations of Theorem 8.1, the attainable set is* $[0, w_{\max}]$*, with* $w_{\max} \geq 1/2$*.*

Proof. In the maximizing construction, input bit t is copied to the second output flag. Each Kraus matrix is block diagonal in that bit. Multiply every Kraus matrix on the left by $I \otimes X$. All resulting matrices are block off diagonal, hence orthogonal to $\mathcal{L}$. The weight is zero. Unitary conjugation on the discarded factor preserves the first-qubit marginal for every input operator, as well as each success rate. Mixing the two operations with weights $\lambda, 1-\lambda$ keeps S fixed and gives weight $\lambda w_{\max}$. Nonnegativity of root fidelity gives $d(\tau_a, \tau_b) \leq s_a + s_b$, so $w_{\max} \geq 1/2$. For these two endpoint constructions, measuring the second flag on a successful input 00 distinguishes them exactly when $s_0 > 0$. □

The same interval holds on $\mathbb{C}^d \otimes \mathbb{C}^2 \to \mathbb{C}^{d'} \otimes \mathbb{C}^2$ for arbitrary positive integers d, d'. Replace the two pair costs by $\sum_{a=0}^{d-1} d(\tau_{a0}, \tau_{a1})$ and take $S = \sum_{a,b} \mathrm{Tr}\, \tau_{ab}$. Here the local subspace consists of $V \otimes I_2$ for rectangular $V : \mathbb{C}^d \to \mathbb{C}^{d'}$. Each input label has one pair whose fidelity can be attained in its own environment sector. The review proves this binary formula including singular marginals and zero-success inputs. For a discarded factor of dimension $q \geq 2$, take all dq basis inputs and keep the complete first-output marginals. For each first-input label a maximize the sum of all real block traces of a positive matrix $X = [X_{bc}]$ with $X_{bb} = \tau_{ab}$, and call that maximum T_a. The full attainable interval is $[0, \sum_a T_a/(qS)]$. Optimal purifications, orthogonal output flags and a cyclic shift exhibit both endpoints. The joint optimization is the published multivariate SDP fidelity up to an explicit normalization [1]. With three pure trine marginals the exact ceiling is $1/2$; independent maximization of all three pairs would give $2/3$. The review gives a dual certificate, a regular-simplex family in every dimension and continuity bounds for imperfect marginals. The data in this theorem are the basis-input marginals; additional calibrated constraints can change the interval.

If first-output tomography also resolves the second output label $b = 0, 1$, write M_{ab} for the subnormalized first-output state on success with the returned label equal to its input value. Put $D = \sum_{ab} \mathrm{Tr}\, M_{ab}$ and $F = \sum_a F_*(M_{a0}, M_{a1})$. The exact interval is $[(D - 2F)/(2S), (D + 2F)/(2S)]$. The matching blocks determine both endpoints. In particular, for orthogonal first outputs within every pair, w is half the fraction of successes returning the input label. A single always-successful pair with perfect return consequently fixes $w = 1/2$. All basis inputs contributing to the normalized Choi weight must be accounted for. The review proves the interval, including zeros and wrong-label outcomes.

For commuting matching blocks, each common eigencomponent has specified purification lengths. Their largest sum occurs when they align. Their smallest resultant

length is the longest length minus the sum of the others, clipped at zero. In a three-label trace-preserving example, even the full joint outputs on all basis inputs allow every weight in $[(2-\sqrt{3})/9, (4+\sqrt{3})/9]$. One coherent-input outcome separates the displayed endpoint channels but still leaves a range of width $2\sqrt{3}/9$. A projection of the returned label onto the prepared equal superposition instead has probability exactly w for every channel in this example. The proof also gives a phase-averaged return identity when labels can change and dual bounds when a calibrated classical readout loses label information. Both coherent preparations and the matched readout have to transfer to the same branch.

Corollary 8.3 (Success totals and resolved outcomes)**.** *Use one common binary effect $E \otimes I_2$ on successful outputs and define per-attempt probabilities r_{iy} with $\sum_y r_{iy} = s_i$. Let $D_{ij} = \sum_y (\sqrt{r_{iy}} - \sqrt{r_{jy}})^2$. Every common branch obeys*

$$D_{01} + D_{23} \le 2(1-w)S.$$

If only s_2, s_3 are measured on the last pair, replace D_{23} by $(\sqrt{s_2} - \sqrt{s_3})^2$. If only their sum is known, their minimum possible contribution is zero. With a fixed common rank-one projective first-qubit readout, all these upper bounds on w are sharp for the stated data.

Proof. Measurement maps the two first-qubit marginals to distributions whose square-root affinity is at least their root fidelity. Equivalently, resolve the aligned purifications with the two square-root effects and apply the reverse triangle inequality separately to their norms. For the success-only pair the inequality reduces to the difference of the two column norms. For a projective readout choose each τ_i diagonal in the measured basis with entries r_{iy}. Then quantum and classical affinities agree and Theorem 8.1 attains equality. If only successes are observed, take proportional states on that pair. If only its sum is observed, use equal successes, subject to $0 \le s_i \le 1$. □

For $s_2 s_3 > 0$ the gain from resolving the last pair is exactly

$$D_{23} - (\sqrt{s_2} - \sqrt{s_3})^2 = 2\sqrt{s_2 s_3}\left(1 - \sum_y \sqrt{q_{2y} q_{3y}}\right), \quad q_{iy} = r_{iy}/s_i.$$

It vanishes when either arm never succeeds or when the two conditional output laws agree. For example, take the four rows $(9,4), (4,9), (16,9), (9,16)$ in hundredths. Both pair discrepancies equal $1/50$ and $S = 19/25$. Success totals alone on the last pair allow $w_{\max} = 75/76$; recording both outcomes gives $37/38$. The equality construction is explicit: choose four orthonormal environment labels e_{by} and set $u_{2b+t} = \sum_y \sqrt{r_{2b+t,y}}\,|y,t\rangle|e_{by}\rangle$. This arithmetic is a test of identification at specified populations.

Why the normalization matters. If only the first two normalized successful states are known, choose the other two successes equal to minimize the unused cost. With $g_i = s_i/p_* > 0$, Theorem 8.1 becomes $g_0 + g_1 - 2\sqrt{g_0 g_1} F(\rho_0, \rho_1) \le 8(1-w_0)$. Equality of g_0 and g_1 alone does not set either to one. With raw successes $(0.02, 0.02, 0.98, 0.98)$ we have $p_* = 0.5$, $g_0 = g_1 = 0.04$. Orthogonal successful first states are compatible with $w = 0.99$. The two remaining inputs account for almost all successes. A calibration of only the two tested arms cannot exclude this explanation.

Consequences for a neighboring output. Let $s_0 = s_1 = p_*$, $w \geq .99$ and take the first-qubit marginal on input 0 to have largest eigenvalue $a = 0.529472934503538\ldots$. Its paired normalized purification has overlap at least $c = 0.96$ after alignment. For a projector with arm-0 probability $x < c^2$, its arm-1 probability is at most $G(x) = (c\sqrt{x} + \sqrt{1-c^2}\sqrt{1-x})^2$, by the angle between the binary square-root probability vectors. The sharp absolute click difference for the fixed first spectrum is $G(1-a) - (1-a) = 0.272953961453098\ldots$. Indeed $G(x) - x$ decreases for $x \geq (1-\sqrt{1-c^2})/2 = 0.36$, and rank-one projectors on arm 0 have $x \in [1-a, a]$. A diagonal pair of marginals and its maximizing projector attain equality in the reduced model.

Let β, α be the second-bit flip probabilities on the two successful inputs. Equation (1) gives $\alpha + \beta \leq .04$. A positive block matrix has top eigenvalue at most the sum of the norms of its diagonal blocks, by block positivity and Cauchy–Schwarz. Applied to the two output flags this bounds the first joint-output eigenvalue by $a+\beta$, and the second by $G(a)+\alpha$. In a pure successful dilation each joint output has the same nonzero spectrum as its environment. Applying G to the environment projector maximizing the second output gives the additional bound $G(a+\beta)$. The maximum of $\min\{G(a+\beta), G(a)+0.04-\beta\}$ occurs at the crossing $\beta = 0.022229575038\ldots$ and is less than 0.810954608659. The first two environment states each contribute $1/4$ to the environment of the Choi purification. The remaining weights sum to $1/2$. Its largest eigenvalue is thus at most $t = (2 + a + G(a) + 0.04)/4$. Since $t > 1/2$, $(t, 1-t, 0, \ldots)$ majorizes its spectrum, giving $\operatorname{Tr} C^2 \leq t^2 + (1-t)^2 < 0.732104302723$. Neither upper ceiling is known to be jointly attained under all these constraints. This distinction matters when a bound on a property is used to infer a particular channel.

9 What a calibrated detector adds

A click on an imperfect detector does not reveal which spectral projector would have fired in an ideal projective measurement. Suppose that the same first-qubit effect $E = LI + (U-L)P_+$ acts in all four arms, with $0 \leq L < U \leq 1$ and $P_- = I - P_+$. Let z_i denote the per-attempt probability of success and a click. The two success-weighted populations $v_{i\nu} = \operatorname{Tr}(P_\nu \tau_i)$ obey $v_{i+} + v_{i-} = s_i$ and $z_i = U v_{i+} + L v_{i-}$, hence

$$v_{i+} = \frac{z_i - L s_i}{U - L}, \qquad v_{i-} = \frac{U s_i - z_i}{U - L}. \tag{2}$$

Corollary 9.1 (Four preparations with a calibrated binary effect)**.** *Suppose* $0 \leq s_i \leq 1$, $S = \sum_i s_i > 0$, *and* $L s_i \leq z_i \leq U s_i$ *for all four inputs. Define*

$$D^{\mathrm{sp}}_{ab} = \sum_{\nu=+,-} (\sqrt{v_{a\nu}} - \sqrt{v_{b\nu}})^2.$$

Among common completely positive, trace-nonincreasing branches with these rates and this local effect, the largest compatible weight is

$$w_{\max} = 1 - \frac{D^{\mathrm{sp}}_{01} + D^{\mathrm{sp}}_{23}}{2S}. \tag{3}$$

Under these observations alone the full compatible interval is $[0, w_{\max}]$.

Proof. The spectral projectors constitute one projective measurement of the first qubit. Equation (2) fixes its successful outcome masses. Corollary 8.3 bounds their discrepancy. To attain equality choose $\tau_i = v_{i+} P_+ + v_{i-} P_-$. The orthogonal flagged columns in the

proof of Theorem 8.1 realize all four states in one branch with success effect $\mathrm{diag}(s_i) \preceq I$. Measuring E on each state reproduces z_i. The construction includes zero columns for zero-success inputs. Flipping the second flag and mixing as in Corollary 8.2 attains each lower weight. □

The raw click/no-click rows are obtained by sending the two spectral outcomes through a binary stochastic matrix with click probabilities U and L. For each reported outcome, Cauchy–Schwarz shows that its square-root affinity is at least the sum of the corresponding weighted spectral affinities. Thus $D_{01}^{\mathrm{raw}} + D_{23}^{\mathrm{raw}} \le D_{01}^{\mathrm{sp}} + D_{23}^{\mathrm{sp}}$. The raw bound is sharp if the common detector is unrestricted and a projective effect is allowed. At fixed E, sharpness is given by (3). If $U = L$, consistency requires $z_i = Ls_i$; the sharp formula then uses only $(\sqrt{s_a} - \sqrt{s_b})^2$ for each pair. A conditional click probability need not be defined on an arm that never succeeds.

Table 1. Four inputs measured with one calibrated first-qubit effect.

Input	s_i	z_i	v_{i+}	v_{i-}
00	0.13	0.080	0.09	0.04
01	0.13	0.050	0.04	0.09
10	0.25	0.146	0.16	0.09
11	0.25	0.104	0.09	0.16

For this exact example, take $(L, U) = (0.2, 0.8)$. All entries are population values per attempted input. The raw discrepancy is $0.014124289562550\ldots$, giving $w_{\max}^{\mathrm{raw}} = 0.990707704235164\ldots$ in the broader class. In the calibrated class each pair contributes $1/50$, $S = 19/25$, and $w_{\max} = 37/38$. The calibrated interval $[0, 37/38]$ excludes $w \ge 0.98$ and includes zero. The same eight observed rates enter both calculations. Calibration changes the set of admissible explanations.

For equal tested successes $s > 0$, put $q_i = z_i/s$, $R = (\sqrt{s_2} - \sqrt{s_3})^2$ when only the other successes are known, and $B = 2(1 - w_0)S - R$. If $B < 0$ the data are infeasible. Otherwise the first states need root fidelity at least $c = \max(0, 1 - B/(2s))$. For a common effect the sharp limit is

$$|q_0 - q_1| \le (U - L)\sqrt{1 - c^2}.$$

The spectral distributions have total variation $|q_0 - q_1|/(U - L)$, bounded by $\sqrt{1 - F^2}$ [28]. The symmetric spectral distributions $((1 + d)/2, (1 - d)/2)$ and its reverse attain equality at $d = \sqrt{1 - c^2}$. If only the untested success sum is prescribed, set $R = 0$ at the optimum. For $s = p_* = 1$, $w_0 = 0.99$ and $(L, U) = (0.2, 0.8)$ the largest difference is 0.168. The witness gives clicks 0.584 and 0.416. In a different example, spectral rows $(0.7, 0.3)$ and $(0.3, 0.7)$ produce clicks 0.62 and 0.38. Their raw discrepancy $0.058454\ldots$ lies below the ceiling 0.08, while the spectral discrepancy $0.166970\ldots$ exceeds it.

Two controls for transfer. The identity channel has $w = 1$ and identical first-qubit outputs on 00 and 01. A single measurement of the *second* qubit perfectly distinguishes these inputs and gives $D_{01} = 2$. The local tensor factor in Corollary 8.3 is essential. A different failure occurs on the correct first qubit if the arm effects are $\mathrm{diag}(0.8, 0.2)$ and $\mathrm{diag}(0.2, 0.8)$. They have the same spectrum but give clicks 0.8 and 0.2 on the identical state $|0\rangle$. A common spectrum alone cannot establish readout transfer.

There is a sharp interpolation if the relative orientation is itself calibrated. For two effects with the same ordered eigenvalues L, U, let θ be the angle between their high-eigenvalue Bloch directions. With the equal-success data above and free first-qubit

states, the exact largest difference is

$$(U - L)\sin(\min\{\pi/2, \theta/2 + \arccos c\}).$$

To prove the upper bound, lift a Bloch vector r to the unit vector $(r, \sqrt{1-|r|^2})$. For two such vectors the qubit identity $F^2 = \operatorname{Tr}\rho_0\rho_1 + 2\sqrt{\det\rho_0 \det\rho_1}$ gives an angle at most $2\arccos c$. Rotating the second detector direction onto the first by θ rotates the second lifted state by at most θ. Projecting half the resulting chord onto the first detector axis bounds the probability difference. Equality is obtained using detector axes $(\cos(\theta/2), \pm\sin(\theta/2))$ and coplanar pure-state directions $(\pm\sin\tau, \cos\tau)$ with $\tau = \min\{\arccos c, (\pi - \theta)/2\}$. Their root fidelity is $\cos\tau$. Theorem 8.1 supplies a single branch realizing them. This bound concerns a known pair of effects. Uncertainty in the axes, spectral gap and successful preparations changes the feasible set. In particular the population inversion amplifies errors as $U - L$ tends to zero.

10 From a population constraint to an observable test

Let $\mathcal{F}_{n-1}$ denote the information available immediately before assignment on eligible attempt n, including its eligibility decision. At each such history there must be one common potential operation across the four trusted inputs and the prescribed common first-qubit readout. The operation and detector may vary between histories within the conditional null. Set $\epsilon = 1 - w_0$. Draw the input privately with known positive predictable probabilities π_i. Failures and both successful outcomes enter each arm's attempt denominator.

Theorem 10.1 (Sequential test for four outcome laws)**.** *At each eligible history choose predictable slopes $t_{by} > 0$, for $b = 0, 1$ and $y = 0, 1$, and put $d_{2b,y} = 1 - t_{by}$, $d_{2b+1,y} = 1 - t_{by}^{-1}$. On a successful assignment i with outcome y, score $X_n = (d_{iy} - 2\epsilon)/\pi_i$. Score a failure by zero. Choose a predictable nonnegative stake η_n such that $1 + \eta_n X_n \geq 0$ for every possible outcome. If $w_n \geq w_0$ at each eligible history, then*

$$M_0 = 1, \quad M_N = \prod_{n \leq N} (1 + \eta_n X_n), \qquad \Pr_{H_0}\{\sup_N M_N \geq 1/\alpha\} \leq \alpha, \quad 0 < \alpha < 1.$$

Proof. For $x, z \geq 0$, $(1-t)x + (1-t^{-1})z \leq (\sqrt{x} - \sqrt{z})^2$ follows from $(\sqrt{tx} - \sqrt{z/t})^2 \geq 0$. The supremum over $t > 0$ equals the right side, including by a limit for a zero rate. Conditional randomization gives

$$\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] = \sum_{iy} d_{iy} r_{iy} - 2\epsilon S \leq D_{01} + D_{23} - 2\epsilon S \leq 0.$$

The last inequality is Corollary 8.3. Each factor is nonnegative with conditional mean at most one. The product is a nonnegative supermartingale. At the first crossing stopped at horizon N, its expectation is at most one and at least $1/\alpha$ times the crossing probability. Letting $N \to \infty$ gives the claim. This is Ville's inequality, as used in sequential statistical inference [29]. □

Threshold 20 gives at most 5% erroneous rejection under this null. For the four-row example in Section 8, choose $(2/3, 3/2)$ and $(3/4, 4/3)$ as slopes for the two pairs, $\epsilon = 1/38$, and $\pi = (3/8, 3/8, 1/8, 1/8)$. Both AM–GM bounds are attained, so $\mathbb{E}X_n = 0$. A

distinct example takes the last two successes equal to one and ignores their outcomes. A common slope v for that pair makes the coefficients $1-v, 1-v^{-1}$ on all successes. For $v=1$ both vanish. With the same first-pair rows and $\epsilon = 1/226$, $D_{01} = 1/50 = 2\epsilon S$. The correctly weighted mean is zero. Omitting the inverse probabilities gives $(3/8)(1/50) - (1/113)[(3/8)(26/100) + 1/4] = 1/226$, positive on the null.

When using calibrated nonprojective effects, the slopes act on the spectral populations. For each measured bit y, use $k_{iy} = d_{i-} + (y-L)(d_{i+} - d_{i-})/(U-L)$ for the successful-arm coefficient. The identity $k_{i1}z_i + k_{i0}(s_i - z_i) = d_{i+}v_{i+} + d_{i-}v_{i-}$ follows from (2). Thus the score $(k_{iy} - 2\epsilon)/\pi_i$ on success and zero on failure gives a conditional mean bounded by $D_{01}^{\mathrm{sp}} + D_{23}^{\mathrm{sp}} - 2\epsilon S \le 0$ under (3). For example, coefficients $(1/2, -1)$ give a score coefficient of 1 on a click and -1.5 on a no-click for $L = 0.2, U = 0.8$. For input 00 in Table 1 their mean is $0.080 - 1.5(0.050) = 0.005 = 0.09/2 - 0.04$. This uses the observed bit without assigning a spectral label to an individual attempt. The predictable stake must keep the factor nonnegative throughout the actual score range, which can widen as $U - L$ decreases. The bound uses the spectra that govern that history; inserting pooled estimates into the score supplies no such guarantee. A learned stake may depend on previous records. Monitoring several products and selecting the first to cross requires multiplicity control or a prespecified mixture.

Power when successes are rare. Take a fixed common detector with eigenvalues $L < U$ and success probability h on every input. In one model all four successful first outputs occupy the high eigenspace. In a second model the odd inputs occupy the low eigenspace. Corollary 9.1 gives maxima of 1 and $1/2$, respectively, at every $h > 0$. For any null $w \ge w_0$ with $1/2 < w_0 \le 1$, the first table has a null realization and the second excludes the null.

Take these branches independently at successive attempts. The observed one-attempt laws agree on even inputs and have total variation $h(U-L)$ on odd inputs. While two records agree, couple their next assignments identically, including under an adaptive assignment rule, and couple the outcomes maximally. Their records then agree through N attempts with probability at least $[1-h(U-L)]^N$. For any test of size α by horizon N,

$$\Pr_{\mathrm{alt}}(\text{rejection by } N) \le \min\{1,\ \alpha + 1 - [1-h(U-L)]^N\}.$$

For $\alpha = 0.05$, $h = 0.001$, $U - L = 0.6$ and $N = 100$, the upper bound is $0.1082524248\ldots$. This coupling bound requires at least 3,837 attempts for 95% power. For uniform independent assignments it requires at least 7,675. A sharper calculation for these two simple independent laws conditions on the number of informative successes. The exact first crossings for simultaneous 5% size and 95% power are 7,065 attempts with an always-odd input and 14,131 with uniform independent four-arm assignment. At the narrower symmetric detector $(L, U) = (0.45, 0.55)$, independent uniform assignment first meets the two targets at 540,254 attempts for the corresponding two laws. The odd-only comparison bounds every adaptive design for its own pair, but is outside a four-arm protocol that requires positive assignment probability for every input. These thresholds use the optimal randomized likelihood-ratio decision for this pair of laws; see the review for the finite-sum enclosures. As $h \downarrow 0$ the two weight ceilings stay separated and the finite-record laws approach one another. There is therefore no finite budget guaranteeing power uniformly above size over this class. Normalized weight is undefined at $S = 0$, outside the domain of the four-input theorem.

History selection. Consider a diagonal Kraus operator. Its squared diagonal entries are $(0.045, 0.005, 0.975, 0.975)$, so it defines a trace-nonincreasing branch. It has $p_* = 0.5$, $w = 0.995$ and identical successful first-qubit outputs on inputs 0 and 1. Every successful first-qubit projective test onto $|0\rangle$ clicks. Under fair two-arm assignment and independent repetitions, the retained arm-0 probability is 0.9. To characterize the consequence of selection, give a click $Z = +Y$ on arm 0 and $Z = -Y$ on arm 1. Under fair assignment and $|p_0 - p_1| \le \delta$, the conditional exponential moment at $\lambda > 0$ is bounded by

$$F_\delta(\lambda) = \tfrac{1}{2}[e^\lambda + (1-\delta)e^{-\lambda} + \delta].$$

Here p_i denote click probabilities. The objective $1+[(e^\lambda-1)p_0-(1-e^{-\lambda})p_1]/2$ is maximized at $p_0 = 1, p_1 = 1-\delta$. Interchanging arms gives the opposite signed score. The two-sided product average at fixed λ is $\cosh(\lambda \sum Z)/F_\delta^n$. If it is wrongly applied to retained successes with $\delta = 0.28$ and $\lambda = \log 2$, at success n it equals

$$\frac{2^{2k-n} + 2^{n-2k}}{2(33/25)^n}, \qquad k \sim \mathrm{Bin}(n, 0.9).$$

At $n = 20$ it exceeds 20 exactly for $k \le 3$ or $k \ge 17$, of probability $0.8670466765656734\ldots$. Removing paths when they first reach 20 and propagating surviving mass with binomial weights gives $0.8986003839221604\ldots$ for crossing by success 20. The first hitting times through that horizon are 9, 13, 16 and 19. These probabilities describe the misapplied test. Theorem 10.1 uses all eligible attempts.

What the past can reveal. Let W be a committed prefix, B the final ideal bit, Y its reported value and $\mathrm{Bern}(s)$ the reference law. Coupling the two complete records and using the triangle inequality gives

$$\mathrm{TV}(\mathcal{L}(W,Y), \mathcal{L}(W)\otimes\mathrm{Bern}(s)) \le \Pr(Y \ne B) + \mathrm{TV}(\mathcal{L}(W,B), \mathcal{L}(W)\otimes\mathrm{Bern}(s)).$$

Dependence in the record alone cannot assign memory to a gate, a readout, or a preparation. Conversely, if the ideal bit is fresh and both directional reporting errors are bounded by e at each history, the reported-one probability lies in $[s(1-e), s+(1-s)e]$. Every value in the interval is realized by suitable one-sided errors. Applying those channels recursively at the positive-probability prefixes proves sufficiency for an entire word law within this reporting model.

11 Physical interpretation

Symmetries can select invariant subspaces, and labelled graph probes can identify a generator. For an instrument, first-output marginals alone always admit zero local weight. Returned labels can exclude zero. Coherent preparations and matched output projections can identify the weight even when complete basis-output tomography leaves an interval.

An instrument test must commit eligibility before randomizing the input, retain every attempt, and characterize success and readout under matched histories. A held-out rejection challenges the model and its calibration together. Varying the detector orientation or success criterion may distinguish the remaining explanations.

Experiments on superconducting processors found detector response sensitive to other active qubits [6], coherent contributions in some measurement bases [30], and

dependence on preceding reports without reset [31]. Multitime tomography can test dependence on prior interventions [32]. Transferring a readout across preparations requires matched observations. Rare successes preclude a uniform trial budget.

Changing a clock scale and its Hamiltonian inversely preserves e^{-itH}. A change of port basis can alter inferred adjacency. Two rotated readouts can produce contrast on an identity channel. A physical correspondence must assign states, operations, units and causal order, and predict a held-out outcome that distinguishes competing assignments. The origin of quantum observables, locality, physical time and spacetime remains open.

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Glossary and notation

Each entry gives the convention used in the paper. Symbols are defined again where a section uses them for a different quantity.

Observation law and identified set.
An observation law assigns outcome probabilities to a specified preparation and measurement. Models are observationally equivalent if those laws agree for every permitted experiment. The identified set contains all property values consistent with them. A sharp upper bound is the least upper bound on this set. Attainment means that some admissible model realizes it.

Calibration and transfer.
Calibration uses reference measurements to constrain a source, unit, operation or effect. Transfer means that the characterized response also applies under the conditions in which it is used. Two detectors can have the same eigenvalues and measure along different axes.

Port and spectral scale.
A port is a labelled input or output mode. The graph Laplacian L acts on the labelled modes, and L^{+} inverts its positive eigenvalues and vanishes on its kernel. In the filter e^{-sL} the scale s has units inverse to those of L. Physical time requires its own calibration.

State, effect and instrument.
A state is positive with trace one. A binary click effect satisfies $0 \preceq E \preceq I$ and gives click probability $\mathrm{Tr}(E\rho)$. A quantum instrument assigns a completely positive, trace-nonincreasing map to each outcome, and their sum preserves trace. Complete positivity means positivity also holds after tensoring with the identity on any auxiliary system. One such outcome map is a *branch*. A channel preserves trace.

Success and marginal.
For input i, s_i is the success probability per attempt, z_i is the probability of success and a click, and $s_i - z_i$ is the probability of success with no click. The first-qubit marginal τ_i is formed by tracing out the second output and has trace s_i. The conditional click probability is z_i/s_i for $s_i > 0$. The mean success on the maximally mixed input is $p_* = \sum_i s_i/4$.

Choi state and local weight.
Vectorizing the Kraus matrices K_j of a nonzero branch gives $C = \sum_j |K_j\rangle\!\rangle\langle\!\langle K_j| / \sum_j \|K_j\|_F^2$. In the output-first convention its input marginal is the transposed success effect divided by its trace. The local weight w is its population in the vectorized subspace $\{V \otimes I_q\}$ for a specified q-dimensional discarded factor. A semidefinite program optimizes a linear functional over positive semidefinite matrices with linear constraints. In the weight theorem one positive block matrix enforces simultaneous compatibility of all overlaps.

Fidelity, discrepancy and distance.
Root fidelity is $F_*(A,B) = \|\sqrt{A}\sqrt{B}\|_1$. For diagonal matrices it is the sum of square roots of products of corresponding entries. For successful outcome masses p, q, the squared discrepancy is $\sum_y (\sqrt{p_y} - \sqrt{q_y})^2$ with no factor $1/2$ and no failure bin. Total variation of normalized laws is $\frac{1}{2}\sum_y |p_y - q_y|$. The trace and Frobenius norms are $\|A\|_1 = \mathrm{Tr}\sqrt{A^*A}$ and $\|A\|_F^2 = \mathrm{Tr}(A^*A)$.

History and time-uniform inference.
History comprises the information available before the next private input assignment. A predictable choice uses only that information. A time-uniform error bound controls the probability of ever crossing a rejection threshold under the conditional model, including when observation stops adaptively.

Sources of the mathematical arguments

The accompanying articles provide the assumptions and proofs for the examples in this synthesis. Each article is cited by title and theorem or section. The contents of this edition locate the complete articles and the visibly typeset supporting proofs.

- *Identification from Restricted Measurements*, Sections 1–5: detector overlap, graph ambiguity, discarded factors and rare-success discrimination.
- *Calibrated Dynamical Identification in Fermionic Systems*, *Clock-Scale Ambiguity, Static CAR States, and Gibbs Modular Equivalence*, and *Continuum and Finite-Lattice Two-Time Identification*: clock and observation assumptions for dynamical inference.
- *Operational Identification and Calibration in Two-Mode Quasifree Systems* and *Restricted Observables and CAR-State Distinguishability*: state classes, readout restrictions and calibrated finite-error inference.
- *Relative Entropy of Finite-Interval Wilson–Dirac Quasifree States*: exact finite decomposition and component bounds. *Finite Part of Wilson–Dirac Relative Entropy*, Theorem 1 and Sections 2–7, assembles the fixed-positive-scaled-mass finite part. *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals*, Theorem 3, Corollary 4 and Theorem 6, gives the stronger modular and nonlinear estimates.
- *Positive Spectral Measures Under Finite Moment and Laplace Constraints*: extremal measures, uncertainty geometry and the distinct numerical-certificate requirements.
- *Compact-Group Transfer Spectra and Ultraviolet Operator Limits*: representation-dependent spectra, semigroup comparisons and perturbation bounds.
- *Four-Preparation Tests for a Conditional Quantum Operation*: the common-instrument population constraints and sequential tests using every eligible attempt.

References

[1] T. Nuradha, H. K. Mishra, F. Leditzky and M. M. Wilde, “Multivariate fidelities,” *J. Phys. A: Math. Theor.* **58**, 165304 (2025). DOI.

[2] A. E. Brouwer, “The Shrikhande graph,” Distance-Regular Graphs. See also S. S. Shrikhande, “The uniqueness of the L_2 association scheme,” *Ann. Math. Statist.* **30**, 781–798 (1959).

[3] T. J. Rothenberg, “Identification in parametric models,” *Econometrica* **39**, 577–591 (1971).

[4] K. Justus, *From Prephysical Hypotheses to Physical Models: Collections*. The contents locate the accompanying articles.

[5] K. Justus, *From Prephysical Hypotheses to Physical Models: Mathematical Supplement*. The contents locate the accompanying articles.

[6] Y. Chen, M. Farahzad, S. Yoo and T.-C. Wei, “Detector tomography on IBM quantum computers and mitigation of an imperfect measurement,” *Phys. Rev. A* **100**, 052315 (2019). DOI.

[7] D. Goderis, A. Verbeure and P. Vets, “Dynamics of fluctuations for quantum lattice systems,” *Commun. Math. Phys.* **128**, 533–549 (1990).

[8] D. K. Hammond, P. Vandergheynst and R. Gribonval, “Wavelets on graphs via spectral graph theory,” *Appl. Comput. Harmon. Anal.* **30**, 129–150 (2011). DOI.

[9] K. Justus, “Equal inverse moments on a labelled graph pair,” exact-rational conductance certificate and verifier, accompanying electronic material.

[10] K. Devriendt, “Effective resistance is more than distance: Laplacians, simplices and the Schur complement,” *Linear Algebra Appl.* **639**, 24–49 (2022). DOI.

[11] M. L. Wagman, “Lanczos algorithm, the transfer matrix, and the signal-to-noise problem,” *Phys. Rev. Lett.* **134**, 241901 (2025). DOI.

[12] P. W. Higgs, “Broken symmetries and the masses of gauge bosons,” *Phys. Rev. Lett.* **13**, 508–509 (1964). DOI.

[13] S. M. Carroll, *Spacetime and Geometry*, Cambridge University Press (2019), Appendix G. DOI.

[14] B. K. Driver, “YM_2: continuum expectations, lattice convergence, and lassos,” *Commun. Math. Phys.* **123**, 575–616 (1989). Author copy.

[15] K. Justus, *Projector-Equivalent Wilson–Dirac Hamiltonians*, mathematical manuscript.

[16] K. Justus, *Positive Spectral Measures Under Finite Moment and Laplace Constraints*, mathematical manuscript.

[17] K. Justus, *Compact-Group Transfer Spectra and Ultraviolet Operator Limits*, mathematical manuscript.

[18] K. Justus, *Operational Identification and Calibration in Two-Mode Quasifree Systems*, mathematical manuscript.

[19] K. Justus, *Relative Entropy of Finite-Interval Wilson–Dirac Quasifree States*, mathematical manuscript.

[20] K. Justus, *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*, mathematical manuscript.

[21] K. Justus, *Cross-Boundary Trace-Class Estimates for Wilson–Dirac Quasifree States*, mathematical manuscript.

[22] K. Justus, *Exact Reference Geometry and the Two-Bond Asymptotic of a Wilson–Dirac Covariance*, mathematical manuscript.

[23] K. Justus, *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals*, mathematical manuscript, Theorem 3, Corollary 4 and Theorem 6.

[24] K. Justus, *Finite Part of Wilson–Dirac Relative Entropy*, mathematical manuscript (2026), Theorem 1 and Sections 2–7.

[25] V. Eisler and I. Peschel, “Analytical results for the entanglement Hamiltonian of a free-fermion chain,” *J. Phys. A* **50**, 284003 (2017). DOI.

[26] E. Nielsen et al., “Gate set tomography,” *Quantum* **5**, 557 (2021). DOI.

[27] A. C. Keith, C. H. Baldwin, S. Glancy and E. Knill, “Joint quantum-state and measurement tomography with incomplete measurements,” *Phys. Rev. A* **98**, 042318 (2018). DOI.

[28] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press (2018), chapters 2–3. DOI.

[29] P. Wills, E. Knill, K. Coakley and Y. Zhang, “Performance of test supermartingale confidence intervals for the success probability of Bernoulli trials,” *J. Res. NIST* **125**, 125003 (2020). DOI.

[30] A. S. Aasen et al., “Readout error mitigated quantum state tomography tested on superconducting qubits,” *Commun. Phys.* **7**, 301 (2024). DOI.

[31] M. Werninghaus, D. J. Egger and S. Filipp, “High-speed calibration and characterization of superconducting quantum processors without qubit reset,” *PRX Quantum* **2**, 020324 (2021). DOI.

[32] G. A. L. White et al., “Non-Markovian quantum process tomography,” *PRX Quantum* **3**, 020344 (2022). DOI.

Spin fluctuation and microscopic ambiguity

Kirtus Justus

I. Standard spin-fluctuation contraction: exact finite-size identities

For each positive integer `N` let the microscopic algebra be `M_2(C)^{⊗N}` with conventional Pauli matrices. Choose the **declared physical spin product state**

$$\rho_{m,N} = \rho_m^{\otimes N}, \qquad \rho_m = (I + m\sigma_z)/2, \quad 0 < m \le 1.$$

Define collective, centered transverse fluctuations

$$Q_N = (2mN)^{-1/2} \sum_{j=1}^{N} \sigma_x^{(j)}, \qquad P_N = (2mN)^{-1/2} \sum_{j=1}^{N} \sigma_y^{(j)}.$$

Then precisely, **at every finite N**,

$$[Q_N, P_N] = \frac{i}{mN} \sum_{j=1}^{N} \sigma_z^{(j)}, \qquad \langle [Q_N, P_N] \rangle = i.$$

Because independent `sigma_z` variables have variance `1-m²` in the product state,

$$\left\langle ([Q_N, P_N] - iI)^\dagger ([Q_N, P_N] - iI) \right\rangle = \frac{1 - m^2}{Nm^2}.$$

Convergence qualification: this is mean-square convergence *relative to the chosen sequence of states*, not operator-norm convergence. In fact `||[Q_N,P_N]-iI|| = 1+1/m` for every `N` (the all-down eigenvector realizes the maximum), so the operator-norm claim is false even for `m=1`.

For real `u,v` let `a=u/sqrt(2mN)`, `b=v/sqrt(2mN)`. Different sites commute and the single-site spin expectation is obtained by multiplying the two Pauli exponentials. Thus the **exact** finite-N ordered characteristic function is

$$C_N(u, v) = \langle e^{iuQ_N} e^{ivP_N} \rangle = \left(\cos a \cos b - im \sin a \sin b\right)^N.$$

For every fixed `m>0`, `u,v`, taking the logarithm and using the uniform small-angle expansion gives

$$\lim_{N\to\infty} C_N(u, v) = \exp\left[-\frac{u^2 + v^2}{4m} - \frac{iuv}{2}\right].$$

The reversed product has the opposite imaginary Weyl phase. More generally, the analogous single-site expansion for any **finite word** in transverse Weyl exponentials yields the standard Gaussian Weyl algebra, with limiting `[Q,P]=iI` in the regular quasi-free fluctuation representation. This reproduces known quantum central-limit results; it is not a new theorem. Its limiting covariance is `V=(1/(2m)) I_2`, with symplectic matrix `Omega=[[0,1],[-1,0]]`, and `V+iOmega/2 >=0` because `0<m<=1`. The corresponding effective oscillator thermal occupancy is `nbar=(1-m)/(2m)`. This does not mean the original spins were thermal harmonic oscillators.

Numerical controls. The accompanying fluctuation calculation builds full `2^N × 2^N` tensor-product spin matrices for `N=2,3,4,6`; their traces of the ordered exponential products

agree with the factorized exact formula to <2e-13. High-precision direct formula checks at N=2,4,16,64,256,1024,4096 approach the displayed limit. For m=.8,u=v=.7, the limit is approximately 0.7142175684315161 -0.17857058381200364 i and the absolute errors are 0.0024151 at N=16, 0.00014976 at N=256, and 9.36e-6 at N=4096. No numerical evidence is being used as a substitute for the elementary limit proof.

The unpolarized/tracial negative control

At m=0, the preceding Q_N,P_N scaling is undefined. The appropriate unrescaled-variance fluctuations are F_x=N^-1/2 sum sigma_x, F_y=N^-1/2 sum sigma_y in the maximally mixed state rho_0=I/2. They obey

$$\langle e^{iuF_x} e^{ivF_y} \rangle = \left[\cos(u/\sqrt{N})\cos(v/\sqrt{N})\right]^N \longrightarrow e^{-(u^2+v^2)/2},$$

and E([F_x,F_y])=0, E([F_x,F_y]†[F_x,F_y])=4/N. The limit is the characteristic function of **two commuting, independent classical Gaussians**, even though the microscopic operators do not commute. This is a statement about this specific Gaussian fluctuation sector, not a no-go for quantum behavior in all unpolarized systems or other scaling limits.

For a state sequence with m_N→0, the normalized CCR commutator variance is (1-m_N²)/(N m_N²), so it is **not** uniform in polarization. Its decay would require N m_N²→∞, and the oscillator covariance 1/(2m_N) diverges regardless. One cannot extrapolate the fixed-m regular oscillator limit all the way to m=0 using the same normalization.

II. Exact microscopic non-identification under complete collective controls

Compare two distinct microscopic systems initialized in their maximally polarized product states:

- Model A: n spin-1 sites, full space dimension 3^n.
- Model B: 2n spin-1/2 sites, full space dimension 2^(2n).

The initial state in both models is the highest-weight state of **total spin J=n**. Under the total-spin generators J_x,J_y,J_z, both cyclic subspaces carry the unique spin-J=n irreducible representation, of dimension 2n+1. The SU(2) raising/lowering relations on a highest-weight vector uniquely determine all matrix elements within this irrep:

$$J_\pm |J, M\rangle = \sqrt{(J \mp M)(J \pm M + 1)}\, |J, M \pm 1\rangle.$$

Therefore an intertwining unitary between these *cyclic irreducible subspaces* maps the initial vectors and every polynomial and bounded functional calculus of the **collective** J_a to each other. All corresponding collective preparations, collective gates (including nonlinear functions such as J_z²), and collective effects give **exactly equal outcome probabilities**, not merely asymptotically equal ones. This is familiar angular-momentum representation theory, not publication novelty.

For n=3, the complete microscopic Hilbert spaces have dimensions 27 versus 64, but all collective experiments restricted to the common seven-dimensional J=3 irrep agree. A full-matrix implementation checks noncommuting rotation sequences and nonlinear collective evolution for n=1,2,3; discrepancies are below 1e-15.

Scope of equivalence: no claim of equality of full observable algebras. A physically calibrated site-resolved intervention, or operations accessing another total-spin sector, can distinguish the models. The comparison demonstrates that knowing the collective bosonic limit, and even far more than that limit, does not identify microscopic constituent number/type without an adequate observation toolbox.

III. Ricker-filtered emergent bosons do not select geometry

This section deliberately reconnects to the program's *negative controls* only; it does not use a wavelet as a particle ontology. Supply a connected graph with real symmetric positive Laplacian `L`, and define the mathematical scale-response operator

$$W_s = sLe^{-sL}, \quad s > 0.$$

Place `N` independent spins at **each already identified graph vertex** and take the `N→∞` polarized fluctuation limit at fixed `m>0`. It yields an independent canonical pair `Q_i,P_i` at every *assumed* vertex. The filtered fields `Q_s=W_s Q`, `P_t=W_t P` satisfy

$$[Q_s(i), P_t(j)] = i(W_s W_t)_{ij}$$

in the limit. The full 4-site spin matrix calculation verifies the same relation for the **finite-N commutator expectation** without invoking the limit. The scale labels remain redundant/noncanonical; the exact Laplacian zero mode is removed.

An aggregate, mode-blind multiscale commutator measures only

$$\mathcal{A}(s,t) = \frac{1}{V}\frac{1}{i}\sum_i \langle [Q_s(i), P_t(i)]\rangle = \frac{1}{V}\,\mathrm{Tr}(W_s W_t),$$

which depends on the eigenvalues of the supplied `L` alone. In the previous exact graph test, the nonisomorphic 16-vertex rook and Shrikhande graphs have identical Laplacian spectra `0^(1),4^(6),8^(9)`. Both therefore yield

$$\boxed{\mathcal{A}(s,t) = st\left[6e^{-4(s+t)} + 36e^{-8(s+t)}\right]}$$

for every pair of mathematical scales, despite different labeled connectivity. At `s=.12,t=.31`, both values are `0.0829085908049164`, with spectral-formula residual `<5e-17`. The full labeled filtered commutator kernels differ in Frobenius norm by `0.0604811368` under the exhibited fixed vertex labeling.

The latter distinction is physically actionable **only if that vertex identification and spatial measurement protocol have independent meaning**. There is no emergent local measurement basis or spatial metric from CCR or the aggregate wavelet result. Thus bosonic algebra plus multi-scale spectral data does not resolve the already identified inverse-geometry ambiguity. This is a synthesis of known quantum-fluctuation and cospectral-graph phenomena, not a novel theorem.

IV. Microscopic dynamics is a separate ingredient, and bosons do not imply Lorentz invariance

As a controlled physical *example*, supply a one-dimensional periodic spin-1/2 chain with independently calibrated hopping `J>0`, on-site energy `Delta=2J+mu` and an independently specified physical evolution parameter:

$$H = \Delta\sum_j n_j - J\sum_j(\sigma_j^-\sigma_{j+1}^+ + \sigma_j^+\sigma_{j+1}^-), \quad n_j = (I - \sigma_z^{(j)})/2.$$

In the **single-down-spin excitation sector**, the exact restricted Hamiltonian is `Delta I-J A_cycle`, giving

$$E(k) = \mu + 2J[1-\cos(ka)] = \mu + Ja^2k^2 + O(k^4a^4).$$

For `J=.75,mu=.2` on a six-site ring, the independent full `2^6` spin matrix has one-excitation energies `[.2,.95,.95,2.45,2.45,3.2]`, matching the formula to `<9e-16`. At `mu=0`, the one-particle

band touches zero with **quadratic** long-wavelength dispersion (`z=2`), not the `z=1` linear spectrum of a relativistic massless scalar. Calling this toy band's touching point a Goldstone boson would require a separate symmetry-breaking argument; none is asserted for the tuned XY model.

In actual symmetry-broken nonrelativistic many-body theories, Watanabe and Murayama (2012, 2014) show that pairs of broken generators with nonzero commutator expectation can describe a single low-energy mode, often with nonrelativistic dispersion. This is structurally suggestive of the same spin commutator mechanism, but **our product-state central limit is not by itself a spontaneous-symmetry-breaking theorem**.

Ricker filtering `W_s` on such an already-defined spatial generator only reweights wave numbers; it neither changes `E(k)` nor supplies the physical space, time, sources or causality needed for a field theory.

Source and gain ambiguity

Kirtus Justus

3. gain-shift ambiguity with arbitrary common source covariance

Let L be a **connected real symmetric weighted graph Laplacian** on n *already labeled* sites, with P0=J/n, Q=I-P0, W_s=sLe^{-sL}. Take distinct **known** scales 0<s<t, a **single shared** zero-mean random source x with covariance Σ strictly positive on the mean-zero subspace, and positive scalar detector gains g_s,g_t:

y_s=g_s W_s x, y_t=g_t W_t x.

The jointly measured covariances are

C_st=g_s g_t W_s Σ W_t, C_tt=g_t² W_t Σ W_t.

All operators W_s and W_t vanish on P0 and are invertible on Q. On Q, C_st C_tt^+ = (g_s/g_t) W_s W_t^{-1}, independent of the generally **noncommuting** covariance Σ. Thus

C_st C_tt^+ = (g_s/g_t) (s/t) exp[(t-s)L]Q.

Consequently the nominal logarithmic reconstruction without calibrated relative gain returns

\hat L = (1/(t-s)) log[P0+(t/s) C_st C_tt^+] = L+αQ,

where α=log(g_s/g_t)/(t-s).

This is an **exact one-parameter ambiguity**, not an additive noise approximation. Unknown Σ by itself cancels for paired measurements; unknown relative gain does not. The identity fails if cross-scale observations do not share the same x, if Σ loses rank on Q, or if the readout covariances contain uncorrected independent additive detector noise.

A conditional self-calibration that weakens the gain premise

A weighted graph Laplacian has offdiagonal entries L_ij=-w_ij ≤ 0. If the candidate is independently known to have **at least one absent edge** in the measured site basis, then max_{i≠j} L_ij=0. As Q_ij=-1/n for i≠j,

\hat L_ij=L_ij-α/n.

Therefore

α=-n max_{i≠j}\hat L_ij,

L=\hat L-αQ, and g_s/g_t=exp[(t-s)α].

The location of the missing edge need not be known. If an approximate shifted generator has entrywise error at most δ, the extracted α has error at most nδ, and the final recovered generator has entrywise error at most nδ. This controls errors **after** the logarithmic inversion; it is not a uniform stability bound from noisy covariance measurements, whose inversion may be severely ill-conditioned. The existence of the missing edge must, however, be justified independently; otherwise the inference has simply inserted sparsity into the premises. This statement is false as unconditional graph reconstruction: for a fully connected positive-weight graph, adding a small complete-graph term generates alternative valid graph Laplacians.

Adversarial source-and-gain transformation

For any c>0, L_c=L+cQ is a valid connected graph Laplacian adding weight c/n to *every* distinct pair. With

T_c=P0 + L(L+cQ)^{-1}_Q Q, x'=T_c x, g'_s=g_s e^{sc},

one has exactly g'_s W_s(L_c)x'=g_s W_s(L)x for **all scales** and individual realizations. Hence if the source and gain laws are insufficiently calibrated, sparse and complete graphs can be observationally identical. The conditional sparsity recovery above **rejects** the complete augmentation only because its prior forbids the augmented model.

Numerical checks

Constructed both 16-vertex rook/Shrikhande Laplacians and a nonwhite, noncommuting positive-definite Σ. Used scales s=log2/4, t=.31, gains g_s=1.73, g_t=.67. The recovered α is 6.93860535328…; the exact target is log(1.73/.67)/(.31-log2/4). Reconstructed each L by the independent no-missing-edge-coordinate self-calibration with entrywise errors 1.25e-13 (rook), 2.26e-13 (Shrikhande). With complete-graph augmentation c=2.3 and transformed sources/gains, four separate filter scales agree to <2.5e-15 in maximum matrix entry. These are floating-point corroborations of the exact formulas, not formal error bars or physical data. The test script and JSON output are retained. A separate verification program constructs both graphs, enumerates all induced three-port types and four-cliques, compares noncommuting three-port control sequences, and verifies an exact π-flux four-cycle caging identity. All assertions pass locally, with three-port controlled response difference 7.33e-16, Krylov ranks 7 in both chosen triangle-port experiments, and π-flux opposite-corner transfer below 6.1e-17.

Prior art boundary: blind graph deconvolution, spectrum/graph learning from structured signals, and system-identification similarity classes are established fields (e.g. Hammond et al. 2011; Ramírez et al. 2017; Burgarth and Yuasa 2014). These conditional formulas are a working synthesis and do not establish publication novelty.

4. Operational readout of full commutation: a simple robustness calculation

Let A and B be Pauli observables with A²=B²=I. After a projective measurement of B, apply U(θ)=exp(iθA), then measure B again. Let b,b’ be the consecutive ideal outcomes. For commuting A,B, E[bb']=1. For anticommuting A,B, E[bb']=cos(2θ). At θ=π/2 they are +1 and -1, respectively, irrespective of the state. This distinguishes the two models without interpreting a global -I phase as an observable by itself.

Under the **specific** error model in which the two recorded bits are independently flipped with probability e, and θ=π/2+δ, the observed contrast is

Δ_corr=2(1-2e)² cos² δ.

Thus a simple instrument noise model retains a finite discrimination signal whenever e<1/2 and |δ|<π/2. This is not a guarantee for arbitrary invasive/nonprojective measurements, source drift, or pulse crosstalk. For e=.08, δ=.07, the predicted contrast is 1.40429640693, checked by full two-qubit matrices. The required B in the dressed model is itself a nonlocal Pauli string under the original site interpretation, so access to its projective measurement is a substantial operational input.

If a faithful state ρ on dimension d has known smallest eigenvalue r>0, then Tr[ρ C†C] ≥ r ||C||_op² for C=[A,B]. A measured small commutator *mean-square* controls an operator-norm cross-commutator only with this faithfulness bound. The weaker expected commutator Tr(ρ[A,B])≈0 does not provide it. If r=0 or scales to zero with d, no uniform operator-level guarantee follows.

Source range

Kirtus Justus

Proposition 1: exact range recovery

Let B be a d-by-n isometry, B* B = I_n. Let Sigma >= 0 have rank r and support Q_s. Define C = B Sigma B. *Then C C^+ = B Q_s B*. If the experiment combines several source preparations with sum of covariances Sigma supported on the entire desired observable subspace Q, then C C^+ = B Q B*. This follows immediately from the range of B Sigma B*. In the specified three-port passive quantum Ricker isometry B=(W_s,W_t,sqrt(I-W_s$^{2-W_t}$2))^T, the first and second diagonal blocks of the latter projection are W_s^2 and W_t^2. Positive square roots recover W_s and W_t. This proof assumes full access to all three output groups, vacuum-floor subtraction, and sufficiently excited source modes.

Theorem 2: quantitative range-projector stability

Let C=B Sigma B* be positive semidefinite and of rank r. Let mu>0 denote its smallest nonzero eigenvalue. If a Hermitian observed matrix Chat=C+E satisfies ||E||_op <= delta < mu/2, and Phat projects onto the r largest eigenvalues of Chat, then, for P=CC^+,

```
||Phat-P||_op <= delta/(mu-delta).
```

Proof: Weyl's theorem places the r signal eigenvalues of Chat above mu-delta and the remainder in [-delta,delta]. Thus Phat is correctly ranked and Chat is invertible on Ran Phat with inverse norm <=1/(mu-delta). Since (I-P)C=0, (I-P)Chat Phat=(I-P)E Phat. Postmultiply by the restricted inverse and take norms. Equal-rank orthogonal-projector geometry gives ||P-Phat||=||(I-P)Phat||. QED.

Interpretation: for an isometric B, mu equals the smallest strictly positive eigenvalue of Sigma. The previously reported rank jump as epsilon approaches zero is therefore compatible with strong instability at finite noise. The proof is a specific Davis-Kahan/Wedin-type estimate, not a claimed novel matrix perturbation theorem.

Theorem 3: exact Gaussian information loss from subspace rotation

Let P_1 and P_2 be real orthogonal projections of equal rank r in dimension d, and let S_i=v I+epsilon P_i for v,epsilon>0. Write G_i=N(0,S_i). Then

```
D_KL(G_1 || G_2) = D_KL(G_2 || G_1)
                 = epsilon^2 ||P_1-P_2||_F^2 / [4 v (v+epsilon)]
                 = epsilon^2 sum_j sin^2(theta_j) / [2 v (v+epsilon)].
```

Proof: det S_1=det S_2=v$^{(d-r)(v+epsilon)}$r and S_i$^{(-1)=v}$(-1)I-epsilon P_i/[v(v+epsilon)]. Substitute into Gaussian KL and use ||P_1-P_2||_F^2 = 2(r-tr(P_1P_2)). QED.

Corollary (fixed dimension and fixed known candidate projectors): M independent preparations yield KL equal to M times the single-preparation expression. If M epsilon^2 -> 0 for fixed v, Pinsker gives asymptotically zero total-variation separation. If M epsilon^2 -> infinity, a likelihood-ratio test distinguishes the two fixed alternatives consistently. The epsilon^-2 statistical threshold pertains to this homodyne Gaussian observation contract, not all quantum probes.

Scale POVM

Kirtus Justus

1. Postulates, definitions, and source of each premise

P1 (supplied graph). The one-particle input space is $\mathbb{C}^n$ with specified orthonormal port labels e_i. Its real symmetric operator L is the Laplacian of a connected positively weighted graph. Therefore $L \succeq 0$, $\ker L = \mathrm{span}\{\mathbf{1}\}$, and $L_{ij} \leq 0$ for $i \neq j$. Let $P_0 = \mathbf{1}\mathbf{1}^T/n$, $Q = I - P_0$, and $K = L^+$.

P2 (supplied measurement). A one-photon scale-register POVM has a separate zero-mode outcome P_0, and positive-scale POVM density $\Pi(ds) = 4sL^2e^{-2sL}\,ds$, $s > 0$. This is the ideal spectral Calderón analysis measurement. Its abstract passive isometric realization is mathematical; no physical continuous-scale apparatus is claimed to have been constructed.

P3 (restricted preparations). For a specified pair of ports $i \neq j$, two possible input density operators ρ_A, ρ_B are supported in $\mathrm{span}\{e_i, e_j\}$ and have identical diagonal populations. Their off-diagonal coherence, phase noise, and purity are not specified. Input relative phase is *not* calibrated. Both preparations are probed with the same normalized POVM and detector calibration.

P4 (statistics). For finite-sample statements, preparations are repeated independently, with independently specified bounds on source spectral gap, scale readout, and detector efficiencies. The distinction between calibrated physical time and mathematical filter resolution is retained throughout.

These postulates import Hilbert space, graph, measurement, input basis and quantum preparation. None is derived from prephysical hypotheses.

2. Elementary moment lemma

Lemma 1. For every integer $k \geq 1$,

$$P_0 + \int_0^\infty \Pi(ds) = I, \qquad \int_0^\infty s^k \Pi(ds) = \frac{(k+1)!}{2^k} K^k.$$

Proof. For a positive spectral eigenvalue λ, the positive-scale measure has density $4\lambda^2 s e^{-2\lambda s}$. The first identity is $\int_0^\infty 4\lambda^2 s e^{-2\lambda s} ds = 1$. For integer $k \geq 1$,

$$4\lambda^2 \int_0^\infty s^{k+1} e^{-2\lambda s} ds = \frac{(k+1)!}{2^k \lambda^k}.$$

The zero eigenvalue contributes zero to the positive-scale integrals. The spectral theorem proves the result. □

A reported *unconditional* scale mean is $\mu(\rho) = \int_0^\infty s\,\mathrm{Tr}(\rho\Pi(ds)) = \mathrm{Tr}(\rho K)$. If a detector instead reports only the conditional mean among positive-scale events, the physically correct conversion is

$$\mu(\rho) = \big(1 - \mathrm{Tr}(\rho P_0)\big)\,\mathbb{E}[s \mid s > 0].$$

Failing to apply this factor biases the proposed reconstruction whenever zero-mode clicks occur.

3. Self-calibrated off-diagonal pseudoinverse

Theorem 2 (phase-contrast ratio). Under P1–P3, define the measured zero-port probabilities $p_X = \mathrm{Tr}(\rho_X P_0)$ and unconditional first scale moments $\mu_X = \mathrm{Tr}(\rho_X K)$ for $X = A, B$. If $p_A \neq p_B$, then

$$\boxed{K_{ij} = \frac{\mu_A - \mu_B}{n(p_A - p_B)}.}$$

No calibration of the relative phase or even of the individual port populations is necessary, provided the populations are the same for the two preparations.

Proof. Equal diagonals imply $\rho_A - \rho_B = z|i\rangle\langle j| + \bar{z}|j\rangle\langle i|$ for some $z \in \mathbb{C}$. Since K is real symmetric,

$$\mu_A - \mu_B = 2\,\mathrm{Re}(z) K_{ij}.$$

Since $(P_0)_{ij} = 1/n$,

$$p_A - p_B = 2\,\mathrm{Re}(z)/n.$$

The asserted ratio follows when its denominator is nonzero. □

Corollary 3 (electrical resistance and full reconstruction). Independently prepare the known basis states e_i and e_j, obtaining $\mu_i = K_{ii}$ and $\mu_j = K_{jj}$. The exact electrical resistance is

$$\boxed{R_{ij} = \mu_i + \mu_j - \frac{2(\mu_A - \mu_B)}{n(p_A - p_B)}.}$$

If this experiment can be performed for every labeled pair with nonzero contrast, all entries of K are determined and $L = K^+$ follows. A direct implementation uses n single-site preparations and two population-matched pair preparations for each unordered pair, totaling n^2 preparation settings. This is a sufficient, not a proven minimal, count for this calibration-light protocol.

Concrete coherent/dephased implementation. Let $p_i + p_j = 1$ and prepare an arbitrarily phase-imperfect pair $\psi = \sqrt{p_i}e_i + e^{i\phi}\sqrt{p_j}e_j$. Compare it with its fully dephased state $\rho_D = p_i|i\rangle\langle i| + p_j|j\rangle\langle j|$. These have the same populations, and $p_D = 1/n$ regardless of the unknown populations. Provided $p_{\mathrm{coh}} \neq 1/n$,

$$K_{ij} = \frac{\mu_{\mathrm{coh}} - \mu_D}{n p_{\mathrm{coh}} - 1}.$$

The same statement holds for an arbitrary phase-drift ensemble, because both outputs depend only on the averaged real coherence.

Nonidentifiability boundary. If $p_A = p_B$, the two corresponding measurement differences vanish in the same real-coherence direction. The ratio is undefined. The two observations alone do not reconstruct K_{ij}. Further independent preparation settings can supply a nonzero contrast.

4. Finite scale-window and readout resolution

Suppose a physical register records only $a \leq s \leq b$, where $0 \leq a < b < \infty$, and assume the independent spectral enclosure $\lambda_2(L) \geq \underline{\lambda} > 0$ and $\lambda_{\max}(L) \leq \overline{\lambda}$.

Lemma 4 (uniform moment-loss bound). If $\tilde{\mu}(\rho) = \int_a^b s\,\mathrm{Tr}(\rho\Pi(ds))$, then for every normalized input state

$$0 \leq \mu(\rho) - \tilde{\mu}(\rho) \leq \boxed{B_{a,b} = \frac{4}{3}\overline{\lambda}^2 a^3 + \frac{e^{-2\underline{\lambda}b}}{\underline{\lambda}}(1 + 2\underline{\lambda}b + 2\underline{\lambda}^2 b^2).}$$

If the observed scale is replaced by a midpoint of a bin of maximum width h, add $h/2$ to this bound.

Proof. For each positive eigenvalue, the missing low-scale first moment is bounded by $4\lambda^2 \int_0^a s^2 ds \le 4\underline{\lambda}^2 a^3/3$. The omitted upper tail is exactly

$$4\lambda^2 \int_b^\infty s^2 e^{-2\lambda s} ds = \frac{e^{-2\lambda b}}{\lambda}(1 + 2\lambda b + 2\lambda^2 b^2).$$

The right-hand side decreases with $\lambda > 0$, so it is bounded by its value at $\underline{\lambda}$. Spectral weights are nonnegative and sum to at most one on the positive subspace. Binning contributes at most $h/2$ per photon. □

Corollary 5. Using the exact zero-mode contrast $d = p_A - p_B \neq 0$ and finite-window scale moments in Theorem 2 gives

$$|\widetilde{K}_{ij} - K_{ij}| \le \frac{2(B_{a,b} + h/2)}{n|d|}.$$

This is a deterministic approximation bound. It assumes no unmodelled detector loss, and the scale moment must be normalized by *injected* photon number or separately correct for events outside the monitored window.

5. Statistical conditioning and explicit finite-sample error

Let each of the four preparations e_i, e_j, ρ_A, ρ_B have M independent trials. A scale observation includes an atom at $s = 0$ for the zero port. With $\lambda_2 \ge \underline{\lambda}$, Lemma 1 gives

$$\mathbb{E}[s^2] \le \frac{3}{2\underline{\lambda}^2}, \qquad \mathrm{Var}(s) \le \frac{3}{2\underline{\lambda}^2}.$$

Hence the union bound, Chebyshev, and Hoeffding imply that all four empirical first moments have absolute error at most η and the two relevant zero-port frequencies have absolute error at most δ, except with probability at most

$$\boxed{\frac{6}{M\underline{\lambda}^2\eta^2} + 4e^{-2M\delta^2}.}$$

These are conservative, not optimal, statistical guarantees.

Theorem 6 (ratio stability). On that event, let $d = |p_A - p_B| > 2\delta$. The empirical ratio $\widehat{K}_{ij} = (\hat{\mu}_A - \hat{\mu}_B)/[n(\hat{p}_A - \hat{p}_B)]$ obeys

$$\boxed{|\widehat{K}_{ij} - K_{ij}| \le \frac{2\eta/n + 2\delta/\underline{\lambda}}{d - 2\delta}.}$$

The derived effective-resistance estimator obeys

$$\boxed{|\widehat{R}_{ij} - R_{ij}| \le 2\eta + \frac{4\eta/n + 4\delta/\underline{\lambda}}{d - 2\delta}.}$$

Proof. Let $h = \mu_A - \mu_B = ndK_{ij}$ with the sign of d retained when dividing. Its empirical perturbation is at most 2η, and the zero-mode contrast perturbation is at most 2δ. Use

$$\widehat{K}_{ij} - K_{ij} = \frac{(\hat{h} - h) - nK_{ij}(\hat{d} - d)}{n\hat{d}}$$

and $|K_{ij}| \le \|K\|_{\mathrm{op}} \le 1/\underline{\lambda}$. The resistance bound follows by adding the two diagonal mean errors. □

These bounds explain why phase calibration can be replaced by *observable* contrast calibration only if the actual contrast is statistically separated from zero.

6. Exact statistical refinement for the rook versus Shrikhande testbench

For the fixed plus-state input on two known ports adjacent in the rook graph and nonadjacent in the Shrikhande graph, the positive-scale densities are

$$p_E(s) = 32se^{-8s} + 96se^{-16s}, \quad p_N(s) = 16se^{-8s} + 160se^{-16s},$$

with the same zero-port mass $1/8$. Their likelihood ratio is

$$\frac{p_E(s)}{p_N(s)} = \frac{2e^{8s} + 6}{e^{8s} + 10},$$

strictly increasing for $s > 0$, and equals one at $s_* = \log 2/4$.

Proposition 7 (one-bit sufficiency for the single-shot optimal *classical* decision). The indicator $X = \mathbf{1}\{s > s_*\}$, setting $X = 0$ for zero-port outcomes, captures the full single-shot total-variation distance of the ideal classical scale-counting distributions:

$$p_E = \Pr_E(X = 1) = \frac{19 + 22\log 4}{128},$$
$$p_N = \Pr_N(X = 1) = \frac{13 + 18\log 4}{128},$$
$$\mathrm{TV}(P_E, P_N) = p_E - p_N = \frac{3 + 2\log 4}{64} \simeq 0.090196698785.$$

Proof. The density difference changes sign exactly once, at s_*. By the classical Neyman–Pearson characterization, its positive region is the optimal single-shot decision set. The gamma(shape 2, rate 2λ) survival function is $(1 + 2\lambda s)e^{-2\lambda s}$; substituting at s_* yields the displayed probabilities. □

For M independent photons, retain only the binary outcomes, and perform the exact binomial likelihood-ratio test. The smallest integer M for which this *particular* equal-prior test has error ≤ 0.1 is **181 photons**; the exact error at 181 is approximately 0.09947986615. At 180 photons it is 0.10012764629. A direct Hoeffding-based sufficient count is 567 photons. This improves the earlier, much looser 960-photon Chebyshev certificate without asserting optimality among collective quantum measurements of the full scale-register states.

The binary threshold is for the exactly specified *plus* input and calibrated graph pair; it is not an invariance under unknown phase drift. For drift-robust effective-resistance reconstruction, use Theorem 2 and its calibration readouts.

Output access and ambiguity

Kirtus Justus

2. Postulates and two different observation contracts

P1. A scalar positive parameter λ is a *supplied* eigenvalue of a supplied real graph generator. The experimenter can prepare its known input eigenmode. This is not derived from prephysical hypotheses.

P2. For calibrated positive scales s, t, the two monitored Ricker signal amplitudes are

$$f_s(\lambda) = s\lambda e^{-s\lambda}, \qquad f_t(\lambda) = t\lambda e^{-t\lambda}.$$

Because $0 \le f_u \le e^{-1}$, the two-signal amplitude stack is passive, with a complementary amplitude

$$h(\lambda) = \sqrt{1 - f_s(\lambda)^2 - f_t(\lambda)^2} > 0.$$

P3. A real coherent source amplitude $a \neq 0$, with mean input photon number $E = a^2$, is unknown but common to both signal paths. Phase-calibrated homodyne produces means $\sqrt{2}af_j$, vacuum variance $1/2$, independently across distinct canonical output modes. The source and ancillary modes are quantum *inputs*, not derived entities.

Contract M (monitored-only): measure the two signal output ports. The third complementary port is discarded.
Contract F (full-output): also measure the complementary canonical port. A physical device must supply the required access, phase reference and sufficiently low excess noise. A mathematical isometric dilation does not independently implement the device.

3. Reconfirmation: original two-monitored-port global optimum

Let $x = s\lambda$, $y = t\lambda$, $f(z) = ze^{-z}$, and $g(z) = z(1-z)e^{-z}$. In Contract M, eliminating the unknown source amplitude by a Schur complement gives

$$\mathcal{J}^{M}_{\lambda|a} = \frac{E}{\lambda^2}\,\mathcal{C}_M(x,y), \qquad \mathcal{C}_M(x,y) = \frac{4x^2y^2e^{-2(x+y)}(y-x)^2}{x^2e^{-2x} + y^2e^{-2y}}.$$

The analysis proves global uniqueness up to exchanging the two scales by reducing the stationary equations to the unique root $u \in (0,1)$ of

$$3\log\frac{1+u}{1-u} - \log u - 2(u + u^{-1}) = 0,$$

whose derivative is strictly positive on the interval. The maximizing scales are

$$\begin{aligned} x_M &= 0.4121145890889544\ldots, \\ y_M &= 2.7010117642659317\ldots, \\ \mathcal{C}_M &= 0.4780676701094133\ldots. \end{aligned}$$

The reported bracketing and differential-evolution calculations agree. This is a *global statement only within Contract M*, with the known source mode, two scale settings, calibrated phase and unit vacuum noise, not a universal quantum Fisher optimum.

4. Theorem 1: full-output amplitude nuisance decouples

Statement. Let $f(\lambda) \in \mathbb{R}^m$ be a continuously differentiable passive signal vector with $\|f\|^2 < 1$. Append $h(\lambda) = \sqrt{1 - \|f(\lambda)\|^2}$, and measure all $m+1$ output modes through the phase-calibrated homodyne contract P3. Put $v = (f, h)$. Then

$$\boxed{\mathcal{J}_{a\lambda} = 0, \qquad \mathcal{J}^F_{\lambda|a} = 4E\left(\|f'\|^2 + \frac{(f^T f')^2}{1 - \|f\|^2}\right).}$$

The right-hand side is also the pure-coherent-state quantum Fisher information for this real-displacement family; complete homodyne achieves it within the specified model.

Proof. The homodyne mean vector is $\sqrt{2}av$ and its covariance is $I/2$, independent of both parameters. Thus the Fisher matrix is four times the Euclidean Gram matrix of v and av'. The identity $\|v\|^2 = 1$ implies $v^T v' = 0$, making the amplitude/parameter cross term vanish. Differentiation of h gives $h' = -f^T f'/h$; substitute in $\|v'\|^2$. The known formula for QFI of real coherent displacements gives the same $4E\|v'\|^2$. □

Numerical design comparison. For the Ricker case with two signal ports, $f = (w_s, w_t)$,

$$\mathcal{J}^F_{\lambda|a} = \frac{E}{\lambda^2}\mathcal{C}_F(x, y), \quad \mathcal{C}_F = 4\left[g(x)^2 + g(y)^2 + \frac{(f(x)g(x) + f(y)g(y))^2}{1 - f(x)^2 - f(y)^2}\right].$$

At the original Contract M maximizer, $\mathcal{C}_F = 0.484225125\ldots$. At equal scaled settings $x = y \approx 2.5212816$, a separate one-dimensional numerical maximization gives $\mathcal{C}_F \approx 0.827935853$. This numerical maximization is restricted to equal scales. The unrestricted proof is given in the full-output design supplement. For comparison, if Contract M uses equal scales, its nuisance-adjusted Fisher information is exactly zero because both signal means are proportional to the unknown amplitude.

The complement's accessibility fundamentally changes the estimation contract. Its contribution is not an automatic free measurement.

5. Theorem 2: high local Fisher information can coexist with exact global aliasing

Statement. At equal scales $s = t$, the full-output model is not globally injective in $\lambda > 0$. Any amplitude $0 < f < e^{-1}$ occurs at exactly two positive scaled eigenvalues

$$x_- = -W_0(-f) \in (0, 1), \qquad x_+ = -W_{-1}(-f) > 1.$$

The full accessible output states coincide exactly for the pair $\lambda_- = x_-/s$, $\lambda_+ = x_+/s$, with the same coherent input amplitude. At the numerically locally optimized equal scale, the alias pair is

$$x_+ \approx 2.5212815653, \qquad x_- \approx 0.2637461880, \qquad f \approx 0.202601510.$$

Proof. The function $x \mapsto xe^{-x}$ is strictly increasing on $(0, 1)$, strictly decreasing on $(1, \infty)$, and equals e^{-1} at $x = 1$. At the two solutions, the two monitored amplitudes both equal f, while the complement amplitude equals $\sqrt{1 - 2f^2}$ in both models. All accessible output coherent-state factors, including their vacuum noise, are thus identical. □

This is global parameter nonidentifiability, not merely a weak statistical signal. A local Fisher matrix alone cannot exclude a disconnected competing parameter value.

6. Theorem 3: splitting nearly equal scales restores uniqueness with a quadratic alias cost

Fix an alias pair $x_+ > 1 > x_-$ satisfying $f(x_+) = f(x_-) = f$, put $s\lambda_\pm = x_\pm$, and set $t = s(1+\delta)$. Let $g(x) = x(1-x)e^{-x}$.

Exact uniqueness for $\delta \neq 0$. For a nonzero common input amplitude and calibrated signal ports,

$$\frac{f_s(\lambda)}{f_t(\lambda)} = \frac{s}{t} e^{(t-s)\lambda},$$

which is strictly monotone in λ if $s \neq t$. Thus the two-scale ideal mean vector is injective up to the unknown amplitude, whereas it is not so for equal scales.

Asymptotic discrimination cost. Let

$$v_{\pm}(\delta) = \left(f(x_{\pm}),\ f((1+\delta)x_{\pm}),\ \sqrt{1 - f(x_{\pm})^2 - f((1+\delta)x_{\pm})^2} \right).$$

Then, as $\delta \to 0$,

$$\boxed{\|v_+(\delta) - v_-(\delta)\|^2 = C_{\text{alias}}\delta^2 + O(\delta^3), \quad C_{\text{alias}} = [g(x_+) - g(x_-)]^2 \frac{1-f^2}{1-2f^2} > 0.}$$

Proof. The first amplitude is exactly equal in the two alternatives. The second differs to first order by $\delta[g(x_+) - g(x_-)]$. Differentiation of the square-root complement produces $-f/\sqrt{1-2f^2}$ times this first-order difference. Sum the two nonzero squared components. Positivity holds because $g(x_+) < 0 < g(x_-)$. □

For a fixed, known coherent amplitude with input energy E, the squared overlap of the two full-output pure coherent states is exactly $\exp[-E\|v_+ - v_-\|^2]$. The equal-prior Helstrom error is at most 10% only if

$$E \geq \frac{-\log(0.36)}{\|v_+ - v_-\|^2} \sim \frac{-\log(0.36)}{C_{\text{alias}}}\delta^{-2}.$$

At the locally optimized equal-scale point, numerical evaluation yields $C_{\text{alias}} \approx 0.2185515$, giving a leading photon requirement near $4.67\delta^{-2}$. The exact threshold for $\delta = 0.01$ is about 47,134 photons. This comparison is for two specified alternative devices and fixed input energy; an unknown source amplitude or additional nuisance parameters cannot be assumed to improve the discrimination guarantee.

Interpretation. Under Contract F, a pair of very similar scales can be locally efficient yet costly for rejecting a distant Lambert-W alias. Under Contract M, the two distinct scales in the original audited global Fisher optimum avoid this specific alias while estimating an unknown amplitude. The designs optimize different scientific tasks. The mathematical tension is between local statistical curvature and global uniqueness, not between quantum theory and classical probability.

Full readout design

Kirtus Justus

1 Global optimum for the full-output Fisher information

The model is the specified scalar eigenmode with unknown nonzero real coherent amplitude a, mean photon number $E = a^2$, and two calibrated scales. Put $x = s\lambda$, $y = t\lambda$, and

$$f(u) = ue^{-u}, \qquad z(u) = u(1-u)e^{-u}, \qquad v = (f(x), f(y), \sqrt{1 - f(x)^2 - f(y)^2}).$$

All three output modes are accessible with phase-calibrated homodyne means $\sqrt{2}av$ and covariance $I/2$. Since $v^T v = 1$, the Gaussian Fisher matrix has $\mathcal{J}_{a\lambda} = 4av^T v' = 0$, and

$$\mathcal{J}_{\lambda|a} = \frac{E}{\lambda^2} C_F(x, y), \qquad C_F = 4\left[z(x)^2 + z(y)^2 + \frac{(f(x)z(x) + f(y)z(y))^2}{1 - f(x)^2 - f(y)^2}\right].$$

The same value is the coherent-state QFI for this real displacement family: the squared overlap of nearby coherent states is $\exp[-E\|dv\|^2]$, giving $4E\|v'\|^2$. This concerns the specified family and available outputs, not all probes and measurements.

Theorem 1. *Over all $x, y > 0$, C_F has a unique global maximum at $x = y = r$, where $r > 3/2$ is the unique upper root of*

$$F(r) = r^2 - 3r + 1 + 2r^3 e^{-2r} = 0.$$

The maximum and maximizing scales are

$$r = 2.52128156583213785047\ldots, \qquad C_F^{\max} = 8r^3 e^{-2r} = 0.82793585316582672742\ldots.$$

Proof. Write $P(u) = f(u)^2$, $Z(u) = z(u)^2$ and $g(u) = Z(u)/(1 - 2P(u))$. The denominators are positive because $P(u) \le e^{-2} < 1/2$. Let $m = \sup_{u>0} g(u)$. The function is continuous, vanishes at 0, 1 and infinity in the relevant limits, and is positive elsewhere, so the supremum is attained at an interior point away from 1.

Cauchy–Schwarz gives

$$\frac{C_F(x, y)}{4} \le \frac{Z(x) + Z(y)}{1 - P(x) - P(y)}.$$

Since $Z(u) \le m(1 - 2P(u))$, adding the two inequalities gives $C_F(x, y) \le 8m$. Equality holds at $x = y = u$ whenever $g(u) = m$. Conversely, equality in the Cauchy step requires the vectors $(f(x), f(y))$ and $(z(x), z(y))$ to be proportional. Because f is strictly positive on this domain and $z(u)/f(u) = 1 - u$, equality requires $x = y$. It remains to prove that g has exactly one global maximizer.

Its derivative factors as

$$g'(u) = \frac{-2u(u-1)e^{-2u}F(u)}{(1 - 2u^2 e^{-2u})^2}, \qquad F'(u) = (2u - 3)(1 - 2u^2 e^{-2u}).$$

Thus F strictly decreases to $3/2$ and strictly increases thereafter. Since $F(0) = 1$, $F(1) = -1 + 2/e^2 < 0$, and $F(u) \to \infty$, it has exactly one root in $(0, 1)$ and exactly one in $(3/2, \infty)$. The derivative signs, together with the zero endpoint values of g, make these the unique maxima on the two branches.

To compare the branch maxima without trusting a numerical optimization, for $0 < u < 1$ use $u^2(1-u)^2 \le 1/16$ and $e^{-2u} < 1$. Since $e > 8/3$,

$$g(u) < \frac{1/16}{1 - 2/e^2} < \frac{2}{23}.$$

On the upper branch, evaluating at $u = 5/2$ and using $e < 11/4$ gives

$$g(5/2) = \frac{225/16}{e^5 - 25/2} > \frac{225/16}{(11/4)^5 - 25/2} = \frac{4800}{49417} > \frac{2}{23}.$$

The elementary bounds on e follow from its series: the sum through $1/4!$ is $65/24 > 8/3$, and the remaining tail is less than $1/100$, giving $e < 65/24 + 1/100 < 11/4$. Therefore the upper critical point is the unique global maximum of g, and equality in the earlier bound forces $x = y = r$. Finally, $F(r) = 0$ implies $(1-r)^2 = r(1-2r^2e^{-2r})$, so $g(r) = r^3e^{-2r}$. This proves the stated coefficient. □

Symbolic differentiation, 80-digit root calculations and a separate two-dimensional numerical optimization agree. The numerical search is corroboration, not the proof of globality. The reported settings are locally tuned in the dimensionless coordinates $s\lambda, t\lambda$; an unknown λ requires a nominal estimate or an adaptive design to implement them.

The earlier two-monitored-output contract still has its different global optimum $(0.4121145891, 2.7010117643)$ and coefficient 0.4780676701. Full-output access raises the best coefficient by about 73.2% within these models. It changes the physical readout contract and is not free information obtainable from discarded output.

Realizable graph moment witness

Kirtus Justus

1. A graph witness with fixed inverse moments

Let the test input be `u=(e0-e1)/sqrt(2)`. On the 16-vertex rook graph, its positive-Laplacian spectral measure consists of weights `1/4` at `lambda=4` and `3/4` at `lambda=8`, hence transformed inverse-eigenvalue moments

`m0=1, m1=5/32, m2=7/256, m3=11/2048, m4=19/16384`.

Construct a connected four-vertex complete weighted graph with fixed weights `w01=w03=w23=1/50`, and unknown `(w02,w12,w13)`. With `S=[e0-e3,e1-e3,e2-e3]`, `B=(S^T S)^{-1}S^T L S`, `G=S^TS`, `h=(1,-1,0)^T`, and `d=det B`, define three integer/rational polynomial equations

`P_j=0.5 h^T G adj(B)^j h - m_j^(rook) d^j=0`, `j=1,2,3`.

Use the full exact rational centre recorded with the certificate and the radius $r = 10^{-34}$. Let C be the inverse Jacobian at that centre. Exact rational polynomial-coefficient bounds for the Hessians on this box give a contraction factor q and centre displacement ρ satisfying $0 \le q < 1$ and $\rho + qr < r$. Their displayed numerical values are approximately $q = 4.479010085739296 \times 10^{-27}$ and $\rho = 3.689057310400891 \times 10^{-63}$; the proof uses the exact inequalities, not those rounded renderings. Banach's fixed-point theorem therefore gives a unique root in the box. Exact characteristic-polynomial sign changes at the rational centre locate its three positive eigenvalues in $(0.084, 0.086)$, $(5.58, 5.59)$, and $(33.81, 33.82)$. Perturbing the three conductances by at most r changes the Laplacian operator norm by at most $6r$, so the root has spectral gap greater than 0.08. A direct polynomial numerator-and-determinant derivative bound, with a strictly positive determinant lower bound on the box, proves exactly $0.006804564875115 < m_4 < 0.006804564875116$.

The computed root is approximately:

$$\begin{pmatrix} 15.689998458283134606411247 7306 \\ 3.968871903781207382760182 1080 \\ 0.023562856373415166885233 0800 \end{pmatrix}.$$

Its positive eigenvalues are approximately `(0.084748449338,5.585135434187,33.814982553350)`, with pair weights `(3.060828e-7,0.84746316616,0.15253652775)`. The slow mode dominates m4 despite tiny weight: its m4 contribution is approximately `0.005933513`.

Sixteen-site lift. Add 12 new vertices; reduce every old four-site edge weight by `3epsilon` with `epsilon=3/500`, connect every old vertex to every new vertex with weight `epsilon`, and connect all new pairs with weight `1/100`. Every edge stays positive (the smallest old adjusted weight is `0.002`). The cross block annihilates original four-site zero-sum vectors. On those vectors, the old-site block agrees with the original L. Hence the selected pair's entire spectral measure is preserved. The additional eigenvalues are `16epsilon=0.096` (one copy) and `0.12+4epsilon=0.144` (eleven copies), so the original low eigenvalue remains the 16-graph spectral gap. Both candidate graphs have 16 vertices, matching m0..m3 for the one specified pair but differing at m4.

This is **exact conditional nonidentifiability** from four transformed moments of one pair, not complete-data indistinguishability, a new physical law, or a first-in-literature claim.

2. A threshold measurement for the graph witness

Let `g_lambda(s)=4lambda^2 s exp(-2lambda s)` be the calibrated scale density for a positive eigenmode. Rook-edge density:

p_R(s)=(1/4)g_4(s)+(3/4)g_8(s).

Alternative density:

p_A(s)=sum_{j=1}^3 w_j g_(lambda_j)(s) with the certified eigenvalue and weight intervals.

Numerical evaluation of the full distributions gives numerical total-variation distance 0.104957402798 (four detected density crossings) and Bhattacharyya coefficient 0.990246324123. These latter integrals are high-accuracy numerical calculations, not outward-certified integral bounds.

A separate **fully certified** binary experiment requires much less: record the bit Y=1{s>3/100}. The exact rational graph eigenvalue/weight enclosures together with rational Taylor bounds for exponentials prove

P_R(Y=1)>930704/1000000, P_A(Y=1)<870032/1000000.

For p>q, the binary Bhattacharyya coefficient sqrt(pq)+sqrt((1-p)(1-q)) is maximized by reducing p and increasing q. Exact rational square comparisons then show B<994759/1000000. Tensorization and the classical Bhattacharyya bound give

P_e(M) <= (1/2) B^M < (1/2)(994759/1000000)^307 < 1/10.

Thus **307 independent photon trials suffice for <=10% equal-prior Bayes error** under this exact scale-threshold measurement contract, for this graph pair, with no detector loss and calibrated known source. This is a conservative sufficient guarantee, not the minimum and not a general physical resource bound.

The submitted 57,291,466 Cantelli guarantee concerns **a different** three-atom positive-measure witness *and* the fourth-moment estimator. Comparing those trial counts as if they were the same alternative/model/estimator would be invalid.

This illustrates a central lesson: matching several moments does not imply closeness of complete distributions. The tiny slow-mode spectral weight makes m4 sensitive to rare long-scale outcomes, while much larger spectral-weight rearrangements cause readily testable differences at short scales.

3. Statistical threshold calculations

The simple equal-phase and antisymmetric rook-edge/nonedge scale densities give the following tests:

- For a single identically repeated upper-threshold bit, equal-phase pair: M=165, t=0.22943, risk 0.099786766829, whereas the numerically optimized M=164 risk is ~0.10046967.
- Antisymmetric pair: M=173, t=0.23077, risk 0.099493720411; numeric optimized M=172 risk ~0.10017110.
- The source's **exact rational sweep certificates** exclude every upper threshold at M=164 and M=172, respectively. The threshold distributions were checked numerically. The claimed global minimum rests on the supplied exact interval-sweep certificate; numerical agreement alone is insufficient.
- Fourier-likelihood inversion reproduces M=116:0.10022951, M=117:0.09926536 for equal phase, and M=126:0.10028047, M=127:0.09939238 for antisymmetric. These full-distribution crossings are **numerical**.
- For the antisymmetric sample-mean Chernoff statistic at threshold 0.1711, the prescribed positive and negative exponential tilts give 0.0998551584 and 0.0998533841 at M=375. The source supplies stronger rational exponential enclosures; The reported floating values agree with those enclosures.

4. Quantum information, phase access and calibration rank

For a_lambda(s)=2lambda sqrt(s) e^(-lambda s) in L^2(ds), the pure-state quantum Fisher information for lambda equals 2/lambda^2; classical measurement of s achieves the same FI. Multiplying the amplitude by a *parameter-dependent* phase exp(i b lambda s) preserves all scale probabilities but changes the QFI to 2(1+b^2)/lambda^2. Thus coefficient two is **not** a universal quantum-physical ceiling. This agrees with Braunstein–Caves 1994. The phase dependence is an additional physical instrument assumption, not a free resource.

Maximizing the specified three-port real-amplitude local-FI objective along equal scales gives `0.8279358531658`. For the quoted equal-scale alias parameters, the ideal continuous-register overlap is `0.3429323662853`, one ideal pure photon has optimal error `0.0303199514`, and a coherent known-amplitude input requires mean energy `0.7774323335` to reach 10% error. These apply to distinct measurement contracts. The source report contains an analytic global-optimum proof of the three-port class; the numerical calculation checks its value and does not replace the analytic global-uniqueness proof.

On the rook and Shrikhande graphs, take `k=128L^+=24I+4A-3J`. The exact calculation constructs all 120 contrast-perturbation residual derivatives, collects 120 off-diagonal components per derivative, and eliminates the resulting **integer** `120 x 120` matrix modulo `1,000,003`. Both ranks equal **119**. Both matrices have zero column sums. This proves rank >=119 over rational numbers and rank <=119 from the exact column sum identity, hence **rational rank=119**. It supports the revised distinction between `d-1` constraints for common spectral functions and locally 119 directions for pair-specific contrast perturbations.

For the deterministic power-law readout $S' = \kappa S^p$, the transformed inverse-moment relation persists under the stated common coherence contrasts; consequently a zero quadratic residual alone does not identify linear scale calibration. A classification of all measurable readout functions preserving that graph-moment relation requires a separate argument. The controlled density-ratio theorem in *Detector Transfer Ambiguity* instead addresses recovery of the dimensionless incident coordinate under one common standard-Borel subprobability response kernel, known control multipliers and per-preparation normalization. Its zero-conditional-variance criterion does not establish the power-law classification.

Detector transfer ambiguity

Kirtus Justus

1 The experiment and its identifiable coordinate

Let $\lambda_\star > 0$ be a reference eigenvalue. The numerical value may be unknown. Assume the same prepared positive eigenmode at every setting, with an independently characterized control $L \mapsto cL$, and assume the ideal Ricker scale law. Write $U = \lambda_\star S$. At coupling $c > 0$ its density is

$$f_c(u) = 4c^2 u e^{-2cu}, \qquad u > 0.$$

The instrument registers a record Z in a standard Borel space. Its common response is a subprobability kernel $R(dz \mid u)$. Its total mass $\eta(u)$ is the probability of detection at scale u. It can include binning, noise, nonlinear response and scale-dependent loss. With a known number of prepared trials, the measured quantities are subprobability measures

$$\nu_c(B) = \int_0^\infty f_c(u) R(B \mid u)\, du.$$

Misses remain in the trial denominator. Common source coupling and common detector response across settings are physical inputs. Unknown setting-specific gains, background counts and source impurity require independent characterization. A mixture of eigenmodes can itself produce unresolved variation in U, even when the detector reports physical scale exactly.

Choose $a > 0$, $\alpha = 1 + a$ and $\beta = 1 + 2a$. Define $X = e^{-2aU}$, and let $\mathbb{E}_1[\cdot \mid Z = z]$ be conditional expectation under the reference joint law restricted to registered outcomes. All the measures ν_c have the same null sets because $f_c > 0$ on the same support.

Change of measure. For $j = 0, 1, 2, \ldots$ and $c_j = 1 + ja$,

$$\frac{1}{c_j^2} \frac{d\nu_{c_j}}{d\nu_1}(z) = \mathbb{E}_1[X^j \mid Z = z] \quad (\nu_1\text{-almost everywhere}).$$

Indeed, $f_{c_j}(u)/f_1(u) = c_j^2 X^j$. Multiply by the common response kernel, integrate over any measurable record set, and disintegrate the reference joint measure. This proves the identity without a smooth-density assumption.

Two settings. Suppose $Z = \phi(U)$ on a detected trial, with ϕ measurable and one-to-one. The first nontrivial density ratio yields the decoder

$$h(z) = \phi^{-1}(z) = -\frac{1}{2a} \log\left[\frac{1}{\alpha^2} \frac{d\nu_\alpha}{d\nu_1}(z)\right].$$

The common efficiency cancels. Once h is known, the reference recorded measure pushed onto the u coordinate is

$$h_\# \nu_1(du) = \eta(u) f_1(u)\, du.$$

This identifies η almost everywhere in that coordinate. The readout mapping itself is determined only where records occur. Dividing a conditional-on-click histogram by a second such histogram introduces the ratio of their total detection probabilities. Counting all prepared trials supplies that normalization directly.

The identified coordinate is $\lambda_\star S$. If $\lambda_\star$ changes to $b\lambda_\star$ and the readout and loss functions of physical scale are composed with $s \mapsto bs$, all these observed measures agree. An independently measured reference value sets the physical unit.

2 A held-out setting and a test of scale resolution

For an arbitrary common response kernel put

$$m(z) = \frac{1}{\alpha^2}\frac{d\nu_\alpha}{d\nu_1}(z), \qquad v(z) = \frac{1}{\beta^2}\frac{d\nu_\beta}{d\nu_1}(z) - m(z)^2.$$

The change of measure gives $v(z) = \mathrm{Var}_1(X \mid Z = z) \geq 0$. Under a recoverable record $v = 0$ almost everywhere, and two settings forecast the third through

$$\nu_\beta(dz) = \beta^2 m(z)^2 \nu_1(dz).$$

Conversely, under the common-kernel contract, equality makes $X = m(Z)$ almost surely among detections. Since X is strictly monotone in U, the decoder $-(2a)^{-1}\log m(Z)$ recovers U. Even if the record contains extra random information, this test concerns recoverability of the incident scale. A strictly positive v quantifies the optimal mean-square error of predicting X from a record. Its integral is the error per prepared reference trial restricted to detections. Divide by $\nu_1(\mathcal{Z})$ for the conditional error per detection.

There is also an admissibility check. Given a proposed exact decoder h, its pushed-forward registered measure must obey $h_\# \nu_1 \leq f_1(u)du$. This condition and the zero-variance forecast are sufficient for a compatible recoverable response when $m \in (0, 1)$ almost everywhere. To see this, disintegrate ν_1 along $u = h(z)$ into its marginal and conditional record law. Multiply that conditional law by the marginal's density relative to $f_1 du$. The resulting kernel has mass at most one and reproduces all three observed measures. Physical verification that the apparatus really uses one common kernel is still an independent step.

For a noisy example, register $Z = U$ with probability 0.55 and $Z = 3.5U$ with probability 0.45 after common loss. At $z = 1$, the conditional variance obtained from the three record densities is 0.02855966327. Direct calculation from the two possible incident scales gives the same value. A finite bin can also have positive variance when a perfectly ordered fine-scale record varies inside it. The finite-bin test below allows for that spread.

3 What finite controls can leave ambiguous

The complete family of exact measures ν_{1+ja} for all $j \geq 0$ identifies the common kernel in dimensionless scale. For each measurable record set B these observations are all moments of the finite measure obtained by pushing $f_1(u)R(B \mid u)\,du$ onto $X \in [0, 1]$. Polynomials are dense in continuous functions on this compact interval, so equal moments imply equal measures. Pull back by the one-to-one coordinate $X(u)$ and divide by f_1. A countable determining class of record sets gives equality of the kernels outside one null set. This is an exact identifiability result. It supplies no stable finite-order inversion of an arbitrary kernel.

An explicit finite counterexample uses $a = 1$, for which $\mathbb{E}_1 X^k = (k+1)^{-2}$. Let

$$H(x) = x^3 - \frac{3105}{2588}x^2 + \frac{5751}{16175}x - \frac{4679}{258800}.$$

This monic polynomial is orthogonal to $1, x, x^2$ under the reference law of X. Write $\epsilon = (2\sum_i |H_i|)^{-1}$ for its coefficient sum bound. Two detectors each return a single possible click, with efficiencies $1/2 + \epsilon H(X)$ and $1/2 - \epsilon H(X)$, respectively. These are genuine probabilities. At couplings 1, 2 and 3 the click probabilities coincide exactly. At coupling 4 their difference is $878769/815843875 > 0$. The unseen difference comes from the next conditional moment. The construction extends to any finite initial progression by using the next orthogonal polynomial.

An all-settings instrument ambiguity. Fix $k > 1$ and $0 < \eta_0 < 1$. Device A reports $z = u$ with efficiency $\eta_0 e^{-2c(k-1)u}$, which changes with the coupling. Device B reports $z = u/k$ with constant efficiency η_0/k^2. Their per-trial densities agree at every coupling:

$$g_c^A(z) = 4\eta_0 c^2 z e^{-2ckz} = g_c^B(z).$$

Their missed-click probabilities agree as well. This is an exact reason to check for coupling-dependent detector changes using an independently characterized reference signal. Additional settings of the same experiment cannot distinguish these devices.

4 Finite counts and two different tests

Fixed numbers of trials. Predeclare disjoint ordered bins B_j. For bins with positive reference mass put

$$m_j = \frac{\nu_\alpha(B_j)}{\alpha^2 \nu_1(B_j)}, \qquad m_j^2 \le \frac{\nu_\beta(B_j)}{\beta^2 \nu_1(B_j)} \le m_j.$$

The lower bound is Jensen's inequality and the upper bound uses $X^2 \le X$. They hold for noisy and nonmonotone responses. For an increasing deterministic response, the m_j decrease. The incident X in bin j is between $l = m_{j+1}$ and $h = m_{j-1}$, using 0 and 1 at the exterior ends. Thus $(h-X)(X-l) \ge 0$ improves the upper bound to $(h+l)m_j - hl$. At the boundary between bins j and $j+1$, the inferred incident coordinate is between $-\log(m_j)/(2a)$ and $-\log(m_{j+1})/(2a)$, assuming the boundary is in the readout range. If it falls in a response gap, the inequalities bracket a separating incident cutoff. This argument gives intervals for scale without estimating a pointwise density.

The study uses three independent sets of M trials, nine positive bins and ordinary two-sided Clopper–Pearson intervals for the marginal per-trial bin probabilities [1]. Allocate total error 0.05 over the 27 intervals by Bonferroni. The within-setting multinomial dependence is allowed. For example, if $[L_{cj}, U_{cj}]$ denotes each interval, a forecast using only settings 1 and α has the universal lower endpoint $\beta^2 L_{\alpha j}^2/(\alpha^4 U_{1j})$ and upper endpoint at most $\min(1, \beta^2 U_{\alpha j}/\alpha^2, \beta^2 U_{1j})$. The ordered test additionally propagates interval bounds for the neighboring means. Empty feasible sets reject their respective hypotheses. On the simultaneous coverage event, every valid hypothesis passes all its necessary inequalities. The familywise probability of a false rejection is therefore at most 0.05. Sparse bins can give an infinite upper bound on incident scale. The numerical beta quantiles are evaluated in floating point.

Independent randomized coupling labels. To test a linear readout through the origin more efficiently, retain the full record. For each prepared trial independently randomize between $c = 1$ and $c = \alpha$ with equal probabilities. Conditional on the registered z, the probability that the label is α equals

$$q_h(z) = \frac{\alpha^2 e^{-2ah(z)}}{1 + \alpha^2 e^{-2ah(z)}}$$

when the scale is recoverable with decoder h. Shared efficiency cancels. Given the registered records and their indices, the labels are independent with these probabilities. The experiment does not condition on a fixed number of prepared or detected trials in each arm.

Let L_h be the conditional likelihood of the labels and let L_{mix} be a preselected, normalized mixture of such joint likelihoods. On the linear null, $h(z) = kz$ for unknown $k > 0$. Reject at $E \ge 20$, where

$$E = \frac{L_{\text{mix}}}{\sup_{k \ge 0} L_{kz}}.$$

For the true k, the mixture likelihood ratio L_{mix}/L_{kz} has conditional expectation one. The displayed statistic is no larger, so Markov's inequality bounds its conditional and unconditional rejection probability by 0.05. This proof allows arbitrary shared efficiency and arbitrary positive k. It uses the same likelihood-ratio principle as finite-sample universal inference [2]. The denominator has a concave one-dimensional log likelihood. The code brackets its maximizer and uses an upper tangent bound. The numerator gives equal weights to the ten fixed decoders $h(z) = kz^{1/p}$ with $p \in \{0.7, 1.25\}$ and $k \in \{0.6, 0.8, 1, 1.25, 1.6\}$. These choices affect power. The bound concerns a predeclared fixed number of prepared trials.

References

[1] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika* **26**(4), 404–413 (1934). https://doi.org/10.1093/biomet/26.4.404.

[2] L. Wasserman, A. Ramdas and S. Balakrishnan, "Universal inference," *Proceedings of the National Academy of Sciences* **117**(29), 16880–16890 (2020). https://doi.org/10.1073/pnas.1922664117.

Sharp two-coupling event region

Kirtus Justus

Postulates

P1. A positive spectral mode at controlled relative coupling `c>0` produces an incident scale `U>0` with density

`f_c(u)=4 c^2 u exp(-2 c u), u>0`.

P2. For a registered event `B`, the detector has a measurable response `eta(u)=R(B|u)` satisfying `0<=eta(u)<=1`, independent of `c`. A general noisy, nonlinear, lossy device is allowed. Record event counts **per prepared trial**, not conditional on successful detection.

P3. Coupling values and their ratios are separately calibrated, and the preparation law P1 is either established directly or generated by an unknown but **coupling-stable** mixture of positive modes whose eigenvalues all scale by the same factor `c`. Zero-mode-only trials, if present, must be handled with their missing-event mass explicitly.

P4. In the finite-shot theorem, trials within each coupling are independent Bernoulli experiments, and the detector event `B` is fixed before observing counts.

These are supplied physical/instrument assumptions, not consequences of the prephysical proposal.

Theorem 1: sharp two-setting ROC region

Write `F_c(t)=1-(1+2ct)exp(-2ct)` and consider couplings `1` and `b>1`. For `0<p<1` let `q_p` be the unique positive root of `F_1(q_p)=p`. Define

`U_b(p)=F_b(q_p)` and `L_b(p)=1-F_b(q_(1-p))`, with continuous endpoint values `U_b(0)=L_b(0)=0`, `U_b(1)=L_b(1)=1`.

If `p_c=Pr(B|coupling c)` under P1-P2, then

`L_b(p_1) <= p_b <= U_b(p_1)`.

Sharpness: for an isolated mode and any `p_1`, the upper boundary is reached by the threshold detector `eta(u)=1_{u<=q_p}` and the lower boundary by `eta(u)=1_{u>=q_(1-p)}`. Every point between these boundaries at that fixed `p_1` is attained by a convex combination of the two threshold kernels.

Proof: `f_b(u)/f_1(u)=b^2 exp[-2(b-1)u]`, strictly decreasing in `u`. Given a fixed `int f_1 eta=p_1`, Neyman-Pearson's likelihood-ratio rearrangement maximizes `int f_b eta` by placing detection weight on the smallest `u`, and minimizes it on the largest `u`. The threshold integrals are `F_b(q_p)` and `1-F_b(q_(1-p))`. Convex mixtures attain the interior. QED.

At `b=2`, `p_1=1/2`, the admissible interval is exactly

`0.151832024412957731889... <= p_2 <= 0.848167975587042268111...`.

In particular, `(p_1,p_2)=(1/2,9/10)` is impossible for a stable detector acting on this known incident law.

Theorem 2: robustness under an unknown stable spectral mixture

Let a latent positive eigenvalue `Lambda` have **any fixed** probability distribution `pi`, independent of the controlled coupling `c`. Given `Lambda=lambda`, suppose

`f_c(u|lambda)=4(c lambda)^2 u exp[-2 c lambda u]`.

The same envelope L_b(p_1)<=p_b<=U_b(p_1) holds for an arbitrary stationary detector eta(u), even though pi and the individual eigenvalues need not be known.

Proof 1 (concavity): conditional on lambda, make the substitution v=lambda u; Theorem 1 applies with the same functions L_b,U_b. Write the conditional event rates p_1(lambda),p_b(lambda). The upper curve is strictly concave because U_b'(p)=b^2 exp[-2(b-1)q_p] is decreasing. The lower curve is convex because L_b(p)=1-U_b(1-p). Jensen's inequality gives

p_b=E[p_b(Lambda)] <= E[U_b(p_1(Lambda))] <= U_b(E[p_1(Lambda)]),

and analogously the lower bound. QED.

Proof 2 (experiment randomization): one can sample V~f_c, draw Lambda~pi independently of c, and set U=V/Lambda. This is a coupling-independent Markov transformation of the original scale experiment. Passing U through the detector is a further randomization and therefore cannot exceed the original Neyman-Pearson ROC envelope. QED.

If the spectral weights change with coupling, or the detector changes with coupling, the conclusion does not follow. If an independently identified zero mode generates no registered positive-scale event, include that missing mass in the per-trial probabilities; the same universal envelope remains conservative by the concavity/convexity argument.

Theorem 3: exact finite-dimensional detector-feasibility dual

For any known finite set of calibrated incident densities f_1,...,f_m, an event-probability vector p=(p_1,...,p_m) is attainable by one unchanged detector kernel 0<=eta(u)<=1 if and only if, for every real vector a=(a_1,...,a_m),

sum_j a_j p_j <= int_0^infty max(0, sum_j a_j f_j(u)) du.

Proof: the map eta -> (int f_1 eta,...,int f_m eta) sends the weak-* compact convex set {eta in L-infinity: 0<=eta<=1} into a compact convex subset of R^m. Its support function in direction a is the stated integral, obtained by choosing eta=1 wherever sum_j a_j f_j>0 and eta=0 where it is negative. By finite-dimensional separating-hyperplane duality, the support inequalities characterize membership exactly. QED.

This theorem is a classical convex feasibility statement for **one event**, not a claim that checking a few selected directions certifies a common response kernel for an entire multibin detector.

At m=2, the continuum of support inequalities reduces to the sharp ROC boundaries above.

Counterexample: all basic three-setting moment checks can pass while the two-setting envelope fails

For couplings 1,2,3, consider the context-dependent binary detector with exact event probabilities

p_click=(1/2,9/10,19/20) and p_miss=(1/2,1/10,1/20).

The normalized moment vectors under the proposed stationary Gamma model would be

- click: (r0,r1,r2)=(1/2,9/40,19/180);
- miss: (r0,r1,r2)=(1/2,1/40,1/180).

In **both** cases, r0*r2-r1^2=31/14400>0 and r0>r1>r2>0. Therefore the commonly used quadratic positivity, monotonicity, and conditional-variance conditions pass. Their total moments also add to (1,1/4,1/9), the correct Gamma reference moments.

Nevertheless, p_click,2=0.9>U_2(0.5)=0.8481679755.... This known-reference bound rejects the stationary detector from **only the first two settings**.

A linear separating certificate is obtained at q=q_(1/2) with slope rho=4exp(-2q)=0.7467292354.... The reference experiment obeys for every stationary detector

p_2-rho*p_1 <= U_2(1/2)-rho/2 = 0.4748033578....

The candidate violates this inequality by exactly 0.9-U_2(1/2)=0.0518320244....

The example demonstrates that treating a small truncated Hausdorff moment test as complete detector certification is unsound when a specific input distribution and domination constraint are already known.

Theorem 4: finite-trial falsification under unknown fixed spectral mixtures

Choose couplings `1,2`, an event `B` selected in advance, and `M` independent prepared trials at each coupling. Let the empirical event fractions be `hat p_1,hat p_2`. For a target familywise error `alpha`, put

`eps=sqrt(log(4/alpha)/(2M))`.

Reject the stationary-detector null if

`hat p_2-eps > U_2(min(1,hat p_1+eps))`

or if

`hat p_2+eps < L_2(max(0,hat p_1-eps))`.

Validity: Hoeffding plus the two-proportion union bound shows that both true event probabilities lie within the displayed intervals with probability at least `1-alpha`. On this event, Theorem 2 prevents rejection. Hence the false-rejection probability is at most `alpha`, uniformly over all fixed spectral mixtures allowed by P3 and all unchanged detectors.

Explicit power guarantee: against the particular coupling-dependent alternative `(p_1,p_2)=(0.5,0.95)`, choose `alpha=0.05`, `M=2500` per setting. Then

`eps=0.029604143746...`,

`0.95-2eps-U_2(0.5+2eps)=0.002437333304...>0`.

On the same simultaneous Hoeffding concentration event, the upper-bound rejection is forced. This proves **type-I error at most 5%, power at least 95%, using 5,000 total independent preparations**, for the specified alternative. It is a conservative sufficient count, not an optimal trial count.

Incomplete detector calibration

Kirtus Justus

2. Independent result: exact detector partial-identification interval

Let trusted density matrices rho_i (i=1,...,m) have exact measured response rates p_i for an otherwise unknown but stable binary measurement effect 0 <= E <= I on a finite-dimensional Hilbert space. For a target state rho_T define

```
L(p) = min Tr(E rho_T),   U(p) = max Tr(E rho_T),
subject to Tr(E rho_i)=p_i for every i and 0 <= E <= I.
```

These are finite semidefinite programs, and by definition are the sharp possible target-response bounds for the entire admissible effect class. Both extrema are attained when the calibration vector is feasible because the effect set is compact. The image is a closed interval by convexity. **They are data dependent**: a target outside the real reference span does not imply ambiguity for every possible calibration dataset.

For any real vector lambda, put A=lambda-weighted subtraction `A=rho_T-sum_i lambda_i rho_i`. Then

```
Tr(E rho_T) = sum_i lambda_i p_i + Tr(E A)
            <= sum_i lambda_i p_i + Tr(A_+),
```

where `A_+` is the positive part of A and the inequality is sharp when E is its positive spectral projector if no other constraints are imposed. Consequently,

```
U(p) <= inf_lambda [sum_i lambda_i p_i + Tr((rho_T-sum_i lambda_i rho_i)_+)].
```

The inequality is equality under the usual finite-dimensional Slater condition: at least one compatible effect satisfies 0 < E < I. At boundary response vectors, rely on the primal SDP or limiting dual, not an unqualified interior-strong-duality assertion. For normalized densities, `L(p)=1-U(1-p)` follows by exchanging E and I-E.

Elementary complete proof for a qubit

Use two trusted reference states |0><0| and |1><1|. Suppose their calibrated click probabilities are p0,p1. A possible effect has the form

```
E = [[p0, z], [conjugate(z), p1]].
```

Positivity of E and I-E is equivalent to

```
|z| <= R := min(sqrt(p0*p1), sqrt((1-p0)*(1-p1))).
```

For the held-out target |+>=(|0>+|1>)/sqrt(2), `q=(p0+p1)/2+Re(z)`. Choosing z=+/-R proves the *exact* interval

```
(p0+p1)/2-R <= q <= (p0+p1)/2+R.
```

In particular, if p0=p1=p, the range is `[max(0,2p-1),min(1,2p)]`. At p=0 the calibration forces E=0 and q=0, despite |+><+| lying outside the real reference span. At p=1/2 the same two reference probabilities are compatible with every target rate from zero to one. These are extreme physical effects, not a numerical fitting artifact.

This is an instance of established quantum partial tomography and set-identification SDPs (e.g. *Physical Review A* 98, 042318, 2018), not a claim of mathematical originality. Its application to the present measurement contract is useful because it avoids inferring universal ignorance from mere failure of reference informational completeness.

Uncertain reference data

When independently justified intervals [l_i,u_i] replace exact p_i, use the same primal with inequalities. Its Lagrangian bound reads

```
U <= inf_{beta_i,gamma_i>=0} [sum_i (beta_i u_i-gamma_i l_i)
      + Tr((rho_T - sum_i (beta_i-gamma_i)rho_i)_+)].
```

Equality follows under appropriate strict feasibility. This is how uncertainty and physicality should jointly enter, rather than treating a noisy point estimate as exact.

3. Cross-check: tetrahedral source-response bound

For four trusted tetrahedral qubit states rho_i=(I+r_i dot sigma)/2, define D_i=(3rho_i-I)/2. They form a dual frame and satisfy sum D_i^2=(5/2)I. The sixteen probabilities p_ij=Tr(E rho_i tensor rho_j) determine E exactly. Under entrywise errors |delta p_ij| <= delta, the *maximum response over separable states*, b_sep(E), changes by at most 4 delta, which is sharp. The proof uses the dual weights

```
w_i(x) = Tr(D_i rho_x)=(1+3r_i dot x)/4.
```

The exact maximum of sum_i |w_i(x)| is two for Bloch vectors |x|<=1, attained at a tetrahedral antipode, with weights (-1/2,1/2,1/2,1/2). Product response weights therefore have absolute sum at most four. Direct construction of a sign-perturbation near E=I/2 attains the bound without leaving the effect set for sufficiently small delta.

Using alpha=.05, Delta=13/1280, and seventeen independent settings of n trials each, the sufficient requirement n > 50 log(34/alpha)/Delta^2 gives n=3,161,479, total 53,745,143. This is a *conservative sufficient* shot budget with trusted preparations, stable effect, and independent Bernoulli trials, not a necessary hardware cost.

Separable-reference detector tests

Kirtus Justus

2. Postulates for a new conditional calibration protocol

1. **Known space and reference controls.** The physical system is two qubits with a known tensor decomposition and trusted local Pauli operations. This decomposition and those operations are supplied; they are not derived from a signed-relational substrate.
2. **Stable binary detector.** For each preparation and measurement condition the same unknown Hermitian effect E acts on the two qubits, with $0 \preceq E \preceq I$. The event complement is $I - E$. No false-click/missed-click parameterization, diagonal assumption, or classical-only noise assumption is made.
3. **Trusted Bell dephasing.** Immediately before every detector readout, independently randomized local gates implement
$$\mathcal{T}(\rho) = \frac{1}{4} \sum_{g \in \{II, XX, YY, ZZ\}} g \rho g^\dagger.$$
This map is an average of product unitaries and hence maps separable states to separable states.
4. **Trusted separable reference preparations.** Four source preparations are physically characterized as $|00\rangle, |++\rangle, |+i, -i\rangle, |+i, +i\rangle$, where $|\pm i\rangle = (|0\rangle \pm i|1\rangle)/\sqrt{2}$. They are not merely assigned labels while their Bloch radii are left unknown.
5. **Independent sampling.** For finite-trial guarantees only, trials within each condition are independent with stable outcome laws. The reference and target settings are interleaved or otherwise characterized for drift. Their outcomes are recorded per prepared pair.

3. Theorem: separable reference tomography of a general binary effect

Let $B_j = |b_j\rangle\langle b_j|$ denote the Bell projectors in the order $(\Phi^+, \Phi^-, \Psi^+, \Psi^-)$. The twirl is self-adjoint and projects operators onto the commuting algebra generated by XX and ZZ. Consequently,
$$E_{\mathcal{T}} = \mathcal{T}^*(E) = \sum_{j=1}^{4} e_j B_j, \qquad e_j = \mathrm{Tr}(E B_j) \in [0, 1].$$
The event probabilities of the four trusted product reference preparations form $r = Me$ with
$$M = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$
Exact symbolic calculation gives
$$\det M = \frac{1}{8}, \qquad M^{-1} = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 1 & -1 & 0 & 1 \\ -1 & 1 & 0 & 1 \\ -1 & -1 & 2 & 1 \end{pmatrix}, \qquad \|M^{-1}\|_\infty = 5.$$
Thus the physically twirled detector response is determined by four separable reference event probabilities. An unrestricted two-qubit Hermitian effect has 16 real parameters, so four settings cannot perform unrestricted effect tomography without the trusted twirl; a full spanning linear tomography would require at least 16 independent input preparations. This is a parameter-count statement about full effect identification, not a universal lower bound for every entanglement witness. **This is conditional tomography, not a measurement-device-independent claim.** The trusted gate and preparation conditions supply the extra physical information.

4. Theorem: exact separable bounds for any twirled binary effect

Write the recovered Bell eigenvalues in sorted order $e_{(1)} \geq e_{(2)} \geq e_{(3)} \geq e_{(4)}$. Then for every separable two-qubit state σ,

$$\boxed{\frac{e_{(3)} + e_{(4)}}{2} \leq \mathrm{Tr}(E\mathcal{T}(\sigma)) \leq \frac{e_{(1)} + e_{(2)}}{2}.}$$

Proof: Local-unitary twirling preserves separability and makes the state Bell diagonal: $\mathcal{T}(\sigma) = \sum_j q_j B_j$. By the two-qubit PPT criterion, Bell-diagonal separability is exactly the polytope $q_j \in [0, 1/2], \sum q_j = 1$. Linear optimization places $1/2$ on the two largest coefficients for the maximum, or on the two smallest for the minimum. These bounds are attained by Bell-diagonal separable mixtures of two Bell states of equal weight. Therefore an observed target event probability outside the interval certifies that its original state was entangled under the stated physical controls. Failure to violate the interval is inconclusive: Bell twirling can destroy entanglement.

This is a standard consequence of Bell twirling, PPT separability and linear optimization, not claimed as a new general quantum theorem.

5. Negative control A: a physically valid off-diagonal detector fakes a violation without twirling

Consider the effect

$$E = 0.1I + 0.8B_{\Phi^+} + 0.05(|\Phi^+\rangle\langle\Psi^+| + |\Psi^+\rangle\langle\Phi^+|).$$

Its eigenvalues are strictly inside $[0, 1]$, so it is a legitimate detector effect. Its Bell diagonal entries are $(0.9, 0.1, 0.1, 0.1)$, whose separable upper envelope **after twirling** is $1/2$. Yet the separable state $|++\rangle$ gives

$$\mathrm{Tr}(E|++\rangle\langle++|) = \frac{11}{20} = 0.55,$$

which would be a false certificate if someone simply treated E as Bell diagonal. Applying the trusted twirl to the same product state changes the event probability to exactly $1/2$. The explicit matrix calculations are checked with exact SymPy arithmetic in `verify_twirl_detector.py`.

For the stipulated model-B Bell-diagonal state with $\Pr(\Phi^+) = 653/1280$, the same valid off-diagonal detector gives event probability $813/1600 = 0.508125$ after twirling, exceeding its exact separable ceiling by $13/1600$. This shows that the calibrated approach can detect the specified entanglement without assuming a classical bit-flip model; it does not claim universal detector efficiency.

6. Negative control B: trusted twirling does not replace trusted reference-state purity

Retain the two original gate-set source radii $a_A = 1/2$ and $a_B = 3/4$, which are unknown to an experimenter using only the native gate-set labels. Let model A's detector effect be $E_A = B_{\Phi^+}$. Let model B use the legitimate effect

$$E_B = \frac{1}{4}\left[I + \frac{XX - YY + ZZ}{(3/2)^2}\right],$$

whose Bell eigenvalues are $(7/12, 5/36, 5/36, 5/36)$. The effects and sources are related by the original similarity gauge, and the Bell twirl commutes with that gauge.

Exact matrix tests verify identical event probabilities for **all 36 ordered product inputs** constructed from the two model-specific labeled six-axis sources, including after twirling. For the corresponding gate outputs, both predict exactly $117/320$ clicks, although one output is separable and the other is entangled. Their respective separable upper bounds are $1/2$ and $13/36$;

the second output exceeds its own bound by $13/2880$. Thus four reference *labels*, or even all native axis reference labels, do not complete tomography when their physical radii remain uncharacterized.

The positive calibration theorem requires trusted physical reference preparations, not just well-conditioned algebra. This is the same gauge issue seen in gate-set tomography, not a contradiction of the positive theorem.

7. Finite-trial theorem and systematic error

Suppose each of the four reference probabilities and one target probability is estimated from M independent Bernoulli trials. For an error level α, define

$$\delta_\alpha(M) = \sqrt{\frac{\log(10/\alpha)}{2M}}.$$

A union bound and Hoeffding imply that all five errors are at most δ_α simultaneously with probability at least $1-\alpha$.

Let $\hat{e} = M^{-1}\hat{r}$ and $b_+(e) = (e_{(1)} + e_{(2)})/2$. Because $\|M^{-1}\|_\infty = 5$ and b_+ is 1-Lipschitz in infinity norm,

$$\left|b_+(\hat{e}) - b_+(e)\right| \le 5\delta_\alpha.$$

A valid one-sided size-α entanglement test is therefore:

$$\boxed{\text{reject separability only if } \hat{p}_{\text{target}} > b_+(\hat{e}) + 6\delta_\alpha.}$$

For a fixed entangled alternative whose true calibrated effect has positive witness margin $\Delta = p_{\text{target}} - b_+(e) > 0$, the same concentration event with failure probability β proves power at least $1-\beta$ whenever

$$\Delta > 6[\delta_\alpha(M) + \delta_\beta(M)].$$

For the ideal Bell effect $E = B_{\Phi^+}$, model B has $\Delta = 13/1280$. With $\alpha = \beta = .05$, $M = 3,698,314$ preparations **per setting** (four references and one target) satisfies the strict sufficient inequality, for 18,491,570 prepared pairs in total. This is deliberately conservative and **not an optimized cost**. It exposes the price of calibrating a largely uncharacterized detector to precision at a 1%-level witness margin. The source's 26,226-pair ideal-detector optimum assumes a different, stronger prior calibration and thus is not directly comparable.

If each of the five settings carries an additional independently bounded systematic probability bias ξ, the same proof yields a valid threshold with $6(\delta_\alpha + \xi)$. Sufficient power against a nominal alternative needs $\Delta > 6(\delta_\alpha + \delta_\beta + 2\xi)$ under worst-case independent biases, or tighter bounds when common-mode drift is characterized. Unbounded setting-dependent drift, state leakage or failures of the twirl invalidate these certificates.

Unitary scaling rigidity

Kirtus Justus

2. Postulates and notation

Let M_2(C) denote one-qubit matrices and M_4(C) two-qubit matrices. For lambda>0, let S_lambda fix the identity and multiply every traceless Pauli component by lambda. Let G_lambda=S_lambda⊗S_lambda, acting on matrices on the *supplied* two-qubit tensor product. For a tensor-product Pauli sigma_a⊗sigma_b, put w(a,b)=1_{a≠0}+1_{b≠0}; then G_lambda(sigma_a⊗sigma_b)=lambda^{w(a,b)}(sigma_a⊗sigma_b).

The channel D(A)=Tr(A) I_4/4 is completely depolarizing. For a two-qubit unitary U write Ad_U(A)=UAU†. A gate set consists of physical preparations, effects and (trace-preserving or outcome-conditioned) completely positive maps with the same supplied product structure. The gauge transformation on operations is Phi -> G_lambda Phi G_lambda^{-1}, with effects transformed dually. This is a predictive similarity, **not** a physical channel G_lambda on arbitrary states if lambda>1.

3. Theorem: a noiseless entangling unitary forbids nontrivial Bloch-scaling gauge

Theorem 1. For lambda>0, lambda≠1, and a unitary U on two qubits, the linear map

Psi=G_lambda∘Ad_U∘G_lambda^{-1}

is CPTP **if and only if**, up to overall phase, U is a product of one-qubit unitaries or a product followed by SWAP. Consequently, if U can entangle some pure product input, Psi is not CPTP for lambda≠1.

Proof. In the normalized Hermitian Pauli basis, Ad_U is a real orthogonal 16×16 matrix R and G_lambda is positive diagonal D, with entries 1, lambda (weight-one Paulis) and lambda^2 (weight-two Paulis). The candidate T=D R D^{-1} is unital and trace preserving. Every unital trace-preserving completely positive map is a Hilbert–Schmidt contraction. If T is CPTP, all its singular values are at most one. But |det T|=|det R|=1; their product is one, so every singular value is exactly one and T^T T=I. This implies R^T D^2 R=D^2, hence R preserves the eigenspaces of D^2. Since lambda≠1, it must preserve the six-dimensional one-body subspace

V_1=span{sigma_j⊗I, I⊗sigma_j : j=x,y,z}.

Conjugation by U therefore normalizes the Lie algebra su(2)⊕su(2). Its two simple ideals can only be individually preserved or exchanged; every automorphism of a single su(2) ideal is implemented by a one-qubit unitary. The remaining unitary commutes with the entire two-qubit matrix algebra and is scalar. Thus U is local or local followed by SWAP. Conversely, those unitaries preserve Pauli weight, commute with G, and yield exactly Psi=Ad_U, which is CPTP. QED.

The proof combines elementary Lie-algebra automorphisms with established Hilbert–Schmidt contractivity for unital CPTP maps. It is not asserted to be an original normalizer theorem.

Direct CNOT witness. For U=CNOT, one has

Psi(X⊗I)=lambda X⊗X, Psi(X⊗X)=lambda^{-1} X⊗I.

For either lambda>1 or lambda<1, a norm-one Hermitian unitary Pauli is mapped to an operator of norm greater than one. A positive unital map cannot do that. This immediately proves nonpositivity for the CNOT instance without Choi analysis.

4. Exact noisy counterexample: physical entangling gates on both sides

Theorem 2 (explicit conditional counterexample). Take `lambda=3/2`, `p=63/100`, and

`Phi_A=(1-p)Ad_CNOT + p D`,

`Phi_B=G_lambda Phi_A G_lambda^{-1}`.

Both maps are CPTP and are entangling channels. The two complete models with the above A/B preparations and effects, their same calibrated one-qubit rotations, and their respectively defined joint gates predict identical probabilities for every circuit assembled from these named operations, including arbitrary repetitions and adaptive classical feedback. Their identically labeled source purities still differ.

Proof of CPTP. Trace preservation and unitality are immediate. Using the unnormalized 16×16 Choi matrix `J(Phi) = sum_ij |i><j| ⊗ Phi(|i><j|)`, exact Gaussian-rational arithmetic gives the spectrum of `J(G Ad_CNOT G^{-1})`:

Eigenvalue	Multiplicity
25/6	1
5/12	4
-5/12	4
0	6
-1/6	1

Because `J(D)=I_16/4`, the transformed depolarized gate has Choi spectrum:

Eigenvalue	Multiplicity
1/300	4
23/240	1
63/400	6
187/600	4
2039/1200	1

All eigenvalues are strictly positive. This proves CP. In fact the exact minimum depolarization making this transformed gate CP is `p=5/8`; the chosen `p=63/100` lies strictly inside that region, with minimum Choi eigenvalue `1/300`. The original Phi_A is trivially CPTP.

Proof of entanglement generation. For the pure product input `|+><+|⊗|0><0|`, the two output partial transposes have eigenvalue spectra respectively `{137/400 (three times), -11/400}` and `{87/200, 137/400 (twice), -3/25}`. Each has a negative eigenvalue, proving that each channel can produce entangled outputs from a product input.

Even the labeled accessible outputs disagree on entanglement. For the identically labeled product preparations `(+x,+z)`, use radii `a_A=1/2`, `a_B=3/4`. The output partial-transpose eigenvalues are

- A: `43/320`, `437/1600` (twice), `511/1600`, all nonnegative, hence separable for two qubits.
- B: `-13/1280`, `1933/6400` (twice), `2599/6400`, hence entangled.

Yet the permitted local preparations and local effects do not distinguish them. To prove the complete equality, observe that the corresponding product states transform by G and effects transform by the dual inverse. Every named gate, including the joint Phi_B, transforms by the same similarity. Thus all G and `G^{-1}` terms cancel telescopically in each circuit probability, including individual adaptive branches. Exact arithmetic additionally checks 36 product source pairs against 36 product effect pairs, all 1,296 single-joint-gate tests, plus all 36 pairs for the uncalibrated SWAP model.

This does **not** mean entanglement is an illusion or that the two full physically calibrated theories are equivalent. It means the restricted gate-set data do not identify the abstract purity and output entanglement assigned to those labels. Independent joint calibration changes the observation contract. QED.

5. Proposition: arbitrarily small nonzero noise can retain some ambiguity

Fix any `0<p<2/3` and choose preparation and effect contrasts strictly inside their positivity intervals. At `lambda=1`, `Phi_p=(1-p)Ad_CNOT+pD` has strictly positive Choi matrix, since `p>0`, and produces an entangled output from `|+0>` because `1-p>1/3`. The Choi matrix and the output partial transpose depend continuously on lambda. Therefore for all lambda sufficiently close to but different from one, the transformed Phi'_p remains CPTP and entangling. The source and effect gate sets also remain physical by continuity. The models remain observationally equivalent under their corresponding uncalibrated gates, yet give different preparation purities.

Thus there is **no universal positive lower noise threshold** below which every nontrivial similarity is automatically ruled out by merely requiring all available gates to be entangling CPTP channels. The allowed gauge differences shrink as noise tends to zero. This is a conditional finite-dimensional continuity argument, not a universal theorem about all noise models.

Predictive dimension and observation

Kirtus Justus

1. Operational and mathematical assumptions

Fix a block length `k >= 2` and an independent reference selector with `P(X=1)=t` and `P(X=0)=q=1-t`, where `0<t<1`. Define `h(1)=1`, `h(0)=-1` and `epsilon_k=min(t,q)^k`. On an aligned k-word x define

$$P_k(x) = \prod_{i=1}^{k} p(x_i) - \epsilon_k \prod_{i=1}^{k} h(x_i).$$

The correction has zero total mass and vanishes when any coordinate is marginalized. It is small enough to preserve nonnegativity. Concatenate independent k-blocks and draw the initial block offset uniformly from `0,...,k-1`, retaining that offset during a continuous run. This yields a stationary selector process. Every marginal with fewer than k sites is exactly the independent Bernoulli law.

Under the quantum-instrument comparison, a selector value 1 implements a known unitary gate; value 0 implements complete depolarization. A calibrated independent probe produces a match bit `Z`. It has `P(Z=1|X=1)=1` and `P(Z=1|X=0)=r`, with `r=1/16` for the ideal generalized Bell measurement. An additional classical detector may flip match bits.

We keep distinct the following experimental contracts: (i) a single continuous stationary controller run; (ii) independently reinitialized fixed-length records; (iii) direct access to the controller versus accessible probe records. Transition-emitting hidden-state counts refer to a model class, not to the microscopic state count of physical hardware.

2. Exact predictive dimension and its positive realization

Proposition 1: zero-run Hankel lower bound

Let `z_m=P(X_1=...=X_m=0)`. Put

$$r_k = 1 - \frac{(-1)^k \epsilon_k}{q^k}.$$

For `m=nk+j`, `n>=1` and `0<=j<k`, phase averaging gives

$$\frac{z_m}{q^m} = \frac{(j+1)r_k^n + (k-j-1)r_k^{n-1}}{k},$$

while `z_m=q^m` for `m<k`. These coefficients have rational generating function

$$\sum_{m\geq 0} \frac{z_m}{q^m} u^m = \frac{\sum_{j=0}^{k-1} u^j + \frac{1-r_k}{k} u^k \sum_{j=0}^{k-2} (k-1-j)u^j}{1 - r_k u^k}.$$

The reduced sequence has Hankel rank `2k-1`, including the special case `r_k=0`. Consequently every linear predictive representation of the full stationary word process has dimension at least `2k-1`.

Proposition 2: positive upper realization

For each proper prefix u of an aligned block, define

$$b(u) = \prod_{i=1}^{|u|} \frac{h(u_i)}{p(u_i)}.$$

For each prefix length j=1,...,k-1, let l_j and u_j be its minimum and maximum over possible j-symbol prefixes. The conditional suffix law is affine in b(u):

$$P(v \mid u) = \prod_i p(v_i) - \epsilon_k b(u) \prod_i h(v_i).$$

Therefore each intermediate conditional law is a convex mixture of the two endpoint laws at that position. Construct one fresh-block hidden state and two endpoint states at each interior position. At interior positions emit x with probability p(x), update b to b h(x)/p(x), and randomly encode the updated b as the convex combination of the two endpoint states at the next position. At the final position emit x with probability p(x)-epsilon_k b h(x) and return to the fresh state. The output probabilities are nonnegative and normalized. Initialize phases uniformly; conditional endpoint weights have mean b zero at each interior position, which makes the initialization stationary.

This is a transition-emitting positive hidden-state model with 1+2(k-1)=2k-1 states. Proposition 1 proves that no smaller model of this class reproduces the complete word law. The equality of minimum positive and linear dimension is **specific to this construction**, not a generic consequence of finite Hankel rank.

Verification

The accompanying verifier constructs exact rational transition probabilities and checks every output word up to lengths 7,8,10,8,7 for k=2,3,4,5,6, respectively. The resulting 2,047 word tests at k=4 agree exactly. The zero-run Hankel ranks are 3,5,7,9,11. Additional checks at t=1/10,1/2,4/5 preserve the claimed positive realization and Hankel ranks. No approximate numerical tolerance is used for these results.

3. Observation channels and predictive rank

Proposition 3: injective per-use readout preserves linear predictive dimension

Suppose the hidden alphabet is {0,1} and an observation kernel K has different conditional output distributions K_0 and K_1. Since both are normalized probability vectors, these columns are linearly independent. There exists a real linear left inverse L with LK=I_2. For any word length m, the output probability tensor is K^{\otimes m} applied to the hidden probability tensor, and the hidden tensor is recovered algebraically by L^{\otimes m}. Thus the full predictive Hankel ranks agree.

The left inverse can contain negative coefficients and is **not asserted to be a physical readout operation**. Statistical error may be amplified severely by it, so exact predictive-rank identification does not imply practical finite-sample identification.

At k=4, t=37/100, r=1/16 and an independent readout flip rate 1%, the leading observed 7x7 zero-run Hankel matrix has exact rank seven and a numerical smallest singular value of approximately 2.75841e-6. Rank is exact, whereas conditioning is poor.

4. New test separating bounded detector-only memory from controller correlations

A one-step calibration of detector error does not establish conditional independence across gate uses. A stronger null model still permits arbitrary within-record correlations of the bit flips.

Null assumption H0: selectors are independent Bernoulli(t), and ideal probe match bits are conditionally independent per use. The detector reports `Y_i=Z_i XOR E_i`, where the four flip variables may have any joint distribution, each has the same Bernoulli(e) marginal, and the entire error vector is independent of the ideal match vector. The distribution may be arbitrary across the four positions, but independent records must be initialized for the statistical guarantee below.

Set `s=t+(1-t)r`, `m=s+(1-2s)e` and the centered four-way statistic

$$T_4 = \prod_{i=1}^{4}(Y_i - m).$$

Theorem 4: a universal correlated-flip bound within H0

Condition on the error vector. Since the ideal match bits are independent,

$$\mathbb{E}(Y_i - m \mid E_i) = (1-2s)(E_i - e).$$

It follows that

$$\mathbb{E}_{H_0} T_4 = (1-2s)^4 \mathbb{E} \prod_{i=1}^{4}(E_i - e).$$

Holder's inequality gives the exact **universal upper bound on the absolute value**

$$\boxed{\left|\mathbb{E}_{H_0} T_4\right| \le |1-2s|^4 \left[e(1-e)^4 + (1-e)e^4\right].}$$

The Holder factor can be attained as a positive fourth moment by perfectly correlated errors `E_1=...=E_4`, though the sharpest negative-side bound could be smaller. The stated symmetric bound is deliberately conservative.

Alternative H1: the stationary four-block correlated selector is used with perturbation amplitude `epsilon_4=t^4`, while detector flips are independent Bernoulli(e). Exactly one of four possible phases aligns a full observed four-word with a block. Conditional-output factorization yields

$$\boxed{\mathbb{E}_{H_1} T_4 = -\frac{t^4}{4}\left[(1-r)(1-2e)\right]^4.}$$

These are different **physically specified model classes**; the theorem is not a general device-independent memory-location witness.

Exact numerical separation

At `t=37/100`, `r=1/16`, `e=1/100`:

$$\begin{aligned} s &= 131/320 = 0.409375, \\ m &= 6579/16000 = 0.4111875, \\ \mathbb{E}_{H_1} T_4 &= -0.003338384177260746\ldots, \\ |\mathbb{E}_{H_0} T_4| &\le 0.000010367004497635\ldots. \end{aligned}$$

Thus the absolute alternative moment is 322.0201 times the conservative detector-only maximum. The signed separation below the null interval is 0.003328017172763111. An independent exact enumeration of all sixteen possible four-bit output patterns reproduces the H1 expectation and zero expectation for the ideal independent model.

Theorem 5: a finite-sample test with declared size and power

Define `A=0.003338384177260746...`, `B=0.000010367004497635...`, `Delta=A-B>0`, and `R=max(m,1-m)^4=0.1202010010523694...`. For N independent and identically initialized four-gate records, compute the average of T4 and reject H0 when

$$\overline{T}_4 < -\frac{A+B}{2}.$$

Each record satisfies `|T4|<=R`; Hoeffding's inequality gives under **every H0 model**, and under the specified H1,

$$P(\text{respective decision error}) \le \exp\left(-\frac{N\Delta^2}{8R^2}\right).$$

For `N=31,264`, this upper bound is below 0.05 for both errors. Thus 31,264 independent four-gate records, totaling 125,056 gate uses, are a conservative sufficient design for at most 5% false rejection and at least 95% power against the named H1 alternative. This is **not** a minimum count. It does not include costs of calibration, randomized preparation, or environmental resets.

Failure conditions: H0 would have to be redefined if detector errors depend on the ideal flags, the nominal per-use flip rate is unknown or nonstationary, the selectors are correlated with detector errors, or there are dependencies across independent records. The source and detector may also both carry memory; this two-class test cannot establish uniqueness of memory location.

Freshness versus reporting error

Kirtus Justus

1. Observation contract

Let `Z=(Z_1,Z_2,Z_3,Z_4)` denote ideal binary match outcomes from four sequential uses of a known gate-or-reset selector, and `Y` the reported record. The independent reference selector has gate probability `t=37/100`; a complete reset produces an ideal match with probability `r=1/16`. Under the memoryless selector, ideal outputs are independent Bernoulli variables with

$$s = t + (1-t)r = \frac{131}{320}.$$

For the comparison process, each reported ideal bit independently flips with probability `e=1/100`. Define

$$m = s + (1-2s)e = \frac{6579}{16000}, \qquad A = \frac{t^4}{4}\big[(1-r)(1-2e)\big]^4.$$

The target four-bit law is

$$Q(y) = \prod_{i=1}^{4} m^{y_i}(1-m)^{1-y_i} - A\prod_{i=1}^{4}(2y_i-1), \qquad y \in \{0,1\}^4.$$

All 16 entries of `Q` are strictly positive and sum to one. Its proper marginals are independent Bernoulli(`m`). These equations are mathematical and dimensionless. A genuine experiment must separately justify ideal-state preparation, gate/controller independence, reporting calibration, trial initialization, and the meaning of observation time.

2. Total-variation freshness theorem

Theorem 1 (error-freshness tradeoff). For any joint law of reported prefix `W=(Y_1,Y_2,Y_3)`, final reported bit `Y_4`, and final ideal bit `Z_4`, fix a reference `s in [0,1]`. Put

$$\begin{aligned} L_s(Q) &= \sum_{w\in\{0,1\}^3} |Q(w,1) - sQ_W(w)|, \\ \epsilon_4 &= \Pr(Y_4 \neq Z_4), \\ \eta &= \mathrm{TV}\big(\mathcal{L}(W,Z_4), Q_W \otimes \mathrm{Bern}(s)\big). \end{aligned}$$

Then

$$\boxed{L_s(Q) \le \epsilon_4 + \eta.}$$

Proof. Since `Z_4` and `Y_4` are coupled on the same probability space, replacing one with the other changes the joint distribution with the prefix by at most `Pr(Y_4 != Z_4)` in total variation. Applying the triangle inequality with the reference product law gives the claim. Because the last variable is binary, `L_s(Q)` is precisely the total variation between `Q_{W,Y_4}` and `Q_W x Bern(s)`. The definition of `eta` also equals `sum_w |Pr(W=w,Z_4=1)-sQ_W(w)|`. No detector-Markov assumption is required. QED.

Interpretation. Exact conditional freshness `Pr(Z_4=1 | W=w)=s` gives `eta=0` and forces reporting error `epsilon_4 >= L_s(Q)`. If an independent experiment establishes `epsilon_4 <= e_max`, then any model reproducing the reported target must satisfy `eta >= (L_s(Q)-e_max)_+`. Dependence can arise because early reports were buffered until after `Z_4`, or because the supposedly independent ideal final outcome is actually statistically predictable. The inequality by itself cannot distinguish those mechanisms.

Theorem 2 (closed form). Suppose a length-`k` target has law `Q(y)=prod_i Bern_m(y_i)-A prod_i(2y_i-1)` with `A>0`, and put `p_m(w)=prod_{i=1}^{k-1}Bern_m(w_i)`. If `A>|m-s| max_w p_m(w)`, then

$$\boxed{L_s(Q) = 2^{k-1}A - (m-s)(2m-1)^{k-1}.}$$

Proof. Each difference `Q(w,1)-sQ_W(w)` equals `(m-s)p_m(w)-A prod_i(2w_i-1)`. Under the stipulated inequality its sign is the negative of the parity product. Summing absolute values yields the formula, since the mean parity product under independent Bernoulli(`m`) variables is `(2m-1)^{k-1}`. QED.

For the supplied four-use law, exact rational arithmetic yields

$$\boxed{L_s(Q) = \frac{875470225373317}{32768000000000000} = 0.026717230998941559\ldots.}$$

Hence if the final bit is misreported at most 1% of the time, every reproducing detector/controller model must have

$$\boxed{\eta \geq \frac{547790225373317}{32768000000000000} = 0.016717230998941558\ldots.}$$

A physically stronger, history-by-history calibration

For the reported prefix `W=111`, its target probability is approximately `0.06952159242` and the final click probability conditional on that prefix is approximately `0.36316811406`. Its distance from the fresh probability `s` is approximately `0.04620688594`. If a laboratory can certify an error cap of 1% **conditional on that particular prefix**, it must also permit at least `0.03620688594` conditional freshness deviation on this prefix to reproduce the target. This stronger statement uses a stronger calibration contract than an average 1% last-bit error rate.

3. Exact sharpness certificate with a buffered detector

The accompanying exact rational transport certificate specifies 36 strictly positive rational joint masses `T(z,y)` on the 16 ideal and 16 reported words. The accompanying verifier checks all of the following with exact rational arithmetic:

1. `sum_y T(z,y)=prod_i Bern_s(z_i)` for every ideal word `z`;
2. `sum_z T(z,y)=Q(y)` for every reported word `y`;
3. for each of the four bits `i` and both true values `b`, `Pr(Y_i != Z_i, Z_i=b)=e Pr(Z_i=b)`;
4. `epsilon_4=e=1/100`;
5. the measured history-future total variation is **exactly** `eta=L_s(Q)-e`.

Thus Theorem 1 is sharp even under all eight calibrated per-bit conditional flip constraints and an independent Bernoulli ideal input process. The joint law defines a valid classical buffered reporting kernel `T(y|z)`, because its ideal-word marginals are positive. It is a mathematical countermodel, not an assertion that any physical device implements the kernel.

The accompanying executable check also evaluates two exponential tail expressions at 75-digit working precision. For the specified 8,982-record test it returns approximately 0.04999520376305833929 and 0.04998587512835677830 8. These are numerical evaluations, not outward-rounded interval certificates. They are separate from, and are not needed for, the exact rational transport identities and saturation proof above.

Information-theoretic corollary. With the certified ideal marginal `Z_4 ~ Bern(s)`, `eta` measures dependence between the reported prefix and the ideal final bit. Pinsker's inequality gives

$$I(Z_4; W) \geq 2\eta^2 \geq 0.0005589316245439451 \text{ nats} \approx 0.0008063678829256 \text{ bits}.$$

This is a *minimum predictive information requirement* for any 1%-accurate reproducing apparatus under the model. It is not evidence of backwards-in-time physical causation: a buffered detector can record earlier labels only after seeing the final outcome.

4. Direct signed-predictability witness

Define `f(w)=sgn[Q(w,1)-sQ_W(w)]`. For the stipulated target, `f(w)=-prod_i(2w_i-1)` and

$$\mathbb{E}_Q\big[f(W)(Y_4 - s)\big] = L_s(Q).$$

For any joint implementation, replacing `Y_4` by `Z_4` changes this expectation by at most `epsilon_4`. Therefore

$$\boxed{\mathbb{E}[f(W)(Z_4 - s)] \geq L_s(Q) - \epsilon_4.}$$

The rational transport certificate saturates the bound at `epsilon_4=0.01`. This single signed statistic is an informative calibration target if a trusted independent path can observe or challenge the final ideal bit. It does not estimate the entire correlation law and is not justified if the reference path changes the detector's behavior.

History-dependent readout tree

Kirtus Justus

Model

Let the null have fresh independent ideal bits $Z_1, \dots, Z_4$, each with success probability $s = 131/320$. Let Y_i be the reported bit, and let $w = Y_1 \cdots Y_{i-1}$ denote a prior reported prefix. At *every* position and every positive-probability reported prefix, assume

$$\Pr(Y_i = 1 \mid Z_i = 0, w) \le \varepsilon, \qquad \Pr(Y_i = 0 \mid Z_i = 1, w) \le \varepsilon,$$

where the probabilities are those of the same null process and the ideal bit is fresh even after conditioning on w: $\Pr(Z_i = 1 \mid w) = s$. Independence of the ideal bit from reported history is a substantive condition. Nothing here bounds a quantum instrument's postmeasurement state or guarantees that the same caps hold in different device time blocks or experimental paths.

At one node, the next reported-one rate $q(w) = \Pr(Y_i = 1 \mid w)$ obeys the sharp interval

$$a_\varepsilon = s(1-\varepsilon) \ \le q(w) \le \ b_\varepsilon = s + (1-s)\varepsilon. \tag{1}$$

The interval is also sufficient, because any $q \ge s$ is realized by false-positive probability $(q - s)/(1 - s)$ and zero false negatives, while any $q \le s$ is realized by false-negative probability $(s-q)/s$ and zero false positives. Choosing such a channel separately after each observed prefix realizes the entire corresponding tree law by induction. These are classical, causal reporting rules; they need not be physically available as a particular quantum readout.

The stipulated alternative is fixed throughout the analysis. Write

$$e_0 = \frac{1}{100}, \quad m = s + (1-2s)e_0 = \frac{6579}{16000}, \quad A = \frac{1}{4}\left[\frac{37}{100}\frac{15}{16}(1-2e_0)\right]^4.$$

For a length-four binary word y containing j ones, its probability is

$$Q(y) = m^j(1-m)^{4-j} - A(-1)^j. \tag{2}$$

All proper marginals of this alternative equal the corresponding product Bernoulli(m) marginals.

Proposition: exact whole-word compatibility threshold

Claim (proved for the specified finite model). The law Q in (2) can be reproduced exactly by a four-step causal reporting process with the independent ideal rate s and a common upper cap ε on both error directions at *every* reported history and position **if and only if**

$$\boxed{\varepsilon \ge \varepsilon_{\text{full}} = \frac{10396360128757}{92107698697800} = 0.1128717824431468\ldots.} \tag{3}$$

Proof. Marginalizing at least one future bit in (2) cancels the alternating term. At depths 0, 1, and 2, each conditional next-one rate is therefore m. At depth 3, for a prefix w with j ones,

$$Q(w) = m^j(1-m)^{3-j}, \quad q(w) = m + \frac{(-1)^j A}{m^j(1-m)^{3-j}}. \tag{4}$$

Every target prefix has positive mass. By (1), a necessary and sufficient global cap is the largest of the one-node requirements

$$\max\left\{0, \frac{q(w)-s}{1-s}, \frac{s-q(w)}{s}\right\}$$

across all 15 prefixes. Among the last-step prefixes, the four distinct one-rates and their required bounds are:

Number of prefix ones	Target next-one rate	Least directional cap
0	0.427540794200	0.030756900233
1	0.387769900872	0.052775814663
2	0.444721046342	0.059845157827
3	**0.363168114062**	**0.112871782443**

Earlier positions require only $(m-s)/(1-s) = 29/9450 \approx 0.003068783$. The largest of all requirements is therefore at $w = 111$. Its next-one probability is exactly

$$q(111) = \frac{81711338569043}{224995905216000} = 0.3631681140623368\ldots,$$

so $(s - q(111))/s$ reduces to (3). Necessity follows because a smaller cap violates (1) at this positive-mass prefix. At the threshold all nodes satisfy (1), and the explicit one-sided flipping construction above realizes every conditional rate in (4), proving sufficiency. □

The proposition is *not* a claim that the supplied alternative itself uses this large error rate. It instead quantifies how much freedom a distinct classical readout model would require to impersonate the same observed data.

What parity misses

Let O_i be the probability of odd parity after i reports under an allowed null, with $O_0 = 0$. At each step, choose reported-one rate b_ε following an even prefix and a_ε following an odd prefix. Since $s < 1/2$, this attains the sharp recurrence

$$O_i = b_\varepsilon + (1 - a_\varepsilon - b_\varepsilon)O_{i-1},$$

and hence

$$O_4^{\max}(\varepsilon) = b_\varepsilon \frac{1-[(1-2s)(1-\varepsilon)]^4}{1-(1-2s)(1-\varepsilon)}. \tag{5}$$

The alternative has $Q(\text{odd}) = 0.526209351956162\ldots$. Equation (5) first reaches that value at

$$\boxed{\varepsilon_{\text{parity}} \approx 0.044121657500400.} \tag{6}$$

At and above this smaller allowance, an allowed null can match the **single odd-parity probability**. Therefore parity-only population observations cannot distinguish the alternative from the null class. Nevertheless, throughout

$$0.044121657500400 \lesssim \varepsilon < 0.112871782443147,$$

the full law remains incompatible with the null because the 111 prefix still violates its conditional lower bound. This is an exact distinction between compressing the four-bit output to one parity bit and retaining the whole word. It does not prescribe a finite-shot optimal full-word test.

A predeclared bounded score detecting this surviving direction is

$$X_\varepsilon(W, Y) = \mathbf{1}\{W = 111\}(a_\varepsilon - Y).$$

Every admissible null satisfies $\mathbb{E}X_\varepsilon \le 0$. Under the target, $\mathbb{E}_Q X_\varepsilon = Q(111)[a_\varepsilon - q(111)]$, positive whenever $\varepsilon < \varepsilon_{\text{full}}$. For example this population excess is 0.00178935619623 at a **5%** directional cap. A small excess and a relatively rare prefix make the corresponding finite-sample efficiency a separate question.

Calibration uncertainty changes the statistical accounting

The four-step tree contains $2(1 + 2 + 4 + 8) = 30$ directional history cells. The familiar 6,343 figure assumes the cap is already valid, uses a fixed 5% hypothesis-test size, and omits the cost of checking reference preparations and readout under those conditions.

For a deliberately optimistic illustration, suppose each cell has n genuinely independent known-ideal reference trials, **zero** observed directional mistakes, stable behavior, and faithful transport to the device's gate path. A one-sided exact binomial upper bound at per-cell failure allocation $\gamma/30$ is

$$u_n = 1 - (\gamma/30)^{1/n}.$$

To place all 30 upper limits at or below 1% using the union bound with total calibration allowance $\gamma = 0.01$, the zero-error case requires at least

$$n = \left\lceil \frac{\log(0.01/30)}{\log(0.99)} \right\rceil = 797$$

observations **in each cell**, or 23,910 classified cell observations across the 30 cells. This is not 23,910 necessarily distinct four-use records; records can contribute observations at several positions, and rare histories need sufficient representation. Even this optimistic criterion is unlikely to succeed when actual error rates sit near the 1% cap, and it does not prove stationarity or instrument transport.

If a total false-rejection budget of 5% is split into 1% for possible calibration-coverage failure and 4% for a held-out fixed-horizon parity test, the latter's first 95%-power crossing moves from 6,343 to **6,758** independent four-use records, assuming the population alternative and all conditional test premises. At 6,757 records the recalculated test power is 0.949974789382334; at 6,758 it is 0.950018042462656. This is a mathematical design illustration, not a physically justified or globally optimal photon or preparation budget. Allocation of coverage and multiple testing must be registered before collecting the held-out data.

Most importantly, even perfect past calibration cannot establish arbitrary future-history bounds without a justified stability or exchangeability/conditional model. For a true error rate exactly at the 1% boundary, a valid upper confidence limit does not typically fall below 1%; a practical certification task needs a real performance margin and directly relevant calibration data.

Two thresholds for the same four reports

The target Q is the supplied sixteen-word law. The target remains fixed while the null error allowance ε varies. With known ideal Bernoulli rate s=131/320, fresh at every prefix w, both directional error caps permit exactly $s(1-\varepsilon)\le q(w)\le s+(1-s)\varepsilon$, where q(w) is the next reported-one probability. Each allowed value is achievable by adjusting just the needed direction. Applying that construction at all fifteen nodes realizes the full law.

Observation retained	First compatible uniform cap
Odd parity alone, s fixed	4.41216575004%
All sixteen words, s fixed	11.28717824431%
All words, one common ideal rate freely fitted	8.15529322792%

For parity, let O be the odd probability. Starting at zero update O to $s+(1-s)\varepsilon+(1-2s)(1-\varepsilon)O$ four times to obtain the maximum. It first reaches the target odd probability 0.526209351956… at the listed cap. The admissible classes are nested. At threshold the maximizing rule matches the target parity, and the same rule remains admissible at larger caps.

For the full word, take the largest of the two directional requirements across positive-probability prefixes. The binding constraint is a false negative after 111: q(111)=0.363168114062… and the minimum cap is $(s-q(111))/s=10396360128757/92107698697800$. For this particular Q, an independently verified upper bound below that threshold on just this direction and prefix would already exclude exact agreement at the population level, provided s applies there. Finite records require uncertainty bounds and the prefix must occur.

If s is allowed to vary but remains common across every node, the admissible interval has width ε. The target therefore requires $\varepsilon\ge q_{max}-q_{min}$. Equality is attained with $s=q_{min}/(1-\varepsilon)=0.39541539935\ldots$. The minimum and maximum target rates are 0.36316811406… and 0.44472104634…. This sensitivity check changes the identified quantity’s calibration assumption, using the same target table.

At a 1% cap, the fixed-prefix minimum total-variation distance is 0.021712152209. Allowing the prefix to vary subject to all four per-position directional caps gives 0.020642673036. The earlier exact constructive and dual certificates and the submitted replay agree. These are distances between distributions. They are distinct from both the cap thresholds and the mismatch percentages discussed in Freshness versus Reporting Error.

What the zero-error counts buy

Assume fixed sample sizes, correctly labelled independent Bernoulli observations within each of 30 cells, fixed rates and valid transfer to the gate path. A cell is defined by position, prior reported history and ideal bit value. No independence between cells is needed for the two guarantees below.

Calibration objective at 99% coverage	Zero errors in each cell	Total cell observations
Thirty separately reported limits using equal Bonferroni allocations	797 per cell	23,910
Single upper limit on the largest of all thirty rates	459 per cell	13,770

For n zero-error trials, the one-sided exact binomial upper limit at failure probability γ is $1-\gamma^{(1/n)}$. Allocating γ=0.01/30 to each cell yields 797. For the common cap, compute a one-sided 99% limit U_i for each rate p_i and report $U_{max}=\max U_i$. For a fixed cell i attaining the true maximum, $U_{max} \geq U_i$, so it covers the maximum with probability at least 99%. This gives one valid common bound for all cells. It does not confer simultaneous coverage on thirty individually tighter limits.

Equivalently, if some cell has rate at least 1%, the chance all cells have zero errors is at most 0.99^n. This quantity is 0.01002119 at n=458 and 0.00992097 at n=459. Other cells can have zero true rate, so a smaller fixed equal sample size cannot certify the same pass-on-all-zeros rule at 99%. Cell observations are not counts of complete records.

Held-out power and the final decision

At 5% size the parity-only power is 0.949970847338 at 6,342 and 0.950004432130 at 6,343. At 4% it is 0.949974789382 at 6,757 and 0.950018042463 at 6,758. Exact integer binomial tails verify the adjacent crossings with randomized decision at the cutoff; an extra record can always be discarded.

The proposed 1%+4% allocation is sufficient. A different fixed rule issues a rejection only when calibration certifies the prespecified cap AND the gate test rejects. If the actual common cap fails, false joint rejection is bounded by γ; if it holds under the null, it is bounded by the gate-test size α. Thus the unconditional size is at most max(γ,α). Keeping γ=1% and α=5% permits the 6,343 held-out crossing. This argument needs stable transferable rates and a fixed decision rule. A 95%-powered held-out test supplies no 95% guarantee that calibration passes and the whole procedure rejects.

Exact binomial limits, NIST; see also Berger & Hsu on intersection–union tests.

What one reference can establish after 111

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> For the specified independent gate model, a calibrated click probability on the ideal match state already gives a lower bound on the gate-path click probability. A separately observed ideal bit on each gate trial is unnecessary for this particular test.

The conditional state supplies the bound

Fix the earlier reported history w=111. Let P be the pure joint output of the known two-qubit unitary acting on its half of a fresh maximally entangled probe. Complete depolarization gives I/16 on the joint output. Under the supplied fresh selector, with unitary probability t=0.37 independent of w and of the apparatus, the conditional state just before the readout is

> $\rho_0 = tP+(1-t)I/16 = sP+(1-s)(I-P)/15$, $s=131/320$.

Let E be the binary click effect applicable at that history. Write $h=\mathrm{Tr}(EP)$ and $q=\mathrm{Tr}(E\rho_0)$. Positivity gives $q \geq sh$, since $\mathrm{Tr}[E(I-P)] \geq 0$. The effect $E=hP$ attains equality for every $0 \leq h \leq 1$. This statement allows an arbitrary quantum effect. It requires that the reference and gate trials at this history admit a common effective readout on the probe. Correlated apparatus side information about the current selector or route can violate that premise.

For the inherited target $q_\star = 81711338569043/224995905216000 \approx 0.36316811406$, a matched h strictly above $q_\star/s = 0.88712821756$ excludes exact agreement under this null. At the previously specified 99% reference response, the lower bound is 0.40528125, which exceeds the target by 0.04211314. These numbers refer to the conditional population; no gate or reference observations were supplied.

The operator condition is explicit. For any state ρ and any reference P, $q \geq \lambda h$ for every effect E if and only if $\rho - \lambda P$ is positive semidefinite. Sufficiency follows from the positive trace pairing. Necessity follows by taking E to project onto any negative eigenvector. Verifying only the scalar $\mathrm{Tr}(P\rho)$ does not certify that operator inequality.

A match probability leaves coherence unknown

Suppose the state is known only to satisfy $\mathrm{Tr}(P\rho)=s$, for the same rank-one reference P. Define $h=\mathrm{Tr}(EP)$ and $q=\mathrm{Tr}(E\rho)$. With the squared convention for fidelity, $F(P,\rho)=s$. Applying the measurement to both states can only increase fidelity, hence $\sqrt{s}\le\sqrt{(hq)}+\sqrt{((1-h)(1-q))}$. Solving the inequality gives the sharp lower envelope

> $q \ge [\max\{0, \sqrt{(sh)} - \sqrt{((1-s)(1-h))}\}]^2$.

For $s+h>1$, take the pure state with amplitudes $\sqrt{s}$ and $\sqrt{(1-s)}$ in a plane spanned by the reference and one orthogonal vector. Project onto the direction with amplitudes $\sqrt{h}$ and $-\sqrt{(1-h)}$. Squaring the overlap attains the bound. For $s+h\le1$, a scaled projector orthogonal to that state gives $q=0$ and reference response h. At $s=1$ the state is P and $q=h$ directly.

Using the same inherited s and $h=0.99$ now gives a minimum of 0.31333675. The target 0.36316811 is attainable: with the displayed projector, varying the state's off-diagonal entry between zero and its maximum magnitude continuously interpolates from 0.4111875 to 0.31333675. Thus an observed 99% reference response and a match population s, without the state structure on page 1, would allow the target. For $q\star<s$, strict exclusion on this information alone requires

> $h > [\sqrt{(s\, q\star)}+\sqrt{((1-s)(1-q\star))}]^2 = 0.99774815803\ldots$.

The distinction is a sensitivity check on the original state assumption. The depolarizing mixture on page 1 already has the stronger operator property. The unrestricted coherent state describes the uncertainty left if experimental characterization supports only the match population. The click-probability inequality is a consequence of standard quantum fidelity monotonicity.

Wilde, Quantum Information Theory, Ch. 9, fidelity under measurement.

The full interval for the reflected click rate

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> With match population s=0.409375, reference response h=0.99 and ordinary click probability q=0.3631681140623368…, the exact range after the joint phase reversal is [0.449704036935…, 0.551029468634…]. All values in this interval are attainable within the stated quantum model.

Fix the reported history 111, a rank-one match projector P, a normalized conditional input ρ, and a common binary effect $0 \le E \le I$. Write $s=\mathrm{Tr}(P\rho)$, $h=\mathrm{Tr}(EP)$, $q=\mathrm{Tr}(E\rho)$ and $r=\mathrm{Tr}(EU\rho U)$, where $U=2P-I$. The earlier lower bound uses the click amplitudes: since $I+U=2P$, the Hilbert–Schmidt triangle inequality gives $\sqrt{q}+\sqrt{r} \ge 2\sqrt{(sh)}$. Applying the same reasoning to no-click outcomes gives a second restriction.

Necessary and sufficient conditions

For $0<s<1$ and $0 \le q,r,h \le 1$, put $b=[(q+r)/2-sh]/(1-s)$ and $c=(q-r)/[4\sqrt{(s(1-s))}]$. The pair of click rates is feasible exactly when

> $0 \le b \le 1$, $c^2 \le hb$, $c^2 \le (1-h)(1-b)$.

To see necessity, purify ρ and split the purification into its normalized P and $I-P$ components. Their amplitudes are $\sqrt{s}$ and $\sqrt{(1-s)}$. Compress E tensored with identity on the purifying space to the two component vectors. This gives a two-dimensional effect with diagonals h and b. The real part of its off-diagonal is c because U changes the relative sign of the two components. Positivity of the effect and its complement gives the two determinant inequalities.

Conversely take the state with real amplitudes $\sqrt{s}$ and $\sqrt{(1-s)}$ and the real effect with diagonals h,b and off-diagonal c. The displayed inequalities make this a valid effect and direct multiplication reproduces q and r. The two-dimensional construction embeds into the joint probe space. If $s=0$ or 1, the reflection leaves the state unchanged, so $r=q$; at $s=1$ both equal h.

Why the upper endpoint matters

At fixed q the click determinant bounds r from below by $(2\sqrt{(sh)}-\sqrt{q})^2$ and from above by $(2\sqrt{(sh)}+\sqrt{q})^2$. Apply the same expressions to $1-r$, $1-q$ and $1-h$ for the no-click determinant. Intersect these ranges with [0,1] and the interval enforcing $0 \le b \le 1$. For the supplied inputs the active lower endpoint comes from the click determinant; the active upper endpoint comes from the no-click determinant.

At fixed q and h	Reflected rate	b	c
Lower endpoint	44.970404%	0.00195526	−0.04399668
Upper endpoint	55.102947%	0.08773340	−0.09551265

At the lower endpoint the effect has eigenvalues 0 and 0.99195526…. At the upper endpoint its complement is singular. The allowed section is an interval because the two-dimensional effect depends affinely on r and the effect set is convex. Averaging the two arms only would give the weaker lower endpoint 0.447394385938…. The sharp minimum increase over q is 0.086535922873…, or 8.65359229 percentage points.

The numerical interval is conditional on the same prepared state, a stable common effect, the specified q and h, and an independently characterized reflection about the joint match state. It does not predict an observed outcome for an uncharacterized local phase pulse. Statistical intervals or bounds on path-dependent errors require propagating those uncertainties through the feasible set. The original four-report table contains no reflected-arm observations.

Published hardware comparison

Aasen et al. found appreciable off-diagonal components in several noisy X- and Y-basis measurements on superconducting qubits. Most Z-basis reconstructions were consistent with statistical fluctuations in their off-diagonal components. Appendix E and Figure 12 identify the location in the preprint. These observations concern the authors’ reconstructed detector effects and do not characterize the sixteen-dimensional reflection considered here.

Aasen et al., Communications Physics 7, 301 (2024).

Preprint, Appendix E and Figure 12.

Four-phase response

Kirtus Justus

1. Inherited observation law and verified reflection bound

The source four-use comparison has ideal single-use rate $s = 131/320$, and the final reported-one probability after the *previously reported* history 111 is

$$q_\star = \frac{81711338569043}{224995905216000} = 0.36316811406233676725\ldots .$$

The conditional history has probability

$$\Pr(W = 111) = \frac{284760442539}{4096000000000} = 0.069521592416\ldots .$$

These are probabilities in a supplied mathematical population law, not observations.

Fix a rank-one projector P on a complex Hilbert space $\mathcal{H}$ of dimension at least two, a state $\rho \geq 0$ of trace one, and an effect $0 \leq E \leq I$. Set

$$s = \mathrm{Tr}(P\rho), \quad h = \mathrm{Tr}(EP), \quad U_\pi = 2P - I, \quad q_0 = \mathrm{Tr}(E\rho), \quad q_\pi = \mathrm{Tr}(EU_\pi \rho U_\pi^\dagger).$$

The Hilbert–Schmidt triangle inequality gives

$$\sqrt{q_0} + \sqrt{q_\pi} \geq 2\sqrt{sh}.$$

Indeed, writing $A = \sqrt{E}\sqrt{\rho}$ and $B = \sqrt{E}U_\pi\sqrt{\rho}$, their squared Hilbert–Schmidt norms are q_0, q_π, and $A + B = 2\sqrt{E}P\sqrt{\rho}$ has squared norm $4sh$.

At $h = 99/100$, this gives

$$q_\pi \geq \left(2\sqrt{sh} - \sqrt{q_\star}\right)^2 = 0.44970403693542629849\ldots ,$$

which is at least $0.08653592287308953124\ldots$ greater than the supplied $q_\star$. This is a sharp *population* inequality under the common-state, common-effect and exact-operation premises.

An equality witness uses a two-dimensional subspace with

$$|\psi\rangle = \sqrt{s}\,|0\rangle + \sqrt{1-s}\,|1\rangle, \quad \rho = |\psi\rangle\langle\psi|, \quad E = \begin{pmatrix} h & -\sqrt{he} \\ -\sqrt{he} & e \end{pmatrix},$$

where

$$e = \frac{(\sqrt{sh} - \sqrt{q_\star})^2}{1-s} = 0.00195526010392640486\ldots .$$

The matrix is positive semidefinite with eigenvalues 0 and $h + e = 0.9919552601\ldots < 1$. The state is pure and positive. Direct substitution gives $q_0 = q_\star$ and q_π equal to the sharp lower bound. It is an algebraic witness, not a claim of a laboratory state.

The ideal unitary–depolarizing mixture $\rho_0 = 0.37P + 0.63I/16$, and more generally any ρ block diagonal with respect to $P \oplus (I - P)$, obey $\rho \geq sP$, so $q_0 \geq sh$ and $q_\pi = q_0$. At $h = 0.99$, the target $q_\star < sh = 0.40528125$ already contradicts this block-diagonal null. Allowing coherent off-diagonal blocks evades the single-arm bound, but not the reflected-arm inequality.

2. New exact four-phase feasibility criterion

Let $R = I - P$, and for phases $\phi \in \{0, \pi/2, \pi, 3\pi/2\}$ define

$$U_\phi = P + e^{i\phi} R, \qquad q_\phi = \mathrm{Tr}(E U_\phi \rho U_\phi^\dagger).$$

Here $0 < s < 1$, $0 \le h \le 1$, and **the same** ρ, E are used in all four arms. The operation is a relative phase between a specified rank-one *joint match sector* and its full orthogonal complement.

Proposition (sharp four-phase feasibility). Four proposed probabilities $q_0, q_{\pi/2}, q_\pi, q_{3\pi/2}$ arise from such a state, effect and phase family if and only if the following hold. Put

$$a = sh, \quad \bar{q} = \frac{q_0 + q_\pi}{2}, \quad b = \bar{q} - a, \quad c = \frac{q_0 - q_\pi}{4} + i\frac{q_{\pi/2} - q_{3\pi/2}}{4}.$$

Then

$$\boxed{q_0 + q_\pi = q_{\pi/2} + q_{3\pi/2},} \tag{F1}$$

$$\boxed{0 \le b \le 1 - s,} \tag{F2}$$

$$\boxed{|c|^2 \le \min\{sh\, b,\ s(1-h)[(1-s) - b]\}.} \tag{F3}$$

The theorem is unchanged if $\mathcal{H}$ has dimension greater than two. The conditions themselves imply that all four probabilities belong to $[0, 1]$, because of the constructive realization below.

Proof of necessity. Put $b = \mathrm{Tr}(ER\rho R) \ge 0$, and $c = \mathrm{Tr}(EP\rho R)$. Directly expanding $U_\phi \rho U_\phi^\dagger$ yields

$$q_\phi = sh + b + 2\,\mathrm{Re}(e^{-i\phi} c).$$

Hence (F1) and $b \le \mathrm{Tr}(R\rho) = 1 - s$. Hilbert–Schmidt Cauchy–Schwarz applied to the diagonal and cross terms gives $|c|^2 \le (sh)b$. Applying the same argument to $I - E$, whose off-diagonal trace term equals $-c$, gives $|c|^2 \le s(1-h)[(1-s) - b]$. Together these prove (F2)–(F3).

Proof of sufficiency and sharpness. Take a two-dimensional space spanned by $|0\rangle, |1\rangle$, set $P = |0\rangle\langle 0|$, and choose the pure state

$$\rho = |\psi\rangle\langle\psi|, \qquad |\psi\rangle = \sqrt{s}\,|0\rangle + \sqrt{1-s}\,|1\rangle.$$

Define

$$E = \begin{pmatrix} h & \bar{c}/\sqrt{s(1-s)} \\ c/\sqrt{s(1-s)} & b/(1-s) \end{pmatrix}.$$

The determinant conditions for $E \ge 0$ and $I - E \ge 0$ are exactly the two inequalities in (F3), while the diagonal bounds follow from (F2) and $0 \le h \le 1$. Therefore $0 \le E \le I$. Direct evaluation produces all four prescribed q_ϕ, using (F1). This constructs a realization for every feasible quartet and proves the criterion is both necessary and sufficient. □

This is an elementary finite-dimensional quantum feasibility result derived from standard effect positivity. **No claim of novelty relative to existing quantum-detector tomography and testing-region theory is made.**

Corollary (sharp phase-contrast ceiling). Maximizing (F3) over b gives

$$|q_0 - q_\pi| \le 4\sqrt{s(1-s)h(1-h)}.$$

The bound is attained by a pure two-dimensional state and a rank-one projective effect (at $b = (1-s)(1-h)$). At the inherited s and $h = 0.99$, its numerical value is $0.19570150261303565\ldots$. It bounds the contrast but does not predict its sign.

What phase cycling identifies. The ideal four response probabilities identify only $\bar{q}$ and the real and imaginary parts of c, for fixed s, h. They do not identify a unique state or measurement effect. In particular every admissible response admits a two-dimensional *pure-state* explanation, so these data alone cannot force a mixed-state interpretation. Conversely a flat response need not mean ρ is block diagonal: the effect may be phase-blind, or the accessible interference product c may vanish.

3. Application, feasibility margins, and negative controls

For the sharp two-arm witness in Section 1, c is negative and real and the four **illustrative witness** probabilities are

Ideal phase arm	An attaining witness (not a universal prediction)
0	0.36316811406233677...
$\pi/2$	0.40643607549888153...
π	0.44970403693542630...
$3\pi/2$	0.40643607549888153...

They satisfy (F1) and saturate the first disk constraint (F3). A different compatible coherent model may yield different quarter-phase rates or a higher reflected-arm rate. **The original four-report probability table contains no experimental responses at the added phase settings.**

If an idealized experiment measured $q_0 = q_\pi = q_\star$, while $s = 0.409375$ and $h = 0.99$ remained valid, then the two-arm inequality would be violated. With fixed, perfectly known population rates, a reflected-arm allowance $\delta + \tau \geq 0.08653592287\ldots$ would be needed to erase that particular discrepancy if all of the allowance were applied to the reflected arm. Finite-shot uncertainty in q_0, q_π, s, h and other systematic uncertainties can close the gap sooner. This number is **not** a tolerated hardware error budget by itself.

For a conservative finite-data test, construct simultaneous lower limits s_L, h_L, upper limits U_0, U_π, and independently justified bounds δ on the reflected state deviation in half-trace distance and τ on the difference between arm effects in operator norm. The strict condition

$$\sqrt{U_0} + \sqrt{\min(1, U_\pi + \delta + \tau)} < 2\sqrt{s_L h_L}$$

is sufficient to reject the joint calibrated model. It is not a mechanism-specific identification of state coherence, gate error, or detector drift. If more phases are measured, one can test whether any quartet inside their simultaneous confidence intervals and systematic allowances satisfies (F1)–(F3); numerical infeasibility needs an independently justified enclosure when a rigorous decision is required.

Two matched negative controls are especially important:

1. **Pure reference:** $U_\phi P U_\phi^\dagger = P$ at every phase. Phase-dependent reference click rates therefore reveal a failure of the common-state/effect/operation idealization, although phase-independent rates do not prove it.
2. **Verified block-diagonal mixture:** $U_\phi \rho_0 U_\phi^\dagger = \rho_0$. Its phase response should be flat. A nonflat response in this control detects some setting-dependent errors or preparation departures, without uniquely assigning their origin.

In a real experiment the phase arm should be chosen only after the preceding reported history has been committed, using a choice isolated from the state source and detector. Each arm needs an adequately populated held-out sample, a declared sampling design, and valid cross-record conditional guarantees. Postselection on history 111 does not automatically deliver a fresh ideal bit or erase apparatus memory. The phase operations here are joint operations on the target/reference probe: local one-qubit phase rotations are **not generally equivalent** to $P + e^{i\phi}(I - P)$.

Angular contrast and rounding

Kirtus Justus

1. Quantum model, notation and sufficient two-dimensional reduction

Let P=|p><p| be a rank-one projector and R=I−P on the specified joint-probe Hilbert space. Let ρ be a normalized state and 0≤E≤I a *common* binary effect, with

$$s = \mathrm{Tr}(P\rho) = \frac{131}{320}, \qquad h = \mathrm{Tr}(EP) = \frac{99}{100}, \qquad q = q(0) = \frac{81711338569043}{224995905216000}.$$

For the phase family U_φ=P+e^{iφ}R, write q(φ)=Tr(E U_φ ρ U_φ†). Put

$$a = sh, \quad d = s(1-h), \quad t = a - q > 0, \quad b = \mathrm{Tr}(ER\rho R), \quad c = \mathrm{Tr}(EP\rho R), \quad z = 2\,\mathrm{Im}\,c.$$

The ordinary readout fixes Re c=−(t+b)/2. Positivity of E and I−E gives

$$|c|^2 \le \min\{ab,\ d(1-s-b)\}, \qquad 0 \le b \le 1-s.$$

These conditions are **sufficient**, not just necessary, because every admissible pair (b,c) is realized by the pure state |ψ>=√s|p>+√(1−s)|u> and the two-dimensional effect

$$E_{b,c} = \begin{pmatrix} h & \overline{c}/\sqrt{s(1-s)} \\ c/\sqrt{s(1-s)} & b/(1-s) \end{pmatrix},$$

embedded in the joint-probe space. Both determinant conditions for E and I−E are precisely the inequalities above. Their joint feasible set is convex.

Equivalently, the allowed (b,z) are the intersection of two disks:

$$(b - C_1)^2 + z^2 \le R_1^2, \quad (b - C_2)^2 + z^2 \le R_2^2,$$
$$C_1 = a + q,\ R_1 = 2\sqrt{aq}, \quad C_2 = -(t + 2d),\ R_2 = 2\sqrt{d(1-q)}.$$

They meet at b_0=(1−s)(1−h) and $z = \pm z_0$, with

$$z_0 = \sqrt{4ab_0 - (t + b_0)^2}, \qquad \kappa = \frac{z_0}{t + b_0} = 1.775489004780283195 84\ldots.$$

2. Sharp two-sided contrast for a deterministic coherent offset

Let the intended relative phase be π; assume that its **only** error is a coherent offset ±δ in the very same family U_φ, with the **same** ρ, E and P in both arms. Then, upon choosing the adverse sign of the imaginary interference,

$$\Delta(\delta) := q(\pi \pm \delta) - q(0) = A(t + b) - Bz, \qquad A = 1 + \cos\delta, \quad B = \sin\delta$$

for 0≤δ≤π (the signs are interchangeable because z can take either sign). Set H=√(A²+B²)=2cos(δ/2).

Sharp minimum. Let

$$\delta_1 = 2\arccos[(C_1 - b_0)/R_1] = 12.7591826805379687\ldots^\circ .$$

For $0 \le \delta \le \delta_1$, the supporting point of the first disk is feasible for both, giving

$$\Delta_{\min}(\delta) = A(t + C_1) - R_1 H = 4a\cos^2(\delta/2) - 4\sqrt{aq}\cos(\delta/2).$$

For $\delta_1 \le \delta \le \delta_c$ the supporting point would leave the disk intersection; the minimum is at its shared upper point (b_0, z_0):

$$\Delta_{\min}(\delta) = A(t + b_0) - Bz_0.$$

Its zero occurs sharply at

$$\delta_c = 2\arctan(1/\kappa) = 58.77860795856966018\ldots^\circ .$$

Every compatible state/effect yields strictly positive contrast for $|\delta| < \delta_c$. At δ_c an admissible two-dimensional **projective** effect gives zero, so this is an open condition.

New sharp upper endpoint. On the same full interval $0 \le \delta \le \delta_c$, maximizing the contrast over the *second* disk gives the simultaneously attainable bound

$$\boxed{\Delta_{\max}(\delta) = A(t + C_2) + R_2 H = 4\sqrt{d(1-q)}\cos(\delta/2) - 4d\cos^2(\delta/2).}$$

Its supporting point is

$$(b, z) = \left(C_2 + R_2\frac{A}{H},\ -R_2\frac{B}{H}\right).$$

This point belongs to the first disk whenever $b \ge b_0$. For the supplied inputs it remains feasible through $\delta = 113.20968269065858...^\circ$, well beyond δ_c. The upper formula therefore holds throughout the guaranteed-positive interval. The constructed two-dimensional $E_{b,c}$ has $0 \le E \le I$ and attains the rate. The image of the convex disk intersection under the linear contrast is a full interval, so **every intermediate contrast is attainable** as well.

Offset	Sharp lower increase (percentage points)	Sharp upper increase (percentage points)
0°	8.653592287309	18.786135457214
10°	8.006123191762	18.720856010456
20°	6.398293220245	18.525231209691
30°	4.697644815950	18.199908655488
45°	2.168778595748	17.471285101295
55°	0.572299788190	16.827620196169
58°	0.116299082295	16.610292664719

These are percentages of theoretical binary *probabilities*, not percentages measured on devices. The decimal entries in this table are rounded descriptions, not lower/upper decimal certificates. The companion verifier emits **downward-rounded lower certificates** for Δ_min separately.

3. Sharp calibration criterion for an independently varying phase offset

Suppose one fixed pre-phase conditional ρ and one fixed E apply to all trials in this comparison, and the distribution of actual offsets ε is transferable to that same response independently of hidden state/effect changes. Define the *population* moments

$$C = \mathbb{E}\cos\varepsilon, \qquad S = \mathbb{E}\sin\varepsilon, \qquad C^2 + S^2 \le 1.$$

No values of these moments are assumed known from experiment. Averaging the fixed response gives exactly

$$\overline{\Delta} = (1 + C)(t + b) - Sz.$$

Put A=1+C, B=|S|, H=√(A²+B²) and b_* = C₁−R₁ A/H when H>0. The **sharp minimum** over all state/effect pairs compatible with the three supplied population inputs is

$$\boxed{M(C,S) = \begin{cases} A(t + C_1) - R_1 H, & b_* \le b_0, \\ A(t + b_0) - Bz_0, & b_* > b_0. \end{cases}}$$

If A=B=0 (the phase is reversed relative to the intended reflection with probability one), take M=0. The proof is the same supporting-line optimization over the two-disk intersection, and both branches are attainable.

The exact **strict sign criterion** is

$$\boxed{1 + C > \kappa|S|.}$$

Necessity as well as sufficiency follows because max z/(t+b)=κ at the disk intersection, and t+b>0 throughout the feasible set. In the symmetric case S=0 the sharp lower contrast simplifies to

$$\boxed{M(C,0) = \frac{1+C}{2}\, 4(a - \sqrt{aq}).}$$

Simultaneous, independently justified confidence bounds C≥C_L and |S|≤S_U therefore give the sufficient physical sign criterion 1+C_L>κ S_U *only when* the common-state/effect and transfer hypotheses have been independently established. The rectangular bound can be conservative if it includes moment pairs that no phase distribution can realize.

Nontransfer control. Two hidden conditional subpopulations can have the same s,h,q but opposite imaginary interference coefficients. If positive and negative angle errors are systematically correlated with these subpopulations, their pooled error histogram can have S=0 while both arms have the adverse response. At the sign boundary their average contrast can be zero. Thus an unlabelled marginal angle histogram is **not** evidence for the independent-transfer condition, and may not be inserted into M(C,S) without justification.

Angle moments and variance

Kirtus Justus

2. Verified sharp purity interval with a fully specified detector

Let $A = Z \otimes I_2$, $B = X \otimes I_2$. Use normalized output-first Choi vectors

$$p = |I_4\rangle, \qquad u = |iA\rangle, \qquad v = |iB\rangle, \qquad |M\rangle = (M \otimes I)|\Phi_4\rangle.$$

These vectors are orthonormal. Define

$$\begin{aligned} k &= \frac{(sh - \sqrt{shq_0})^2}{sh(1-h)} = 0.115482549888153\ldots, \\ d &= s + k = 0.524857549888153\ldots, \\ r &= 1 - d = 0.475142450111847\ldots. \end{aligned}$$

$w = \sqrt{s}\,p - \sqrt{k}\,u$, $\eta = w/\sqrt{d}$, and the **fixed** detector

$$E = |e\rangle\langle e|, \qquad e = \sqrt{h}\,p + \sqrt{1-h}\,u.$$

Let $F = |v\rangle\langle v|$ be the added, orthogonal outcome, and let $D = I - P - |u\rangle\langle u| - F$, a rank-13 projector. The model's complete lower-endpoint phase curve, together with the known s, h, fixes the compression of ρ onto $\mathrm{span}\{p, u\}$ to $|w\rangle\langle w|$. This **requires the complete curve**, not merely q_0. Positivity then forces zero support on the orthogonal direction inside that span.

For $f = \mathrm{Tr}(F\rho)$, $0 \le f \le r$, the sharp purity lower envelope is

$$\boxed{J(f) = d^2 + f^2 + \frac{(r-f)^2}{13}}, \qquad \boxed{J(f) \le \mathrm{Tr}(\rho^2) \le 1}.$$

The minimum is attained by

$$\rho_\star(f) = |w\rangle\langle w| + fF + \frac{r-f}{13}D.$$

It is a mixture of normalized unitary Choi vectors, hence is CPTP and unital. A pure unitary Choi state also attains the same phase curve, f, and the specified product-input $Z = +1$ probability

$$\boxed{p_0(f) = d + \frac{6(r-f)}{13}}.$$

The source's construction uses a first-qubit Hermitian unitary and a second-qubit X, with anticommuting Pauli terms. Its unitarity, input and output Choi marginals, and probabilities can be checked by direct 16×16 multiplication. Mixing the pure state with $\rho_\star(f)$ continuously fills $[J(f), 1]$, while preserving the complete phase curve, f, and $p_0(f)$.

f	$J(f)$, minimum purity	$p_0(f)$, output first-qubit $Z = +1$
0	0.292841628282	0.744154065324

f	$J(f)$, minimum purity	$p_0(f)$, output first-qubit $Z=+1$
$r/14$	0.291601186810	0.728490028508
$r/2$	0.336257079801	0.634505807606
r	0.501235795573	0.524857549888

The two named endpoint Choi states have half-trace distance 0.766538917119 and optimal **state** discrimination success $(1+T)/2 = 0.883269458559$ at equal priors. That is discrimination of two specified normalized Choi states with a fixed input; unrestricted channel discrimination is a distinct optimization, as explained by Watrous.

3. Additional result: an exact purity-excess identity

Proposition (purity projection identity). Under the exact phase-curve and F-population assumptions of §2, every compatible density operator, with or without the Choi marginal constraint, obeys

$$\boxed{\mathrm{Tr}(\rho^2) = J(f) + \|\rho - \rho_\star(f)\|_2^2.} \tag{1}$$

Proof. Let $P_\eta = |\eta\rangle\langle\eta|$. The measured full phase response and positivity give

$$\mathrm{Tr}(P_\eta\rho) = d, \quad \mathrm{Tr}(F\rho) = f, \quad \mathrm{Tr}(D\rho) = r - f.$$

As $\rho_\star = dP_\eta + fF + [(r-f)/13]D$, the three fixed populations imply

$$\mathrm{Tr}(\rho\rho_\star) = d^2 + f^2 + \frac{(r-f)^2}{13} = \mathrm{Tr}(\rho_\star^2) = J(f).$$

Expand $\mathrm{Tr}[(\rho-\rho_\star)^2]$ and rearrange. No commutativity between ρ and $\rho_\star$ is assumed. Equality $\mathrm{Tr}\,\rho^2 = J(f)$ holds exactly at $\rho = \rho_\star$. □

This identity gives a **model-independent-within-the-specified-feasible-class** quantitative interpretation of additional measurements: deviations from the minimum-purity model are controlled by the Hilbert–Schmidt norm.

Corollary (an additional product-input probability). Let p be the first-qubit $Z=+1$ probability on product input $|00\rangle$, computed from the **same** trace-preserving channel. Then

$$\boxed{\mathrm{Tr}(\rho^2) \geq J(f) + \frac{13}{276}[p - p_0(f)]^2.} \tag{2}$$

Proof. In the output-first normalized Choi convention the probability observable is

$$M = 4(|0\rangle\langle 0| \otimes I_2)_{\mathrm{out}} \otimes |00\rangle\langle 00|_{\mathrm{ref}}.$$

Let $\theta = (\sqrt{k}\,p + \sqrt{s}\,u)/\sqrt{d}$. The exact phase curve forces $\rho\theta = 0$ and $\rho_\star\theta = 0$. Set $B_\theta = I - |\theta\rangle\langle\theta|$. The Hilbert–Schmidt projection of $B_\theta M B_\theta$ orthogonal to the three fixed-population operators P_η, F, D is

$$M_\perp = B_\theta M B_\theta - P_\eta - \frac{6}{13}D.$$

Direct finite-rank identities give $\mathrm{Tr}\,M^2 = 32$, $\langle\theta, M\theta\rangle = 1$, $\langle\theta, M^2\theta\rangle = 4$, $\langle\eta, M\eta\rangle = 1$, $\langle v, Mv\rangle = 0$, and $\mathrm{Tr}(DM) = 6$. Consequently

$$\|M_\perp\|_2^2 = (32 - 8 + 1) - 1 - \frac{36}{13} = \frac{276}{13}.$$

Since $\rho - \rho_\star$ is orthogonal to the three fixed-population operators and supported in B_θ, Cauchy–Schwarz yields

$$|p - p_0(f)| = |\operatorname{Tr}[M_\perp(\rho - \rho_\star)]| \le \sqrt{276/13}\,\|\rho - \rho_\star\|_2.$$

Insert (1). □

The coefficient in (2) is a **certified, generally non-sharp** lower bound. Trace-preservation and positivity can strengthen it further. When $p = p_0(f)$, the bound correctly reduces to $J(f)$; explicit pure and mixed channels show that equality of these rates cannot identify purity.

For finite-data intervals $f \in F_0$, $p \in P_0$, with the other input quantities known exactly, a valid population purity bound is obtained by minimizing the right side of (2) over $F_0 \cap [0, r]$ and $P_0 \cap [0, 1]$. The input quantities themselves must be jointly calibrated and their uncertainties propagated before applying this to experiment.

4. Verified angle-dependent contrast and a variance refinement

The phase-calibration source studies the **general joint-density-operator class**, without imposing a Choi marginal. Its conclusions are sharp in that class. A stronger CPTP or instrument-specific conclusion needs a separate proof.

For a coherent error angle ε about the intended reflection, put $x = \cos\varepsilon$, $a = sh$, $t = a - q_0$, $d_1 = s(1 - h)$, and define

$$b_0 = (1 - s)(1 - h), \quad u_0 = a + q_0, \quad R_0 = 2\sqrt{aq_0},$$
$$z_0 = \sqrt{4ab_0 - (t + b_0)^2} = 0.0852578917486220\ldots.$$

The source proves that if the state may depend on ε but the **same fixed effect** and exactly the same s, h, q_0 apply in every angle stratum, the sharp minimum contrast at a specified cosine is

$$m(x) = \begin{cases} (t + b_0)(1 + x) - z_0\sqrt{1 - x^2}, & -1 \le x \le x_1, \\ (t + u_0)(1 + x) - R_0\sqrt{2(1 + x)}, & x_1 \le x \le 1, \end{cases} \tag{3}$$

where $x_1 = 2[(u_0 - b_0)/R_0]^2 - 1 = 0.9753069372438004\ldots$. Its branches agree with matching first derivatives. In contrast, if a *single conditional response* applies across all realized angles, the strict-increase criterion is $1 + C > \kappa|S|$, with $\kappa = 1.775489004780283\ldots$, $C = \mathbb{E}\cos\varepsilon$ and $S = \mathbb{E}\sin\varepsilon$.

For states depending on angle, the worst-case average with an **entire specified angle law** is $\mathbb{E}m(\cos\varepsilon)$, whereas with only its first sine/cosine moments the worst case is $m(C)$, by convexity and an attaining two-point angular distribution. The sharp strict-rise condition in this broader class is

$$C > \frac{\kappa^2 - 1}{\kappa^2 + 1} = 0.5183463333912298\ldots.$$

For equal +60° and −60° offsets, the source's two population calculations reproduce:

shared response: $+\,0.064901942155\ldots$, angle-dependent states: $-\,0.001806421221\ldots$.

Further proposition (variance-strengthened guarantee). If the entire angle law is relevant to the same group-specifically calibrated instrument, put $X = \cos\varepsilon$, $C = \mathbb{E}X$ and $V = \operatorname{Var}(X)$. Then

$$\boxed{\mathbb{E}[\Delta(\varepsilon)] \geq \mathbb{E}[m(X)] \geq m(C) + \frac{z_0}{2}V.} \tag{4}$$

The constant z_0 is the **largest valid global strong-convexity constant** for the specific function m in (3).

Proof. In the first branch,

$$m''(x) = \frac{z_0}{(1-x^2)^{3/2}} \geq z_0,$$

and in the second branch,

$$m''(x) = \frac{R_0}{2\sqrt{2}(1+x)^{3/2}} \geq R_0/8 = 0.09591181732043\ldots > z_0.$$

The first derivatives match at x_1. Hence $m(x) - (z_0/2)x^2$ is convex on $[-1, 1]$. Jensen's inequality applied to that function proves the second inequality in (4); the first holds separately in every calibrated angle stratum. Because $m''(0) = z_0$, no larger uniform curvature constant holds on the full domain. □

Equation (4) uses a **measured or otherwise independently established** distribution; it does not invent such a distribution. Its variance correction can improve a lower contrast guarantee when full angle information is available. The condition $m(C) + (z_0/2)V > 0$ is sufficient, not generally necessary, for strict positivity. This refinement does not apply if only pooled s, h, q_0 are known or if the angle distribution transfers incorrectly to the gate path.

Trace-preserving channel constraints on phase contrast and purity

Research direction and responsibility · Kirtus Justus. Mathematical and computational assistance · ChatGPT (OpenAI). Port Coquitlam, British Columbia, Canada · 7 November 2026.

In the supplied four-use probability model, does the mean click rate necessarily rise after a balanced coherent error of ±60° about the commanded reflection? General density states can give either sign. A two-qubit channel has an additional constraint: its normalized Choi state has reference marginal $I_4/4$. Here the sign for the full channel class at ±60° remains open. The following checks give a narrower bracket.

Offset and model	Result (percentage points)
±60°, one shared response, general states	Sharp interval ≈ [+6.49019, +14.08960].
±60°, offset-specific general states, common detector	Sharp interval ≈ [−0.18064, +16.45926].
±60°, offset-specific channels, common detector	Minimum strictly exceeds the exact density-state lower endpoint. A feasible unitary pair gives +0.11820700… .
±61°, same channel class	The same unitary pair gives −0.02364942688… . A decrease is possible.

Fix P as a maximally entangled unitary match, h=0.99 as its detector click probability, s=131/320 as its population and q=81711338569043/224995905216000 as the ordinary click probability in each offset group. The readout phase is $U(\varphi)=P+\exp(i\varphi)(I-P)$. The ordinary and phase arms must use the same group weights. All tabulated contrasts are model probabilities. No matched outcomes after history 111 were supplied.

First offset at which a uniform rise can fail



The left endpoint is excluded for this channel class. The right endpoint is an upper bound from an explicit unitary pair. Its displayed position is rounded. The diagram brackets a mathematical threshold. No calibration measurements enter it.

A separate channel constraint strengthens the product-readout purity bound from 13/276 to 91/1230. The new coefficient is locally sharp for trace-preserving channels at each interior orthogonal-outcome population f (p. 4).

Why the density-state endpoint is excluded

All statements about the channel minimum here require one common effect and a rank-one, maximally entangled match. The input and output dimensions of the channel are both four.

Write $a=sh$, $t=a-q$, $b_j=\mathrm{Tr}[E(I-P)\rho_j(I-P)]$ and $z_j=2\,\mathrm{Im}\,\mathrm{Tr}[EP\rho_j(I-P)]$. With balanced signs the mean contrast is $(1+\cos\delta)A-\sin\delta\,B$, where $A=t+(b_+ +b_-)/2>0$ and $B=(z_+ -z_-)/2$. The attainable set is compact. Its largest ratio B/A, denoted κ_{ch}, is positive by the example on p. 3. The first zero of the worst-case contrast is $\delta_{ch}=2\arctan(1/\kappa_{ch})$ for $0<\delta<\pi$.

Equality would force incompatible unitary channels

The two positivity disks bound each $|z_j|/(t+b_j)$ by $\kappa=z_0/(t+b_0)$, with $b_0=(1-s)(1-h)$, $z_0=\sqrt{[4ab_0-(t+b_0)^2]}$. Equality requires the common corner. At 60° the unique minimizing support point is also that corner. At the corner, $|\mathrm{Tr}[EP\rho_j(I-P)]|^2=s(1-s)h(1-h)$. Equality in the vector Cauchy–Schwarz bounds forces both states pure, with orthogonal component along the same direction fixed by E. Indeed the positive Schur-complement remainder has zero trace. The two corner signs give opposite phases on this direction.

Writing normalized Choi kets as $|M\rangle=(M\otimes I)|\Phi_4\rangle$, take the match ket to be $|I\rangle$. Both operators $\sqrt{s}\,I+\sqrt{(1-s)}\exp(\pm i\gamma)V$ would have to be unitary, with $\mathrm{Tr}\,V=0$. Subtracting their unitarity equations forces V Hermitian. Adding them gives $V^2-\lambda V-I=0$, with $\lambda=(t+b_0)/[(1-s)\sqrt{(h(1-h))}]$. Trace zero with j positive eigenvalues and $4-j$ negative ones forces $\lambda^2=(4-2j)^2/[j(4-j)]$ in $\{0,4/3\}$. The prescribed λ^2 is 0.667689209796… . Thus both corner conditions cannot hold. Compactness excludes equality both for the ±60° density minimum and for $\kappa_{ch}=\kappa$.

> The exact general-state sign boundary $\delta_D=2\arctan(1/\kappa)$ is strictly below δ_{ch}. Page 3 gives $\delta_{ch}\le\delta_w$ with $60.83257915540° < \delta_w < 60.83257915542°$. Neither exclusion alone decides the sign at 60°.

What the variance correction adds

For $X=\cos\varepsilon$, let $m(x)$ be the angle-specific density-state lower support. Its corner branch has $m''(x)=z_0/(1-x^2)^{3/2}$. On the smooth branch $m''(x)\ge 2\sqrt{(aq)}/8>z_0$. The first derivatives match at the switch. Jensen applied to $m(x)-z_0x^2/2$ proves $E\Delta\ge m(EX)+z_0\,\mathrm{Var}(X)/2$. Since $m''(0)=z_0$, the uniform curvature constant is optimal. Balanced ±60° has $\mathrm{Var}(X)=0$. The variance result gives no improvement for that particular pair.

For a constant absolute offset, the density-state interval can also use pooled s and q if h is common and the two arms have matched group weights. Conjugate the interference coefficient in groups with negative offset to align the sine terms. Weighted Cauchy–Schwarz then gives both disk bounds on the pooled coefficients. The channel exclusion above uses s,h,q fixed in each group.

A certified unitary pair at 61°

These two diagonal unitaries are feasibility witnesses in the channel optimization. The construction gives an upper bound on the minimum; it makes no global optimality claim.

Let $z_4=(112+15i)/113$, $H=4\sqrt{s}-z_4$ and $m=(|H|^2-3)/(2|H|)$. Set $z_1=(H/|H|)(m+i\sqrt{(1-m^2)})$, $z_2=(H-z_1)/2$. The four diagonal entries of U are z_1,z_2,z_2,z_4. They all have modulus one: $|H-z_1|^2=4$ by construction. Their sum is $4\sqrt{s}$, so the unitary Choi state has match population s. Assign U to the negative offset and its complex conjugate to the positive offset.

Within the diagonal Choi subspace, p has coordinate vector $(1,1,1,1)/2$. Define orthonormal real vectors $v_1=(2,-1,-1,0)/\sqrt{6}$ and $v_2=(-1,-1,-1,3)/\sqrt{12}$. Put $u(y)=(v_1+y\,v_2)/\sqrt{(1+y^2)}$, $e=\sqrt{h}\,p+\sqrt{(1-h)}u(y)$, $E=|e\rangle\langle e|$. The effect is the same in both groups. Write $\alpha(y)=\langle u(y)|U\rangle$, using the normalized Choi vector $(z_1,z_2,z_2,z_4)/2$.

> The continuous scalar $|\sqrt{(sh)}+\sqrt{(1-h)}\alpha(y)|^2-q$ changes sign between $y=-0.014452126001853375$ and -0.014452126001853374. A root therefore sets the ordinary rate exactly in both groups. The verifier bounds the contrasts over this whole root enclosure.

Here $b=(1-h)|\alpha|^2$, $z=2\sqrt{[sh(1-h)]}\ \mathrm{Im}\ \alpha>0$. The gain in each group is $(1+\cos\delta)(t+b)-\sin\delta\, z$. Outward interval evaluation gives the strict brackets below, in percentage points.

Offset	Lower endpoint	Upper endpoint
60°	+0.11820700	+0.11820701
61°	−0.02364943	−0.02364942

The gain changes sign at $2\arctan[(t+b)/z]$. The interval checks at the two ends of the bracket on p. 2 have opposite signs. The diagonal-unitary form was found by exploring a restricted numerical family. The exact construction and interval checks establish feasibility independently of that search. The optimization over all channel states and effects remains unsolved.

A postselected operation need not have the input marginal $I_4/4$. The channel restriction applies to an apparatus only if that marginal, the same effective detector and the history-conditioned transfer assumptions can be established independently.

Kliesch and Roth, Theory of Quantum System Certification, PRX Quantum 2, 010201 (2021). Source ↗

White et al., Non-Markovian Quantum Process Tomography, PRX Quantum 3, 020344 (2022). Source ↗

A locally sharp product-readout purity bound

Separate fixed-detector problem. The complete specified phase curve is required. The ordinary click rate alone is insufficient.

Use $p=|I_4\rangle$, $u=|iZ\otimes I\rangle$ and $v=|iX\otimes I\rangle$. Put $k=(sh-\sqrt{[shq]})^2/[sh(1-h)]$, $d=s+k$, $r=1-d$, $\eta=(\sqrt{s}\,p-\sqrt{k}\,u)/\sqrt{d}$ and $\theta=(\sqrt{k}\,p+\sqrt{s}\,u)/\sqrt{d}$. The fixed detector projects onto $\sqrt{h}\,p+\sqrt{(1-h)}u$. The phase curve fixes the η population to d and forces θ into the kernel. Write $F=|v\rangle\langle v|$, $D=I-|p\rangle\langle p|-|u\rangle\langle u|-F$ and $f=\mathrm{Tr}(F\rho)$. Its remaining thirteen directions have total weight $r-f$.

The source gives $\rho_\star=d|\eta\rangle\langle\eta|+fF+(r-f)D/13$, $J(f)=\mathrm{Tr}(\rho_\star^2)$, and $p_0(f)=d+6(r-f)/13$ for the first-qubit Z-positive probability on input $|00\rangle$. The three fixed populations imply $\mathrm{Tr}(\rho\rho_\star)=J(f)$, hence $\mathrm{Tr}(\rho^2)=J(f)+\|\rho-\rho_\star\|_2^2$, where $\|\cdot\|_2$ is the Hilbert–Schmidt norm. All purities through one are possible on the supplied line $pZ=p_0(f)$.

> For a trace-preserving channel that transfers to the product input, $\mathrm{Tr}(\rho^2) \geq J(f)+(91/1230)[pZ-p_0(f)]^2$. At every interior $0<f<r$ the coefficient is attained for all sufficiently small deviations in either direction.

Proof. The probability observable is $M=4(|0\rangle\langle 0|\otimes I)_{\mathrm{out}} \otimes |00\rangle\langle 00|_{\mathrm{ref}}$. Let $B=I-|\theta\rangle\langle\theta|$. The population-only projection $M_1=BMB-|\eta\rangle\langle\eta|-6D/13$ has $\|M_1\|_2^2=276/13$. This gives the valid 13/276 coefficient in the supplied study. Its output partial trace is $L=6(|00\rangle\langle 00|-I/4)$: in BMB the marginal is $6|00\rangle\langle 00|+I/4$, and the subtractions remove $7I/4$. Thus $\mathrm{Tr}\,L^2=27$.

For traceless Hermitian H put $A(H)=B(I\otimes H)B$. It is orthogonal to the three population operators: η, v and θ are maximally entangled, and D is the remaining projection. Expanding B at θ gives $\mathrm{Tr}[A(H)A(G)]=(4-2/4)\mathrm{Tr}(HG)$ for traceless H,G. The trace-preserving direction is $M_2=M_1-(2/7)A(L)$. It has zero output partial trace and squared norm $\nu=276/13-(2/7)27=1230/91$. Thus $pZ-p_0=\mathrm{Tr}[M_2(\rho-\rho_\star)]$, and Cauchy–Schwarz gives the bound.

For sharpness set $\rho=\rho_\star+[(pZ-p_0)/\nu]M_2$. On the support B, $\rho_\star$ is strictly positive for interior f, so both signs of a sufficiently small perturbation preserve positivity. All constrained populations and the Choi marginal are unchanged. More explicitly, $|pZ-p_0|\leq\sqrt{\nu}\,\min\{d,f,(r-f)/13\}$ suffices. Equality is asserted in the trace-preserving class. No unital or mixed-unitary restriction is needed here.

The accompanying calculations check the channel constraints, the angular inequalities and the local purity coefficient. These are deterministic mathematical checks of model populations, not experimental observations.

K. Justus, Angle Moments and Variance, accompanying mathematical notes.

K. Justus, Purity After an Additional Readout, accompanying mathematical notes.

How similar must the two conditional states be?

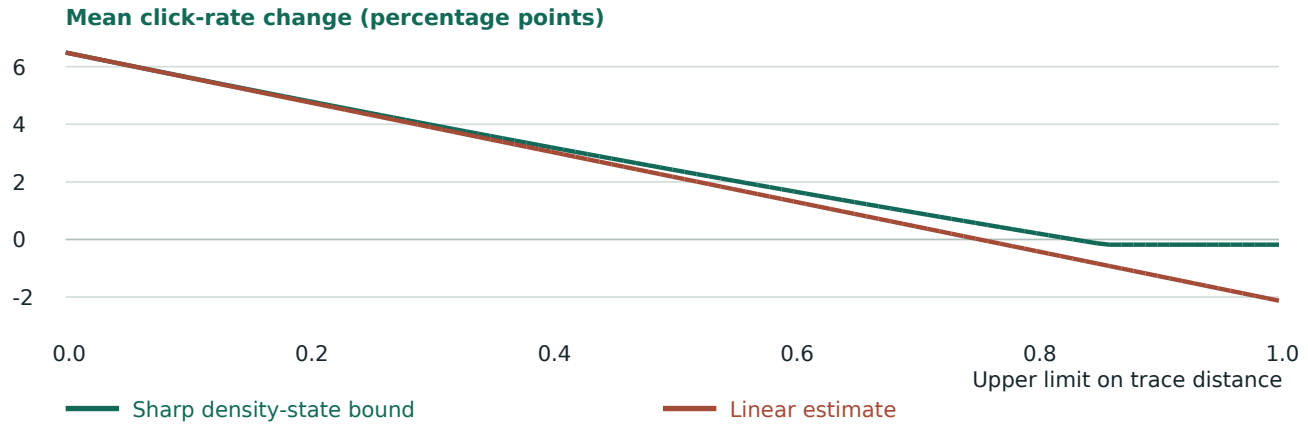
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Consider equal groups at offsets +60° and −60° from a commanded joint reflection after history 111. Let $D=\tfrac{1}{2}\|\rho_{+}-\rho_{-}\|_{1}$. Given an upper limit $D\le\tau$, what change in click probability must every compatible pair produce? The source gives a valid linear estimate. The exact minimum follows by intersecting the readout positivity constraints with the distance constraint.

> For general density states, the sharp strict-rise condition based on the cap alone is τ below 0.830970598587409… . At that value zero is attainable. For two normalized trace-preserving channel states with a maximally entangled match, the minimum is still strictly positive at that same cap. Its unconstrained sign at 60° remains open.

Distance cap τ	Source linear bound (pp)	Sharp minimum (pp)
0	+6.49019	+6.49019
0.50	+2.18177	+2.42395
0.80	−0.40328	+0.21771
0.84	−0.74795	−0.06312

Rounded to nearest. All rows concern model populations and the same detector across groups. The sharp column allows any detector effect consistent with h. It also applies in the specified sixteen-dimensional probe, and the examples attaining the minima use only three dimensions.



No trace distance, gate-path offset distribution, or history-matched phase result was measured in the supplied files. These curves describe what a separately justified upper limit could establish.

The exact distance-constrained minimum

Common rank-one $P=|p\rangle\langle p|$, binary effect E, $R=I-P$ and $U(\varphi)=P+\exp(i\varphi)R$. The direct and phase arms use matched weights. The match population s and ordinary click rate q are pooled.

Set $s=131/320$, $q=81711338569043/224995905216000$, $h=\langle p|E|p\rangle=0.99$, $a=sh$, $t=a-q$, $g=\sqrt{[h(1-h)]}$. In group j let $b_j=\mathrm{Tr}(E\,R\rho_j R)$ and $c_j=\mathrm{Tr}(E\,P\rho_j R)$. Write $b=(b_+ +b_-)/2$ and $c'=(c_+ +\mathrm{conj}(c_-))/2=-(t+b)/2+iz/2$. Direct expansion gives $\Delta=(1+\cos\delta)(t+b)-\sin\delta\, z$. Take $0\le\delta\le\pi$.

Positivity of E and its complement yields $|c_j|^2\le\min\{hs_jb_j,(1-h)s_j(1-s_j-b_j)\}$. Applying triangle and Cauchy–Schwarz after conjugating the negative group gives the same two inequalities for c' with pooled s,b. In particular $(b-u)^2+z^2\le R_1^2$, with $u=a+q$ and $R_1=2\sqrt{(aq)}$. The two circles meet at $b_0=(1-s)(1-h)$, $z=\pm z_0$ where $z_0=\sqrt{[4ab_0-(t+b_0)^2]}$. The first circle's allowed height increases up to b_0; the second circle's height decreases beyond it. The maximum $|z|$ is z_0.

Put $G=REP$ and $\sigma=(\rho_+-\rho_-)/2$. Then $z=\mathrm{Tr}[i(G\dagger-G)\sigma]$, so $|z|\le\|G\|D\le g\tau$. Here $\|G\|^2=\langle p|E^2|p\rangle-h^2\le h(1-h)$. Replacing b by its absolute minimum $u-R_1$ and z by $g\tau$ reproduces the submitted linear bound. The first disk gives a stronger simultaneous constraint: at a given $0\le z\le z_0$ the smallest feasible b is $u-\sqrt{(R_1^2-z^2)}$. This lies in the second disk because $b\le b_0$.

> Choose $z=\min\{g\tau,\, z_0,\, R_1\sin(\delta/2)\}$ and $b=u-\sqrt{(R_1^2-z^2)}$. Then $\Delta_{\min}=(1+\cos\delta)(t+b)-\sin\delta\, z$. This follows by differentiating the scalar objective on $0\le z\le\min(g\tau,z_0)$; at $\delta=\pi$ the contrast is identically zero.

Attainment. In basis p,v,w use $E=|e\rangle\langle e|$ with $e=\sqrt{h}\,p+\sqrt{(1-h)}v$. Both states have diagonals $s,b/(1-h),1-s-b/(1-h)$. Only the p,v entry and its conjugate are off diagonal. Set $\rho_{pv}=[-(t+b)\pm iz]/(2g)$. The disk bound gives positivity, and $b\le b_0$ ensures nonnegative spectator mass. Direct multiplication gives the required q in each group and the displayed contrast. The state difference has eigenvalues $\pm z/g$ and zero, so $D=z/g\le\tau$.

At $\delta=60°$ the first zero occurs on the first circle with $z=\sqrt{3}(t+b)$. Thus $(t+b)^2=ab$ at the smaller root $b_\star=(a-2t-\sqrt{[a(a-4t)]})/2$. It obeys $0<b_\star<b_0$. The exact critical cap is $\tau_\star=\sqrt{3}(t+b_\star)/g$. Both the cap and the angular sign threshold refer to exact expressions; printed decimals are approximations.

Why channels cannot attain that zero

Here P is a maximally entangled unitary match and both normalized states are Choi states of trace-preserving maps on a four-dimensional system. The common E is otherwise unrestricted.

At $\tau=\tau^\star$ a nonpositive contrast would attain the unique general-state optimum $(b^\star,g\tau^\star)$. This requires $D=\tau^\star$, $\|REP\|=g$ and equality in the trace-norm inequality for z. Put $v=REPp/g$. From $(E-E^2)p=0$ one gets $Ev=gp+(1-h)v$. The Hermitian observable $i(G^\dagger-G)$ has simple extreme eigenvectors $(p\pm iv)/\sqrt{2}$. Equality places the positive and negative parts of σ on these opposite eigenvectors. Tr σ=0 fixes their weights to be equal. Thus the group difference is purely imaginary off-diagonal in span{p,v}, with zero difference on all other blocks.

The compression of either group to p,v has the same diagonals s,k. Write its upper off-diagonal entry as $x+i(m\pm n)$; then $gx=-(t+b^\star)/2$ and $gn=z/2$ (up to sign convention). Positivity gives $sk\geq x^2+m^2+n^2$ by averaging the two determinants. Also $b^\star\geq(1-h)k$. Equality on disk 1 gives $x^2+n^2=sb^\star/(1-h)$. Hence $k=b^\star/(1-h)$, $m=0$, and both compressed blocks have rank one. Any coupling to the spectator space must have columns in that block's range. The coupling is the same in the two groups, yet the two ranges are distinct, so the coupling vanishes.

In the output-first Choi convention let $p=|I_4\rangle$ and $v=|V\rangle$. Tr V=0 and $\mathrm{Tr}(V^\dagger V)=4$. The vanishing output partial trace of the imaginary off-diagonal state difference forces $V=V^\dagger$. Each group contains a positive Choi summand with operator $K=\sqrt{s}\,I+\alpha V$, where $|\alpha|^2=k$ and $2\sqrt{s}\,\mathrm{Re}\,\alpha=-(t+b^\star)/g$. Trace preservation of the full state therefore requires $K^\dagger K\leq I$. Each eigenvalue λ of V obeys $k\lambda^2-(t+b^\star)\lambda/g-(1-s)\leq 0$.

> Both roots lie inside $[-0.684,1.537]$. Four real eigenvalues in that box with zero sum have mean square at most 0.8316605. The normalized Choi vector requires mean square one. This contradiction excludes zero at the critical cap.

For the numerical box, interval arithmetic gives a positive polynomial at each endpoint and a negative value at zero. Maximizing the convex sum of squares over the box with zero sum puts at least three coordinates at endpoints. The only admissible pattern, up to permutation, is $(-0.684,-0.684,1.537,-0.169)$. The common-effect and normalized-channel feasible set is compact, so the minimum is strictly positive at $\tau^\star$. This proof supplies neither its size nor its sign without the distance cap.

Calibration and the separate purity check

No numerical distance, fidelity, second-output expectation or offset response has been assigned to an apparatus.

A possible calibration target

The norm argument uses only the signed quadrature $z=\mathrm{Tr}[i(G^\dagger-G)(\rho_+-\rho_-)/2]$. If the effect and projector can be characterized, this Hermitian observable can in principle be measured on each matched group. A certified upper limit on z then cuts the first disk directly. A trace-distance upper limit controls every observable; in particular it gives $z\le g\tau$. Conversely, agreement on a single unrelated observable generally does not upper-bound the distance between two states.

For the two groups, the source's sufficient linear threshold $D<0.753198530485524\ldots$ checks out, as does the adverse example $D\approx0.85687405$. Using squared fidelity Fsq and $D\le\sqrt{(1-\mathrm{Fsq})}$, $\mathrm{Fsq}>1-\tau_\star^2\approx0.30948786$ is sufficient with the supplied inputs [1]. This last conversion is not asserted to be a sharp fidelity threshold. Common effect, equal weights, matched ordinary arms and the same joint phase action are still required. If s,h,q or the angles are uncertain, the lower bound has to be minimized over their simultaneously justified region.

What the second-qubit Y expectation identifies

This is a different experiment on the particular two-sign unitary family. Its orthogonal Choi components have populations $1-\ell$ and ℓ and off-diagonal $t'\sqrt{[\ell(1-\ell)]}$. The sign of that off-diagonal disappears from the first-qubit output on $|00\rangle$ when the second qubit is traced out. For the calibrated auxiliary unitary N, $a_0=\langle0|N|0\rangle>0$. The second-qubit marginal gives $y=2a_0t'\sqrt{[\ell(1-\ell)]}$, so the purity along this family is $(1-\ell)^2+\ell^2+y^2/(2a_0^2)$. At $\ell=0.01$ the endpoints give $\pm0.10229823305\ldots$ and purity one; their equal mixture has $y=0$ and purity 0.9802. This measurement is local to the second output. A two-qubit correlation measurement is unnecessary for this identity.

At exact f=r the previously derived first-qubit transverse identity $d^2+r^2+o^2/2$ is a separate boundary result. Its inference needs a normalized trace-preserving Choi state and transfer to the different product preparation. The linear and sharp distance guarantees on pp. 1–2 concern general states. Only the additional exclusion on p. 3 requires the Choi marginal. White et al. give an experimental framework for examining intervention-dependent dynamics [2]. None of the submitted results measures this gate path.

Mathematical checks
The accompanying calculation encloses the contrast threshold, compares Born probabilities and trace distances with the formula, tests pooled-state controls and verifies the spectral exclusion box. These deterministic checks concern model populations.

K. Justus, Angle Moments and Variance, accompanying mathematical notes.

[1] J. Watrous, The Theory of Quantum Information, ch. 3, Theorem 3.33. Source ↗

[2] G. A. L. White et al., Non-Markovian Quantum Process Tomography (2022). Source ↗

What the additional outcome identifies

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> The orthogonal outcome identifies purity along the submitted interpolation path. Removing that restriction gives an exact interval. Even the additional first-qubit Z reading from a product input leaves the whole interval attainable at the values predicted along the path.

The normalized Choi ket $|M\rangle$ is $(M\otimes I)|\Phi_4\rangle$ for the four-dimensional input and output. Use $A=Z\otimes I$, $B=X\otimes I$ and the orthonormal kets $p=|I\rangle$, $u=|iA\rangle$, $v=|iB\rangle$. Fix the input probabilities $s=131/320$, $h=0.99$ and $q=81711338569043/224995905216000$. Put $k=(sh-\sqrt{[shq]})^2/[sh(1-h)]=0.115482549888\ldots$, $d=s+k$, $r=1-d=0.475142450112\ldots$, and $w=\sqrt{s}\,p-\sqrt{k}\,u$. The fixed binary detector projects onto $e=\sqrt{h}\,p+\sqrt{(1-h)}u$. The second outcome is $F=|v\rangle\langle v|$.

The minimum-purity endpoint has weight d on the unitary Choi ket $w/\sqrt{d}$, and weight $r/14$ on each of the 14 remaining orthogonal Pauli kets. The pure endpoint $w+\sqrt{r}\,v$ corresponds to a unitary by anticommutation of A and B. Both channels and every convex mixture are mixed-unitary, trace-preserving and unital.

Quantity for the two named endpoints	Recomputed value
Choi half-trace distance	0.766538917119
Optimal one-copy equal-prior success	0.883269458559
Three-outcome total variation	0.441203703675
Product-input Z-positive probabilities	0.728490028508 and 0.524857549888

In the basis $w/\sqrt{d},v$, the Choi difference has diagonals $0,13r/14$ and off-diagonal $\sqrt{(rd)}$. The other 13 spectator eigenvalues are $-r/14$, and there is one zero direction. The resulting trace norm gives the quoted Helstrom success for these two Choi states. Optimization over all channel inputs is a separate discrimination problem.

The sharp interval after observing F

The premises here are the particular complete lower-endpoint phase curve, the known match population s, and the fully specified detector E above. The ordinary rate alone would supply less information. Write f=Tr(Fρ). The full curve fixes the compression of ρ onto span{p,u} to |w⟩⟨w|. Indeed its mean gives sh+(1−h)⟨u|ρ|u⟩, and the two quadratures fix the complex off-diagonal entry. Positivity eliminates the direction in this span orthogonal to w. The complementary subspace has population r, so 0≤f≤r.

> For each feasible f, the exact interval of normalized Choi purities is [J(f),1], with J(f)=d²+f²+(r−f)²/13. Every value occurs even within mixed-unitary channels.

Let D be the projection onto the 13-dimensional space perpendicular to p,u,v. In the orthogonal decomposition along w, v, and this remaining space, discarding off-diagonal blocks lowers the squared Hilbert–Schmidt norm. The last block has trace r−f, giving Tr(ρ²)≥J(f) by Cauchy–Schwarz. The state ρ(f)=|w⟩⟨w|+fF+(r−f)D/13 attains it. Its spectral terms are normalized unitary Choi kets with nonnegative weights, using the 13 Pauli operators other than I,A,B.

With L=d²+r²/14=0.291601186810187…, the lower endpoint is L+(14/13)(f−r/14)². In contrast, the submitted path gives L+(1−L)[(14f/r−1)/13]², and that formula applies only for r/14≤f≤r on that path.

Extra probability f	Minimum possible purity	Maximum possible purity
0	0.292841628283	1
r/14 = 0.03393875…	0.291601186810	1
r/2 = 0.23757123…	0.336257079801	1
r = 0.47514245…	0.501235795573	1

The maximum and all intermediate values are attained by the following construction. It preserves not only the full phase curve and f but also the Z-positive rate d+6(r−f)/13 for the product-input test. The latter is exactly the value associated with f on the submitted interpolation path.

Why the product-input test can stay blind

The minimum-purity state on page 2 gives $p_+=d+6(r-f)/13$ for input $|00\rangle$ and first-qubit Z readout. Its w component preserves the first-qubit value and its v component flips it. Among the other 13 tensor Paulis, six preserve that value. The supplied interpolation has the same formula since both p_+ and f are affine in its mixing parameter and the formula holds at its two endpoints.

A pure channel with these same probabilities

Use first-qubit Pauli matrices X and Z. Let $U=(\sqrt{s}\,I-i\sqrt{k}\,Z+i\sqrt{f}\,X)/\sqrt{(d+f)}$. Anticommutation gives $U\dagger U=I$. Choose a Hermitian unitary N on the first qubit such that $|\langle 0|UN|0\rangle|^2=6/13$. Such an N exists: apply U† to $\sqrt{(6/13)}|0\rangle+\sqrt{(7/13)}|1\rangle$ and choose its overall phase so its first entry a is real. For resulting entries (a,b), take N with rows (a,conj(b)) and (b,−a).

> The four-dimensional operator $V=(U\otimes I)[\sqrt{(d+f)}I_4+i\sqrt{(r-f)}(N\otimes X)]$ is unitary because $N\otimes X$ is a Hermitian involution and $d+r=1$.

The part of $|V\rangle$ with second-qubit factor I is $w+\sqrt{f}\,v$. The rest has second factor X and is orthogonal to p,u,v. Its ordinary phase curve and f are therefore correct. Applied to $|00\rangle$, the two terms of V end in orthogonal second-qubit states. The Z-positive probability is $d+(r-f)|\langle 0|UN|0\rangle|^2$, the required value.

Write $\rho_1=|V\rangle\langle V|$. Since the two states agree on the w population d, the v population f, and the remaining total r−f, $\mathrm{Tr}[\rho(f)\rho_1]=J(f)$. Therefore the mixture $(1-t)\rho(f)+t\rho_1$ has purity $J(f)+(1-J(f))t^2$ for $0\le t\le 1$. Its outcome probabilities and both Choi marginals are fixed. This proves both attainability and completeness. It does not give the same interval for arbitrary independent values of f and p_+.

For exact s,h and phase curve, an interval of admissible f values yields a purity lower bound by minimizing J over that interval intersected with [0,r]. Real uncertainty in the other inputs also has to be propagated. Plugging a measured frequency into an exact equality would not do so.

A local check at the upper F rate

Consider the exact boundary f=r. There is no population outside span{η,v}, η=w/√d. Positivity forces ρ to have the two-by-two form with diagonal entries d,r and off-diagonal γ, $|\gamma| \le \sqrt{(dr)}$. This yields $\mathrm{Tr}(\rho^2)=d^2+r^2+2|\gamma|^2$. The Choi marginal $I_4/4$ forces γ to be real: the two operators giving η,v are W=(√s I−i√k A)/√d and iB; W†(iB) is a nonzero traceless anti-Hermitian operator. In the partial trace the cross term therefore vanishes exactly for real γ.

If the same channel transfers to the product preparation |00⟩, W and iB give the first-qubit states ξ=(√s−i√k)|0⟩/√d and ν=i|1⟩, respectively. Their second-qubit state is |0⟩ in both cases. The Z probability is d for every γ. A calibrated first-qubit measurement of $O=|\xi\rangle\langle\nu|+|\nu\rangle\langle\xi|$ instead has expectation 2γ. Hence

> At f=r, with the specified exact phase curve and justified channel transfer, $\mathrm{Tr}(\rho^2)=d^2+r^2+\langle O\rangle^2/2$. This is a single transverse spin component with known orientation.

This last identity uses exact support saturation. A finite estimate close to r requires bounds on the residual population and calibration errors before using a two-dimensional reduction. None of the F, Z or transverse outcomes here are measured hardware values.

A physical requirement on conditioning

For a nonzero trace-nonincreasing instrument branch B, define H=B†(I). The input success probability on state τ is Tr(Hτ). The input marginal of its normalized Choi state is $H^{T}/\mathrm{Tr}\,H$, so it equals $I_4/4$ precisely when H is proportional to I. Testing this property uses characterized preparations. If the history precedes the new input, common subsequent dynamics across preparations is a distinct transfer question. White et al. demonstrate experimental characterization of multitime processes with interventions.

Keith et al., Physical Review A 98, 042318 (2018), incomplete measurement and compatible states.

Watrous, The Theory of Quantum Information, chs. 3–4, state discrimination and mixed-unitary channels.

White et al., PRX Quantum 3, 020344 (2022), non-Markovian quantum process tomography.

Three-axis purity certificate

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2. Model and population constraints

Use output-first vectorization on $\mathcal{H} = \mathbb{C}^4 \otimes \mathbb{C}^4$, with normalized Choi vectors $|M\rangle = \mathrm{vec}(M)/2$. Let

$$p = |I_4\rangle, \quad u = |i(Z \otimes I_2)\rangle, \quad v = |i(X \otimes I_2)\rangle,$$

where X, Y, Z are the Pauli matrices. Put

$$a = sh, \quad k = \frac{(a - \sqrt{aq_0})^2}{sh(1-h)}, \quad d = s + k, \quad r = 1 - d,$$

$$\eta = \frac{\sqrt{s}\,p - \sqrt{k}\,u}{\sqrt{d}}, \qquad \theta = \frac{\sqrt{k}\,p + \sqrt{s}\,u}{\sqrt{d}}, \qquad F = |v\rangle\langle v|,$$

and $D = I - |p\rangle\langle p| - |u\rangle\langle u| - F$, of rank 13. The **exact complete lower-endpoint phase curve** with its fixed effect forces the orthogonal vector θ into the kernel of each compatible positive Choi state. For the additional-outcome probability $f = \mathrm{Tr}(F\rho)$, the fixed populations are $d, f, r - f$. The normalized CPTP condition is $\mathrm{Tr}_{\mathrm{out}}\,\rho = I_4/4$.

The minimum-purity state in this compatible class is

$$\rho_\star(f) = d|\eta\rangle\langle\eta| + fF + \frac{r-f}{13}D, \qquad J(f) = d^2 + f^2 + \frac{(r-f)^2}{13}.$$

All vectors spanning the three sectors can be chosen as maximally entangled Choi vectors; hence $\mathrm{Tr}_{\mathrm{out}}\,\rho_\star = I_4/4$. For any compatible ρ, the populations give the exact identity

$$\mathrm{Tr}\,\rho^2 = J(f) + \|\rho - \rho_\star(f)\|_2^2.$$

This setup is restricted to that exact phase curve and detector. Ordinary readout probability alone does not fix θ as a null direction.

3. Joint first-qubit X/Y/Z measurement theorem

Theorem (three-axis locally sharp purity certificate). Fix the model of section 2, with exact s, h, q_0, the full specified phase curve, and the extra-outcome population $0 \le f \le r$. Assume a normalized CPTP Choi state ρ transfers unchanged to a fresh product input $|00\rangle$. For each $Q \in \{X, Y, Z\}$, denote by p_Q the probability that a projective measurement of the **first output qubit** in the corresponding Pauli basis returns the positive eigenvalue. Set

$$p_Z^\star(f) = d + \frac{6(r-f)}{13}, \qquad p_X^\star = p_Y^\star = \tfrac{1}{2}.$$

Then

$$\boxed{\mathrm{Tr}(\rho^2) \;\ge\; J(f) + \frac{91}{1230}\big[p_Z - p_Z^\star(f)\big]^2 + \frac{7}{97}\Big[(p_X - \tfrac{1}{2})^2 + (p_Y - \tfrac{1}{2})^2\Big].}$$

The inequality holds globally over all compatible normalized CPTP Choi states. At every **interior** $0 < f < r$, equality is attainable for every sufficiently small triple of deviations from the indicated reference probabilities, using an explicitly positive normalized CPTP Choi matrix. This is *local* simultaneous sharpness; it does not assert attainability of arbitrary triples far from the reference probabilities, nor impose unital or mixed-unitary constraints.

Proof. Define the actual output-probability observables

$$M_Q = 4\,[((I_2 + Q)/2) \otimes I_2]_{\text{out}} \otimes [|00\rangle\langle 00|]_{\text{in}}.$$

Let $B = I - |\theta\rangle\langle\theta|$. For any Hermitian M, first remove the three fixed population directions:

$$M_1 = BMB - \text{Tr}(BMB|\eta\rangle\langle\eta|)|\eta\rangle\langle\eta| - \text{Tr}(BMBF)F - \frac{\text{Tr}(BMBD)}{13}D.$$

This operator is supported on B, orthogonal to the three fixed populations, and traceless. Put $L = \text{Tr}_{\text{out}}\, M_1$, a traceless Hermitian reference operator. Since θ is maximally entangled, for traceless Hermitian H, G,

$$\text{Tr}[B(I \otimes H)B(I \otimes G)] = \frac{7}{2}\,\text{Tr}(HG).$$

The operators $B(I \otimes H)B$ are already orthogonal to the three fixed populations. Therefore

$$\widehat{M} = M_1 - \frac{2}{7}B(I \otimes L)B$$

is the Hilbert–Schmidt projection onto the additional tangent condition $\text{Tr}_{\text{out}}\,\Delta = 0$. It is also orthogonal to θ, and to the three population constraints.

Direct Pauli-matrix multiplication in the specified 16-dimensional vectorization gives the following **exact** Gram matrices for the ordered observables (Z, X, Y):

$$G^{(1)}_{PQ} = \text{Tr}(M_{P,1}M_{Q,1}) = \begin{pmatrix} 276/13 & 9 & 9 \\ 9 & 49/2 & 21/2 \\ 9 & 21/2 & 49/2 \end{pmatrix},$$

$$G^{(L)}_{PQ} = \text{Tr}(L_P L_Q) = \begin{pmatrix} 27 & 63/2 & 63/2 \\ 63/2 & 149/4 & 147/4 \\ 63/2 & 147/4 & 149/4 \end{pmatrix}.$$

For example, the X/Y population subtraction is $M_{Q,1} = BM_QB - B/2$; $M_Q^2 = 4M_Q$ and $\text{Tr}\, M_Q = 8$ give $\text{Tr}\, M_{Q,1}^2 = 49/2$. Their partial traces have diagonal $7(|00\rangle\langle 00| - I_4/4)$ and an off-diagonal magnitude 1/2 with X and Y in orthogonal quadratures, giving $\text{Tr}\, L_Q^2 = 149/4$ and $\text{Tr}\, L_X L_Y = 147/4$. The entries involving Z follow by the same rank-one contractions, with $L_Z = 6(|00\rangle\langle 00| - I_4/4)$.

Subtracting the trace-preservation projection gives

$$G_{PQ} = \text{Tr}(\widehat{M}_P\widehat{M}_Q) = G^{(1)} - \frac{2}{7}G^{(L)} = \text{diag}\left(\frac{1230}{91}, \frac{97}{7}, \frac{97}{7}\right).$$

For $\Delta = \rho - \rho_\star(f)$, all imposed constraints are homogeneous and $\text{Tr}(M_Q\Delta) = \text{Tr}(\widehat{M}_Q\Delta)$. Orthogonal Bessel inequality in Hilbert–Schmidt space yields

$$\|\Delta\|_2^2 \geq \frac{[p_Z - p_Z^\star(f)]^2}{1230/91} + \frac{[p_X - 1/2]^2 + [p_Y - 1/2]^2}{97/7}.$$

Together with the purity-excess identity, this proves the inequality. For a sufficiently small deviation vector $(\delta_Z, \delta_X, \delta_Y)$, define

$$\rho = \rho_\star(f) + \frac{91\delta_Z}{1230}\widehat{M}_Z + \frac{7\delta_X}{97}\widehat{M}_X + \frac{7\delta_Y}{97}\widehat{M}_Y.$$

It preserves every linear phase, population and trace-preservation constraint, reproduces the desired three deviations, and achieves equality. For interior f, the base state is strictly positive on the allowed fifteen-dimensional support, so every sufficiently small Hermitian perturbation on that support remains positive. This completes the constructive local-sharpness proof. □

Restriction to the longitudinal reading. Setting $p_X = p_Y = 1/2$ reduces exactly to the coefficient $91/1230$. Measuring additional X/Y axes need not make purity identifiable: if all probabilities equal their reference values, compatible states can still have a wide range of purities.

4. Numerical witness and finite-data limits

At $f = r/2$, the verifier constructs the positive channel state with deviations $(\delta_Z, \delta_X, \delta_Y) = (0.005, 0.007, -0.004)$. It checks the full Choi marginal, positivity, phase response at six settings, exact extra-outcome probability, and equality in the three-axis bound within floating-point tolerance. It prints an attained purity of approximately 0.336263620116203.

The theorem assumes exact population parameters. For a simultaneous confidence region on (f, p_X, p_Y, p_Z), a valid conservative lower purity certificate is obtained by **minimizing the right-hand side of the theorem over that entire region**, together with any independently justified region for (s, h, q_0). For known f and marginal probability intervals, this is achieved by substituting each interval's distance from its corresponding reference probability. For uncertain f, minimize over its allowed range; the objective is convex quadratic in f when the other inputs are fixed. An empirical claim requires calibration transfer, finite-sample uncertainty allocation, and an independently supported full-phase compatibility premise. Merely inserting observed frequencies into exact equations provides no coverage guarantee.

The mutually noncommuting X, Y, Z output observables would generally be measured on different preparations. The normalized Choi marginal applies to a **trace-preserving channel**. A history-selected trace-nonincreasing instrument branch need not have that marginal, and applying the same process to product input $|00\rangle$ is an additional physical transfer claim. Nothing here verifies those premises experimentally.

When three output measurements identify channel purity

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A fixed joint detector supplies a complete model phase curve and an extra-outcome rate f. Separate X, Y and Z measurements on the first output qubit use a product input $|00\rangle$. What follows if all three probabilities equal their minimum-purity reference values? The answer changes at the upper endpoint of f.

> For every $0 \le f < r$, at least two distinct purities fit the reference triple. At $f=0$ the full interval [0.29284162828..., 1] fits. At $f=r$ the reference triple fixes purity to 0.501235795572884... . These are population-level implications of the specified trace-preserving model.

What was supplied and what was checked

Use the normalized Choi ket $|M\rangle = \mathrm{vec}(M)/2$, output indices first. Put $p=|I_4\rangle$, $u=|iZ\otimes I\rangle$ and $v=|iX\otimes I\rangle$. With $s=131/320$, $h=0.99$, $q=81711338569043/224995905216000$, set $k=(sh-\sqrt{[shq]})^2/[sh(1-h)]$, $d=s+k$, $r=1-d$. The specified lower-endpoint phase curve fixes the p,u compression to $|w\rangle\langle w|$, $w=\sqrt{s}\,p-\sqrt{k}\,u$. Write $\eta=w/\sqrt{d}$, $\theta=(\sqrt{k}\,p+\sqrt{s}\,u)/\sqrt{d}$, and D for the projection perpendicular to p,u,v. The θ direction has zero population. The remaining populations on η,v,D are $d, f, \ell = r-f$.

The minimum state $\rho_\star = d|\eta\rangle\langle\eta| + f|v\rangle\langle v| + \ell D/13$ has purity $J(f) = d^2+f^2+\ell^2/13$. Its probabilities are $pZ_\star = d + 6\ell/13$ and $pX_\star = pY_\star = 1/2$. Write $\Delta j = pj - pj_\star$. The supplied global lower bound is $J(f) + (91/1230)\Delta Z^2 + (7/97)(\Delta X^2 + \Delta Y^2)$. I reconstructed its coefficients exactly over the rationals in a rotated frame. At $f=r/2$, the stated shifts (0.005, 0.007, −0.004) in lab order Z,X,Y give a positive trace-preserving state with purity 0.336263620116203 to the precision shown.

Observed f	Matching all three reference probabilities
0	Every purity in [J(0),1] occurs. An explicit pure unitary is on p. 3.
$0<f<r$	Some ambiguity always survives. At $f \approx 0.075590554093$ the full interval [J(f),1] occurs.
r	The only compatible state is $\rho_\star$ at the reference triple. Away from it the transverse component gives purity exactly.

The normalized input marginal and transfer of the same channel to $|00\rangle$ must hold to apply any of these results. The physical gate path after the reported history 111 was not measured in the supplied packet.

Supplied three-axis proof, sections 2–4. Source ↗

Keith et al. (2018), Joint quantum-state and measurement tomography with incomplete measurements. Source ↗

Projection on the actual support

Positive states with zero population in a direction annihilate it. The allowed linear perturbations therefore change at f=0 and f=r.

Define $A=\sqrt{(s/d)}$, $B=\sqrt{(k/d)}$, so $A^2+B^2=1$. The rotated transverse deviations are $\Delta_1=A\Delta X+B\Delta Y$ and $\Delta_2=-B\Delta X+A\Delta Y$. To see the rotation explicitly, write $A=\cos\alpha$, $B=\sin\alpha$, $W=\exp(-i\alpha Z)\otimes I$ and $T=\exp(-i\alpha Z/2)\otimes I$. The basis change $M\mapsto TW\dagger MT\dagger$ maps η,θ,v to $|I_4\rangle$, $|iZ\otimes I\rangle$, $|iX\otimes I\rangle$. It fixes the product input up to phase and rotates its first-qubit Bloch vector by $-\alpha$. This is a change of coordinates in the proof.

Support	Gram diagonal in order Z, Δ_1, Δ_2	Rank
0<f<r	1230/91, 97/7, 97/7	3
f=0	159/13, 25/2, 254/21	3
f=r	0, 0, 1/2	1

At f=0, take the reciprocals to get the globally valid excess bound $(13/159)\Delta Z^2+(2/25)\Delta_1^2+(21/254)\Delta_2^2$. This is locally sharp in all three directions. At f=r only Δ_2 can vary: $pZ=d$, $\Delta_1=0$ and $|\Delta_2|\le\sqrt{(dr)}$. In the surviving two-dimensional Choi block the diagonals are d,r and trace preservation makes the coherence γ real. Applying the channel to $|00\rangle$ gives $\Delta_2=\gamma$. Thus purity is exactly $d^2+r^2+2\Delta_2^2$. Every allowed γ gives a positive normalized Choi state.

Gram calculation

For each support S, compress each half-Pauli observable $2(Q\otimes I)_{out}\otimes|00\rangle\langle 00|_{ref}$ to S, then remove its components along the fixed population projections. Using half-Paulis gives the same probability deviations because the reference marginal is fixed. Call the resulting observable K and its output partial trace L. To impose zero marginal, project away $S(I\otimes H)S$ for traceless Hermitian H. The companion constructs the exact 15×15 Gram matrix on the traceless Pauli basis and uses its Moore–Penrose inverse.

For the interior, the Gram matrix on H is $(7/2)I$ in an orthonormal basis. At f=0 the two missing directions give the form $3\mathrm{Tr}(HG)+(1/8)\mathrm{Tr}(Y_1H)\mathrm{Tr}(Y_1G)$, with $Y_1=Y\otimes I$. The raw half-Pauli norms squared are 159/13,25/2,25/2; their reference partial traces have squared norms 0,0,5/4 with $|\mathrm{Tr}(Y_1L)|=1$ on the last axis. Subtracting 5/12−1/84 from the last entry gives 254/21. All transverse cross entries vanish in this frame.

For nonsingular Gram G, the equality candidate is $\rho^\star$ plus the sum of the projected observables with coefficients $G^{-1}\Delta$. This state meets all linear constraints. The lower bound is attained exactly when this candidate is positive semidefinite. At an interior point of its support small changes preserve positivity. For distant triples the certificate alone does not determine the optimum.

Which purities fit the reference triple?

The following constructions preserve all three output rates, f, the fixed-detector phase curve and both normalized Choi marginals. Work in the rotated coordinates of p. 2.

A nonzero interval for every f<r

The unresolved thirteen two-qubit Pauli directions contain six that preserve the first bit of $|00\rangle$ and seven that flip it. Keep the d and f weights on η and v. Starting with $\rho_\star$, put the six preserving weights entirely on $|I\otimes X\rangle$ and the seven flipping weights on $|X\otimes X\rangle$. Every component gives a computational-basis first-qubit output. The same three reference probabilities follow, and the state is mixed-unitary. Its purity is $J(f)+72\ell^2/169$. Mixing with $\rho_\star$ realizes every value over that interval, since $\mathrm{Tr}(\rho\rho_\star)=J(f)$. Its width vanishes only at f=r. This proves ambiguity at every positive residual without assuming the top endpoint is one.

A pure channel at the lower endpoint

At f=0 put $c_0=\sqrt{(d+6r/13)}$, $c_1=2\sqrt{d}-c_0$. On ordered pairs (00,11) and (01,10) use the unitary blocks $c_jI+i\sqrt{(1-c_j^2)}X$. Both c_j lie in $[-1,1]$. The total trace divided by four is $\sqrt{d}$, and the coefficients on $Z\otimes I$ and $X\otimes I$ vanish. On $|00\rangle$ the components carry distinct second-qubit flags, so the first-qubit density is diagonal with Z-positive rate $c_0^2=pZ_\star$. This pure Choi state and $\rho_\star$ agree on all specified outcomes. Their overlap is J(0); their mixtures cover [J(0),1].

An explicit interior example

Let f be the unique root in (0,r) of $df=(42/169)(r-f)^2$. Then $f\approx0.075590554093$. Put $U=(\sqrt{d}\,I+i\sqrt{f}\,X)/\sqrt{(d+f)}$, a first-qubit unitary. Choose a Hermitian involution N taking $|0\rangle$ to $U\dagger[\sqrt{(6/13)}|0\rangle-i\sqrt{(7/13)}|1\rangle]$ up to phase. Make the first entry a real by a phase choice and call the second entry b; the matrix with rows (a,b*) and (b,−a) has this property. The two-qubit unitary $V=(U\otimes I)[\sqrt{(d+f)}I_4+i\sqrt{\ell}(N\otimes X)]$ has the required visible Choi component.

Its two product-output branches have orthogonal second-qubit flags. The first branch gives first-qubit off-diagonal $-i\sqrt{(df)}$; the second gives $+i\ell\sqrt{42/13}$. The root condition cancels them. The Z-positive weight is $d+6\ell/13$. This proves all purities through one fit the reference triple at this interior f. The construction supplies no general pure endpoint for other values of f.

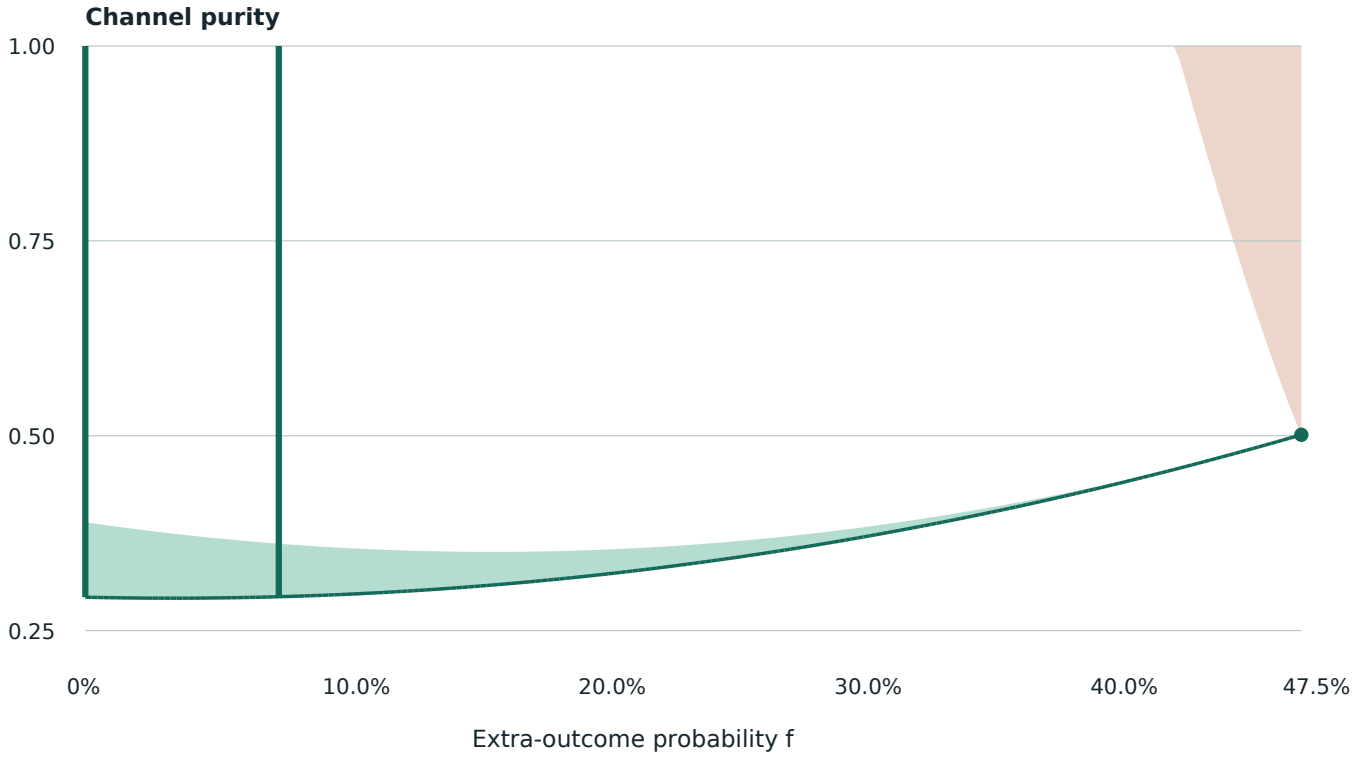
Near the upper boundary the opposite can happen. For example at $\ell=0.01$ and the reference triple, the preceding transverse bound gives purity ≤0.58577 (rounded upward), while the constructed sign-varying family at $\ell=0.01$ spans 0.9802 to 1 for a different first-qubit output. The data vectors must be specified to compare purity ranges.

For that upper check the reference transverse expectation is zero. With $\beta=2\sqrt{[3(1-\ell)\ell]}+4\ell$, the channel bound gives $d^2+f^2+(\beta^2+\ell^2)/2+2(1-\ell)\ell+\ell^2$ as one valid upper estimate. This is a conservative exclusion of purity one for the indicated data.

Transverse readout and channel purity near a support boundary, p. 2. Source ↗

Purities compatible with the three reference rates

This plot applies to one curve of observations: pX=pY=1/2, pZ=d+6(r−f)/13, and the same complete fixed-detector phase curve.



Green shows values with explicit channels. The vertical green lines give the full range at f=0 and f≈7.5591%. The dot gives the unique value at f=r. The rust region is excluded by the conservative transverse estimate; all values below the green lower curve are excluded by J. Other white values above the curve are unclassified by these constructions.

Using the bound with finite observations

The three Pauli components require distinct measurement settings on comparably prepared outputs. A valid confidence statement needs one simultaneous region for the underlying probabilities and calibrated inputs. Minimize the applicable certificate over that region and all compatible models. A confidence interval merely touching f=0 or f=r does not prove exact zero population, so using a boundary coefficient for the entire interval is unwarranted. Uncertainty in the detector or phase curve also releases the exact support constraint unless separately bounded.

White et al. (2022), Non-Markovian quantum process tomography. Source ↗

Channel purity from a single prepared input

Research direction and responsibility · Kirtus Justus, Port Coquitlam, British Columbia. Mathematical and computational assistance · ChatGPT (OpenAI).

On a continuous interval $0 \le f \le f\star$, the three reference readings of the first qubit admit every normalized Choi purity from J(f) to one. At f=0, measuring the complete two-qubit reference output instead bounds that same purity above by 0.812504.

What the first-qubit data imply

Here f is the probability of a specified additional outcome, $r \approx 0.4751424501$ is the available population, and $J(f)=d^2+f^2+(r-f)^2/13$ with $d=1-r$. The detector and full phase curve are fixed. All three first-qubit probabilities are set to their reference values. A pure channel has a rank-one normalized Choi matrix.

Extra-outcome probability	Compatible Choi purities
$0 \le f \le f\star \approx 0.105322985$	Exactly [J(f),1]. Both endpoints and every intermediate purity are attained.
$f\star < f \le 19r/32 \approx 0.282115830$	J(f) is attained. Distinct purities exist throughout. The maximum over all channels is unresolved here.
$19r/32 < f < r$	Purity one is excluded. A nonzero interval above J(f) is attainable. The sharp upper endpoint is unresolved here.
$f = r$	The three reference readings identify a single state with purity $d^2+r^2 \approx 0.501235795573$.

The exact $f\star$ is the positive root of $7f^2+(39d-7)f-6d(1-d)=0$. Its stated precision is a numerical evaluation. It bounds a constructive family. It supplies no impossibility result for arbitrary channels above $f\star$.

What changes when both qubits are observed

The two-qubit reference output at f=0 is mixed even though its first-qubit marginal can be reproduced by a unitary. A unitary sends a pure input to a pure joint output. Page 4 quantifies that distinction for any specified joint output. It gives a sharp interval when that output is the only datum and a valid upper bound with additional observations imposed.

A continuous set of unitary witnesses

Use output-first normalized Choi kets $|M\rangle=\mathrm{vec}(M)/2$. The match is $p=|I_4\rangle$, the other visible direction is $u=|iZ\otimes I\rangle$, and the extra outcome projects onto $v=|iX\otimes I\rangle$. A fixed effect projects onto $\sqrt{h}\,p+\sqrt{(1-h)}u$. The supplied values are $s=131/320$, $h=99/100$ and $q=81711338569043/224995905216000$. Put $k=(sh-\sqrt{(shq)})^2/[sh(1-h)]$, $d=s+k$ and $r=1-d$. The full specified phase curve fixes the p,u compression to the outer product of $w=\sqrt{s}\,p-\sqrt{k}\,u$.

Take $\cos\alpha=\sqrt{(s/d)}$, $\sin\alpha=\sqrt{(k/d)}$, and $T=\exp(-i\alpha Z/2)\otimes I$. Writing a laboratory matrix as TUT rotates the calculation into coordinates where $w/\sqrt{d}$ is $|I_4\rangle$, its null direction is $|iZ\otimes I\rangle$ and v is $|iX\otimes I\rangle$. The reference first-qubit output is diagonal in both frames. The required amplitudes in the construction below are $\sqrt{d}$ on $|I_4\rangle$, zero on $|iZ\otimes I\rangle$ and $\sqrt{f}$ on $|iX\otimes I\rangle$.

Construction

Write $\ell=r-f$, $c=\sqrt{(d+f)}$, and choose β by $c\cos\beta=\sqrt{d}$ and $c\sin\beta=\sqrt{f}$. With $P_\pm$ the first-qubit X eigenprojectors, take $U=P_+\otimes V_+ + P_-\otimes V_-$, where $V_\pm=\exp(\pm i\beta)[cI+i\sqrt{\ell}\,(n_\pm\cdot\sigma)]$ for real unit vectors $n_\pm$. Both $V_\pm$ are unitary. Taking their traces gives the three required amplitudes.

On input $|00\rangle$ the first-qubit density in the X basis has diagonals ½ and upper off-diagonal $\langle 0|V_-^\dagger V_+|0\rangle/2$. To obtain the reference output it is enough to make this overlap the real number $z=2(d+6\ell/13)-1$. After removing $\exp(2i\beta)$, the relative unitary has form $aI+i\,m\cdot\sigma$ with $a=c^2+\ell\,n_-\cdot n_+$ and $|m|^2=1-a^2$. Hence a ranges over $[1-2\ell,1]$. Jointly rotating the axes gives every imaginary part in $[-\sqrt{(1-a^2)},\sqrt{(1-a^2)}]$ for its expectation on $|0\rangle$.

> Set $a=z\cos(2\beta)$. The imaginary part $-z\sin(2\beta)$ is feasible since $|z|\le 1$. Thus this construction works precisely when $z(d-f)/(d+f)\ge 1-2\ell$. Expanding gives $7f^2+(39d-7)f-6dr\le 0$.

The polynomial is strictly increasing for $f\ge 0$ and changes sign inside $(0,r)$. Interval arithmetic puts its root strictly between 0.10532298498594176 and 0.10532298498594177. For each f in the interval, the unitary has purity one. Its overlap with the minimum state $\rho\bigstar$ is $J(f)$, since they share the sector populations. At mixture weight t on the unitary, purity is $J(f)+t^2[1-J(f)]$. This proves the entire interval.

What remains when the support changes

Let D be the thirteen-dimensional projection orthogonal to p,u,v and F=|v⟩⟨v|. The minimum state is ρ★=|w⟩⟨w|+fF+ℓD/13. Positivity annihilates its prescribed null direction in every compatible channel. The first-qubit reference rates on |00⟩ are pX=pY=½ and pZ=d+6ℓ/13. Fixing the sector populations gives Tr(ρ²)=J(f)+‖ρ−ρ★‖₂².

Support	Gram diagonal for Z and two rotated transverse axes
Interior, 0<f<r	1230/91, 97/7, 97/7
Lower endpoint, f=0	159/13, 25/2, 254/21
Upper endpoint, f=r	0, 0, ½

For each nonzero Gram entry, the reciprocal multiplies the squared deviation along that axis in the lower bound. At f=0 the two transverse axes are rotated by α relative to the laboratory axes. At f=r the only free component is a real coherence γ. The second rotated transverse probability deviation equals γ and the attainable purity is d²+r²+2γ², |γ|≤√(dr). At the reference triple γ=0.

Ambiguity and a necessary test

For any ℓ>0, six unresolved Pauli directions preserve the first bit of |00⟩ and seven flip it. Redistributing the two fixed group weights between Pauli directions leaves all first-qubit observations intact. Mixing the uniform state with a state that concentrates each weight on one direction attains the entire interval from J(f) to J(f)+72ℓ²/169. This is an attainable interval within the full model.

For a pure channel U, the visible operators act trivially on the second qubit. The residual operator has squared Frobenius norm 4ℓ. The two off-diagonal blocks of U in the second-qubit basis have equal squared norms by unitarity, so the block that flips |0⟩ has squared norm at most 2ℓ. On the pure joint output, the first and second marginals share the same eigenvalues. The first marginal must therefore have largest eigenvalue at least 1−2ℓ. At the reference readings it equals d+6ℓ/13. Purity one requires ℓ≥13r/32, equivalently f≤19r/32.

At f=0 the sharp purity floor is 0.292841628282…. The interior example at f≈0.075590554093 solves df=42ℓ²/169 and lies inside the interval on page 2. For the separate, nonreference interior triple with deviations (0.005,0.007,−0.004) in Z,X,Y order at f=r/2, the three-axis lower bound is attained at 0.336263620116203.

Complete output on one pure input

A sharp interval from the joint output state

Let Φ be any trace-preserving channel from four dimensions to four dimensions. The only datum in this theorem is its entire output σ on one specified pure input. Write P=Tr(σ²), λ=λmax(σ), and C for the trace-one Choi matrix of Φ.

> The exact interval is [(P+3/4)/16, (9+6λ+P)/16]. Both endpoints and every intermediate purity occur. Under extra detector or phase constraints these bounds remain valid, with separate attainability questions.

Lower endpoint

Choose an input basis with the measured input first. The blocks of 4C are Φ(|i⟩⟨j|). Squaring the Hilbert–Schmidt norm adds their squared norms. The first diagonal block contributes P, each of the other three density matrices contributes at least ¼, and all other contributions are nonnegative. Measure the input in this basis, send the first outcome to σ and the other outcomes to $I_4/4$. This channel attains the lower endpoint.

Upper endpoint

Take a Stinespring isometry. For each basis input j let τj be the environmental marginal of its pure joint output. Then τ_0 has the nonzero spectrum of σ. The maximally entangled input purifies C with environmental marginal $(\tau_0+\tau_1+\tau_2+\tau_3)/4$. Writing $\omega=(\tau_1+\tau_2+\tau_3)/3$ gives 16 $\mathrm{Tr}(C^2)=P+6\mathrm{Tr}(\tau_0\omega)+9\mathrm{Tr}(\omega^2)\le P+6\lambda+9$.

In a Schmidt purification of σ choose a system and environment Schmidt pair for λ. For the other three inputs choose orthonormal system vectors perpendicular to this system vector, all paired with that same environment vector. The four images are orthonormal, so they define an isometry. Here ω is pure in the top eigenspace of τ_0 and the upper bound is attained. Interpolation between the two channels keeps σ fixed and, by continuity, attains all intermediate purities.

Application to the fixed phase experiment

For ρ★ the full output on |00⟩ is diagonal with weights $d+2\ell/13$, $4\ell/13$, $f+3\ell/13$, $4\ell/13$ in binary order. At f=0 its purity ceiling from the joint output alone is approximately 0.812503777157. The first-qubit data alone at that f permit purity one. At f=r/2 the complete-output ceiling is approximately 0.798737595195. All these joint reference outputs are mixed, including both endpoints. No unitary reproduces any of them on the pure input.

Purity range at f=0 with the phase observations fixed



Solid green denotes attained purities. Dots mark values below the joint-output ceiling whose attainability is unresolved in the fixed phase model.

How much of the channel is determined?

On the allowed support, every phase effect is a multiple of the same rank-one projector. Extra phase settings consequently cannot recover independent directions in the channel. The table gives exact real dimensions of the local set of channels at the reference state. “Joint” fixes the complete two-qubit output on the same input. Unitality means $\Phi(I)=I$.

Extra outcome	First qubit	First, unital	Joint	Joint, unital
$0<f<r$	204	189	192	177
$f=0$	176	161	164	149

To certify the ranks, use the rotated Pauli frame of page 2 and compress the constraint matrices to the support. Include sixteen reference-marginal rows, the independent sector-population rows and three or fifteen output rows. Add fifteen output-marginal rows for unitality. Clearing the dyadic denominators gives integer rows. Elimination modulo 101 gives full row rank in each case. The reference state is positive definite on its support, so every sufficiently small tangent perturbation is physically allowed. These dimensions apply to the specified reference outcomes.

Even among mixtures of unitaries, the four output groups of the unresolved Pauli directions have sizes 2,4,3,4 on $|00\rangle$. Holding their group weights fixed gives a product of simplexes of dimension nine. At $f=r/2$ it realizes purities from approximately 0.336257079801 to 0.346943960175 with identical joint output. Thus the upper bound on page 4 can exclude purity one without determining the channel.

What an observation could establish

A mixed joint output for an independently pure input excludes a unitary channel on that input path. For measured data, maximize the upper bound over a simultaneous confidence region for the joint output. Additional phase and detector information can restrict the feasible set further. An instrument conditioned on history 111 can be trace-decreasing. Its success probability and transfer across preparations need independent characterization before applying a normalized-channel theorem. No matched hardware trials are part of these calculations.

Mathematical inputs · Endpoint constraints in three-axis channel tomography.

Mathematical inputs · Operational identification and observation fibers.

Watrous, The Theory of Quantum Information, ch. 2 · channel representations.

Ziman (2008) · incomplete process tomography.

Keith et al. (2018) · compatible state sets and bounds.

White et al. (2022) · experimental multitime process characterization.

Conditional quantum measurement

Purity bounds from local output measurements

Research direction and responsibility · Kirtus Justus, Port Coquitlam, British Columbia.
Mathematical and computational assistance · ChatGPT (OpenAI). All probabilities in the numerical example are supplied model values.

> On one known pure product input, first-qubit tomography and one binary measurement on the second qubit give a sharp interval for channel purity. A unitary channel fits these local readings exactly when the two possible second-qubit probabilities both lie between the eigenvalues of the first-qubit state.

Let C be the normalized Choi matrix of a trace-preserving two-qubit channel. Its purity is $\mathrm{Tr}(C^2)$. On input $|00\rangle$ let Ω be the output, and let a be the larger eigenvalue of its first-qubit marginal. Write β for the probability of a specified outcome of a rank-one projective measurement on the second qubit. Set $b=\min(\beta,1-\beta)$, the less frequent population in that basis.

Exact upper endpoint

Put $m=\min(1,a+b)$. Over all such channels, the sharp upper endpoint of purity is $[9+6m+m^2+(1-m)^2]/16$. It reaches one precisely when $b\geq 1-a$. If $b<1-a$, the measured local data exclude every unitary channel. There is no need for a joint output measurement to apply this result.

Exact lower endpoint

Work in the first marginal's eigenbasis and order the second outcomes by decreasing probability. Let x be $(a+b)/2-¼$ clipped to $[\max(0,a+b-1), \min(a,b)]$. Define $L=(a-x)^2+x^2+(1-a-b+x)^2+(b-x)^2$. The sharp lower endpoint is $(L+¾)/16$. Every intermediate purity occurs. Sharpness here concerns precisely the specified local data; extra Choi observations can restrict both endpoints.

Numerical distinction

At the supplied 1% residual point, $a=0.529472934503538\ldots$. A unitary fits the local readings only if the second-qubit probability belongs to $[1-a,a]$, approximately $[0.47052707,0.52947293]$. No such second-qubit rate was measured. Page 3 uses the independently specified mathematical detector to obtain a one-sided bound on that rate.

Extremal states and channels

For a fixed joint output Ω on one pure input, put P=Tr(Ω²), λ=λmax(Ω). The exact Choi purity endpoints are (P+¾)/16 and (9+6λ+P)/16. For the minimum, the sixteen blocks of 4C are Φ(|i⟩⟨j|). Their squared Hilbert–Schmidt norms add. Send the measured input to Ω, the other three basis states to $I_4/4$ and set the coherences to zero. This attains the lower endpoint.

For the maximum, let τj be the environment state after a channel isometry on input j. The spectrum of τ_0 equals that of Ω up to zeros. The environment state on the maximally mixed input has the spectrum of C up to zeros. Writing $\tau=(\tau_1+\tau_2+\tau_3)/3$ gives 16 $\mathrm{Tr}(C^2)=P+6\mathrm{Tr}(\tau_0\tau)+9\mathrm{Tr}(\tau^2)\le P+6\lambda+9$. Choose a Schmidt purification of Ω for input 0, and map the other inputs to mutually orthogonal system vectors perpendicular to its top system Schmidt vector, all with that top environment vector. These orthonormal images attain equality.

Optimizing over unobserved joint correlations

In the measured basis of the second qubit, write Ω as a positive two-by-two block matrix with diagonal blocks A,D. Then A+D is the known first marginal and Tr D=b. Positivity bounds any cross-block expectation by the geometric mean of the two diagonal expectations. The Rayleigh quotient gives $\lambda\max(\Omega)\le\|A\|+\|D\|\le a+b$. It is also at most one. Since m≥½, the spectrum (m,1−m,0,0) majorizes every compatible spectrum, giving $P\le m^2+(1-m)^2$.

For b<1−a, these two bounds are attained together. Use $|\varphi\rangle=\sqrt{a}\,|00\rangle+\sqrt{b}\,|11\rangle$ and $\Omega=|\varphi\rangle\langle\varphi|+(1-a-b)|10\rangle\langle 10|$ in the chosen bases. For b≥1−a, a pure Schmidt state of weights a,1−a with a rotated second basis realizes the requested probability b. The two marginal spectra then coincide.

For the minimum, dephasing in both selected bases keeps the measured data and can only lower joint purity. The four diagonal probabilities are a−x, x, 1−a−b+x and b−x. Minimizing their sum of squares gives the clipped value on page 1. Convex mixtures of endpoint channels preserve the local data. Continuity of purity proves all intermediate values occur.

Related marginal compatibility results · S. Bravyi, quantum marginal inequalities.

Quantitative exclusion at the supplied model point

Normalize Choi kets as $|M\rangle=\mathrm{vec}(M)/2$. The projections onto $p=|I_4\rangle$, $u=|iZ\otimes I\rangle$ and $v=|iX\otimes I\rangle$ are orthogonal and lie in the subspace of $|A\otimes I\rangle$. Let w denote its Choi population. The measured populations of p,v are s,f. The ordinary effect is the projection onto $\sqrt{h}\,p+\sqrt{(1-h)}u$. Its probability q implies $w\geq s+f+k$ with $k=(\sqrt{q}-\sqrt{(hs)})^2/(1-h)$, by the reverse triangle inequality. Without that calibrated effect one still has $w\geq s+f$.

For a unitary U, its off-diagonal blocks that change the second bit have equal squared Frobenius norms. Their sum is bounded by $4(1-w)$. The flip probability on $|00\rangle$ is thus at most $2(1-w)$. Both marginals of its pure joint output have the same spectrum, so $a\geq 2w-1$. This reproduces the exclusion thresholds 0.749034900224 and 0.98 using two and three sector population bounds respectively, against $a\approx 0.529472934504$.

Ceiling on purity without a full phase curve

For any trace-preserving channel with Kraus matrices K_j, the sum of squared Frobenius norms of their components outside the local subspace is $4(1-w)$. Each local component acts trivially on the second bit, so the probability β of finding that output bit equal to one on input $|00\rangle$ satisfies $\beta\leq 4(1-w)$. This inequality uses neither unitality nor a pure-channel hypothesis.

> For the specified s,h,q,f the lower bound on w is 0.99. Hence $\beta\leq 0.04$ and $\mathrm{Tr}(C^2)\leq 0.807905661518$, rounded upward. This ceiling uses the ordinary effect and first-qubit tomography. Its sharpness under all the supplied Choi constraints is unresolved.

For contrast, the minimum-purity reference state of the full phase model has $\beta=8(0.01)/13$. If that rate were independently established along with the same first marginal, the interval based only on those local data would be approximately [0.07783103,0.79476870]. The full phase-model state supplies a prediction for β. Additional Choi data can further restrict that interval.

With only the five rates s,f,pX,pY,pZ in a common box of half-width 0.03 around the reference values, the unitary lower estimate is 0.629034900224 and the largest permitted a is 0.573054978875, rounded to nearest. Their difference exceeds 0.05597. The crossing of this particular box certificate lies strictly between 0.03990195042113343 and 0.03990195042113345. These box widths describe population uncertainties and carry no confidence level without an independently justified sampling procedure.

What a positive exclusion margin proves

Put $W=2(s+f)-1-a$. The submitted distance bound $W/6$ is valid. The specific projectors give a stronger certificate. For every normalized unitary Choi state C_u, its conventional half-trace distance from C satisfies $\frac{1}{2}\|C-C_u\|_1 \geq W/(3+\sqrt{3})$. At the supplied point the lower bound exceeds 0.04639890. This is a Choi-state distance. No optimality of the coefficient is asserted.

Proof

Let M project onto $\mathrm{span}\{p,v\}$. For C_u take any top eigenvector of its first output marginal on $|00\rangle$. Let Q be the projection onto its orthogonal first-qubit direction, arbitrary second output, and reference input $|00\rangle$. The unitary inequality and the variational formula for a give $W \leq \mathrm{Tr}[(2M+4Q)(C-C_u)]$. The added input-only term cancels since both Choi matrices have reference marginal $I_4/4$.

The compression MQM has eigenvalues $\frac{1}{4}$ and 0 on M. Indeed Q sends p,v to collinear vectors on the chosen first direction and second bit zero, with total squared norm $\frac{1}{4}$. In the two-dimensional principal-angle block, $2M+4Q$ has top eigenvalue $3+\sqrt{3}$. The other eigenvalues are smaller and the minimum is zero. For a traceless Hermitian difference the expectation is bounded by this spectral range times the half-trace norm. This proves the certificate.

Interpretation

The bounds constrain a set of channels. They give no probability law over compatible channels and do not identify a microscopic source of mixing. An application requires the same calibrated input and output measurements, trace preservation on the tested path, and a justified simultaneous uncertainty region. A branch selected after history 111 can have input-dependent success probability. The supplied materials contain no matched hardware records for this path.

At the interior reference state, first-qubit measurements leave 204 local real channel dimensions, or 189 with unitality imposed. The exhibited nine-dimensional family has the same full two-qubit output on $|00\rangle$. In the separate $\pm 60°$ contrast experiment, zero is attainable for general states at the exact distance cap ≈ 0.83097060; normalized trace-preserving channels have positive contrast at that cap. Their sign without the cap is unresolved.

Supplied mathematical note · Pure-channel exclusion with incomplete phase data.

A. Higuchi, A. Sudbery and J. Szulc (2003) · one-qubit marginals of pure states.

J. Watrous, The Theory of Quantum Information, ch. 2 · channel representations.

G. A. L. White et al. (2022) · experimental multitime process characterization.

Local Choi population and qubit flip probability

A sharp constraint from trace preservation and its application to incomplete output data

Statement

Let C be the trace-one Choi matrix of a completely positive, trace-preserving two-qubit channel. Normalize vectorization by $|K\rangle = \mathrm{vec}(K)/2$. Let w be the Choi population in the subspace spanned by operators $V \otimes I_2$, and β the probability of measuring 1 on the second qubit following the input $|00\rangle$. Then

$$4(1-w) \geq \beta + \tfrac{1}{2}(1-\sqrt{(1-\beta)})^2.$$

For every β in [0,1] a channel attains equality. This is a sharp constraint on w and β considered by themselves.

Proof

Write each Kraus matrix in blocks A_j, B_j, C_j, D_j in row order, using the second bit for the block index and the first bit within each block. Orthogonal projection onto $V \otimes I_2$ replaces both diagonal blocks by $(A_j+D_j)/2$ and sets the other blocks to zero. With Frobenius norms it follows that

$$4(1-w) = \Sigma_j \left[\tfrac{1}{2}\|A_j-D_j\|^2 + \|B_j\|^2 + \|C_j\|^2\right].$$

Define $\alpha = \Sigma_j \|B_j|0\rangle\|^2$. Likewise $\beta = \Sigma_j \|C_j|0\rangle\|^2$. Trace preservation on $|00\rangle$ and $|01\rangle$ gives $\Sigma_j \|A_j|0\rangle\|^2 = 1-\beta$ and $\Sigma_j \|D_j|0\rangle\|^2 = 1-\alpha$. Restrict the Frobenius norms to their first columns, concatenate over j, and apply the reverse triangle inequality. The right-hand side is at least

$$\beta + \alpha + \tfrac{1}{2}(\sqrt{(1-\alpha)} - \sqrt{(1-\beta)})^2.$$

For fixed β this expression is increasing in α. On $0 \leq \alpha < 1$ its derivative is $\tfrac{1}{2} + \sqrt{(1-\beta)}/(2\sqrt{(1-\alpha)})$, and the endpoint follows by continuity. Substituting $\alpha = 0$ proves the inequality.

Attainment

In computational basis order take $K_0 = \mathrm{diag}(\sqrt{(1-\beta)},1,1,1)$ and $K_1 = \sqrt{\beta}\,|01\rangle\langle 00|$. They satisfy $\Sigma_j K_j^\dagger K_j = I_4$. The off-diagonal blocks contribute β and the diagonal-block difference contributes exactly $\tfrac{1}{2}(1-\sqrt{(1-\beta)})^2$.

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Consequence for the supplied model

Application to a calibrated rank-one effect and first-qubit tomography on the same channel

Local observations and purity

Let a be the larger eigenvalue of the first output marginal on $|00\rangle$, $b = \min(\beta,1-\beta)$, and $m = \min(1,a+b)$. When those local observations are the only data, a unitary is possible exactly for $1-a \le \beta \le a$. Necessity follows from the equal marginal spectra of a pure bipartite output. Sufficiency follows by rotating the second Schmidt basis to give β and extending the resulting output vector to a unitary.

The local-data upper bound is $H(m) = [9+6m+m^2+(1-m)^2]/16$. To see this, let Ω be the joint output. The positive block-matrix inequality bounds its maximum eigenvalue by the sum of the two diagonal-block norms, at most a+b. Thus (m,1−m,0,0) majorizes its spectrum. For a channel isometry, the environment states τ_j on the four basis inputs obey $16\,\mathrm{Tr}(C^2) = \mathrm{Tr}[(\Sigma_j \tau_j)^2]$. Here τ_0 has the spectrum of Ω up to zeros. Bounding the other three states by purity one gives $16\,\mathrm{Tr}(C^2) \le \mathrm{Tr}(\Omega^2)+6\lambda\max(\Omega)+9$.

Conditional numerical ceiling

In the supplied normalization, let $p = |I_4\rangle$, $u = |iZ \otimes I_2\rangle$, $v = |iX \otimes I_2\rangle$ and E project onto $\sqrt{h}\,p + \sqrt{(1-h)}u$. Write s, f for the p,v populations and $q = \mathrm{Tr}(EC)$. The reverse triangle inequality gives $w \ge s+f+(\sqrt{q}-\sqrt{(hs)})^2/(1-h)$. For $s=131/320$, $h=99/100$, $q=81711338569043/224995905216000$ and the specified 1% residual point, this yields $w \ge 0.99$ and the reference $a \approx 0.529472934504$.

Put $x = \sqrt{(1-\beta)}$. Page 1 gives $(x+1)^2 \ge 4-8(1-w)$, whence $\beta \le 1-(\sqrt{3.92}-1)^2 < 0.039797974644667$. H is increasing for the relevant m. Substitution using the exact inputs gives the outward-rounded ceiling

$$\mathrm{Tr}(C^2) \le 0.807826398288.$$

This improves the previous ceiling 0.807905661518. Attainment on page 1 fixes only the tradeoff between w and β. Its example does not establish equality under the separate detector, projector-population and first-marginal constraints. The sharp purity maximum for the full data is open. No matched device measurements or calibrated transfer across preparations are supplied.

Mathematical inputs: Purity Bounds from Local Output Measurements, pp. 1–3. Channel representation: J. Watrous, The Theory of Quantum Information, chapter 2. Pure marginal spectra: A. Higuchi, A. Sudbery and J. Szulc, Phys. Rev. Lett. 90, 107902 (2003).

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Paired-input constraints on quantum channels

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The supplied population values imply $\mathrm{Tr}(C^2) \le 0.742865465969$ for the normalized Choi matrix of the specified trace-preserving channel. The calculation also predicts a bound on the unmeasured joint output for a neighboring input. Both conclusions require the same channel to describe the two preparations.

Let w be the normalized Choi population of the four-dimensional space of operators $A\otimes I_2$, with normalized vectorization $|K\rangle=\mathrm{vec}(K)/2$. On input $|00\rangle$, a is the largest eigenvalue of the first output marginal, and β is the probability of finding the second qubit in state 1. At the supplied model point $w\ge0.99$ and $a\approx0.529472934504$. These are model populations. No device readings or measured input-transfer errors were supplied.

Quantity	Conditional result at the model point
Second-qubit flip on $\|00\rangle$	$\beta \le 0.039797974645$ (outward).
Largest joint-output eigenvalue on $\|01\rangle$	At most 0.824617080524 (outward). This prediction can be held out.
Choi purity using both input constraints	At most 0.742865465969 (outward). Global optimality is open.
Choi half-trace distance to any unitary	At least 0.173002789690 (outward). Exact local-data result on page 3.

Input values and the single-input result

Use match population $s=131/320$, reference response $h=99/100$, ordinary probability $q=81711338569043/224995905216000$, and orthogonal-projector population f. The detector ray is $\sqrt{h}|I_4\rangle+\sqrt{(1-h)}|iZ\otimes I\rangle$. The additional outcome projects onto $|iX\otimes I\rangle$. Positivity gives $w\ge s+f+(\sqrt{q}-\sqrt{(hs)})^2/(1-h)$. At the specified 1% residual point the right side is 0.99.

For local readings alone put $m=\min(1,a+\min(\beta,1-\beta))$. The sharp purity upper endpoint is $H(m)=[9+6m+m^2+(1-m)^2]/16$. The lower endpoint and unitary criterion are supported. Combining the sharp flip relation on page 2 with H gives the valid outward ceiling 0.807826398288. The tighter bound in the table uses another consequence of w.

What the local Choi population constrains

Write each Kraus matrix K_j in second-bit blocks A_j,B_j,C_j,D_j. Orthogonal projection onto operators of the form $V\otimes I_2$ gives $4(1-w)=\Sigma_j[\tfrac{1}{2}\|A_j-D_j\|F^2+\|B_j\|F^2+\|C_j\|F^2]$. Let β be the flip probability on $|00\rangle$ and α the reverse flip on $|01\rangle$. Keeping the first columns and applying the reverse triangle inequality to the concatenated diagonal blocks bounds the right side below by $\beta+\alpha+\tfrac{1}{2}(\sqrt{(1-\alpha)}-\sqrt{(1-\beta)})^2$. This expression increases in α, giving $4(1-w)\geq\beta+\tfrac{1}{2}(1-\sqrt{(1-\beta)})^2$.

Equality for each β is achieved by $\mathrm{diag}(\sqrt{(1-\beta)},1,1,1)$ and $\sqrt{\beta}\,|01\rangle\langle 00|$ as Kraus operators. This proves sharpness when w and β are the only constraints. At $w\geq 0.99$, inversion yields $\beta\leq 1-(\sqrt{3.92}-1)^2$.

Comparison of two inputs

Let V be the Stinespring isometry from these Kraus matrices. In a common environment compare $\psi=V|00\rangle$ with $\varphi=(X_2\otimes I_{env})V|01\rangle$, where X_2 flips the second output bit. Their squared difference is $\Sigma_j[\|(A_j-D_j)|0\rangle\|^2+\|(C_j-B_j)|0\rangle\|^2]\leq 8(1-w)$. Consequently $\mathrm{Re}\langle\psi|\varphi\rangle\geq 4w-3\geq c=0.96$. X_2 is a mathematical change of output coordinates in this argument.

The largest eigenvalue of the joint output on $|00\rangle$ is at most $m_0=a+\beta_{max}\approx 0.569270909148$. Indeed its two positive diagonal blocks have operator norms at most a and β, and positivity bounds the off-diagonal contribution by their geometric mean. The same eigenvalue cap holds in the environment, because ψ is pure.

For any environmental ray let p_0,p_1 be its probabilities in ψ and φ. Then $p_0\leq m_0$ and $c\leq\sqrt{(p_0p_1)}+\sqrt{[(1-p_0)(1-p_1)]}$. Writing each square-root pair as a unit vector gives $p_1\leq m_1=(c\sqrt{m_0}+\sqrt{[(1-c^2)(1-m_0)]})^2$ since $m_0<c^2$. This is uniform over rays. The environment and joint output on $|01\rangle$ thus have largest eigenvalue at most $m_1\approx 0.824617080524$.

> The environment on a maximally mixed input is the average of its four basis-input states and has the nonzero spectrum of C. Therefore $\lambda_{max}(C)\leq t=(m_0+m_1+2)/4$. Since $t\geq\tfrac{1}{2}$, $(t,1-t,0,\ldots)$ majorizes the spectrum of C and $\mathrm{Tr}(C^2)\leq t^2+(1-t)^2<0.742865465969$.

A sharp distance from the local readings alone

For the specified first marginal and second outcome β, the minimum of $\frac{1}{2}\|C-P_u\|_1$ over compatible trace-preserving channels C and unitary Choi states P_u is the following function of $m=\min(1,a+\min(\beta,1-\beta))$.

> $D_{\min}(m) = [1-m + \sqrt{((1-m)^2+12(1-\sqrt{m})^2)}]/8$. The formula also applies when the complete joint output Ω is fixed, with $m=\lambda_{\max}(\Omega)$. It is a normalized Choi-state distance.

Lower bound

Fix a candidate unitary U. Its Choi vector in the input basis is $(g+\sqrt{3}\,v)/2$, where $g=U|0\rangle\otimes|0\rangle$ and $v=\Sigma_{j=1}^{3} U|j\rangle\otimes|j\rangle/\sqrt{3}$. Here $|0\rangle$ denotes the tested four-dimensional input $|00\rangle$. The g diagonal of C is at most m/4 from the output bound. The v diagonal is at most ¾ from trace preservation on the three-dimensional complementary input subspace. Positivity bounds the magnitude of the cross entry by $\sqrt{(3m)/4}$.

On any unit vector $xg+yv$ with x,y real nonnegative, the expectation of P_u-C is therefore bounded below by the Rayleigh quotient of the matrix with entries $[(1-m)/4, \sqrt{3}(1-\sqrt{m})/4; \sqrt{3}(1-\sqrt{m})/4, 0]$. Its top eigenvector can be chosen nonnegative and its top eigenvalue is $D_{\min}(m)$. Half-trace distance bounds the expectation of a rank-one test for a traceless difference. This proves the lower bound for every compatible C,U.

Attainment

Choose the joint output attaining eigenvalue m. For local data with $a+b<1$ and $b=\min(\beta,1-\beta)$, one such output is $|z\rangle\langle z|+(1-a-b)|10\rangle\langle 10|$, where $z=\sqrt{a}|00\rangle+\sqrt{b}|11\rangle$ in the corresponding bases. For $a+b\geq 1$ use a pure joint output with the specified marginals and second probability. For a fixed joint output take its eigenvectors u_j with eigenvalues λ_j in descending order.

Set $U|j\rangle=u_j$. Use Kraus matrices $K_0=U\,\mathrm{diag}(\sqrt{m},1,1,1)$ and $K_j=\sqrt{\lambda_j}\,|u_j\rangle\langle 0|$ for j=1,2,3. The operators preserve trace and give the chosen joint output. The residual Choi components are perpendicular to g,v. Thus P_u-C has only one positive eigenvalue when $m<1$, exactly the one computed above. At $m=1$ take $C=P_u$.

The model gives an upper limit on m. $D_{\min}$ decreases with m. Substitution gives distance at least 0.173002789690 to every unitary Choi state. With the weaker purity ceiling 0.807905661518 alone, the sixteen-dimensional spectral argument gives approximately 0.101546767054 instead. Both apply to a mathematical observation class.

Selected-branch convention

Kirtus Justus

1. Notation and normalized instrument branch

Let $\mathcal{B} : M_4 \to M_4$ be a **nonzero** completely positive trace-nonincreasing map. Write $H = \mathcal{B}^*(I_4)$ for its positive success effect, so $0 \le H \le I_4$. Define $p_* = \operatorname{Tr} H/4 > 0$. The trace-one Choi state C of the branch is the Choi operator of $\mathcal{B}$ acting on the output half of the normalized maximally entangled state, divided by p_*. Equivalently, it has Kraus representation

$$C = \sum_j |K_j\rangle\!\rangle\langle\!\langle K_j|, \quad |K\rangle\!\rangle = \tfrac{1}{2}\operatorname{vec} K, \quad \sum_j K_j^\dagger K_j = G := H/p_*, \quad \operatorname{Tr} G = 4.$$

Thus $C \ge 0$ and $\operatorname{Tr} C = 1$; its input-reference marginal is $G^{\mathsf{T}}/4$. A postselected branch is not generally represented by a normalized CPTP Choi state, because G need not be I_4.

All computational basis vectors use the order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. Let $\mathcal{L}$ be the four-dimensional Hilbert-Schmidt subspace of matrices $V \otimes I_2$. Its normalized Choi-population is $w = \operatorname{Tr}(\Pi_{\mathcal{L}} C)$. Let $g_0 = \langle 00|G|00\rangle > 0$ and $g_1 = \langle 01|G|01\rangle \ge 0$. Let β be the **conditional** probability that the second output qubit is 1 given input $|00\rangle$ and success of this same branch.

2. Theorem: sharp branch-relative flip bound

Theorem 1. For the stated branch, without a trace-preservation or unitality assumption,

$$\boxed{4(1-w) \;\ge\; g_0\beta + \tfrac{1}{2}\left(\sqrt{g_0(1-\beta)} - \sqrt{g_1}\right)^2.} \tag{1}$$

For every triple $g_0 > 0$, $g_1 \ge 0$, $g_0 + g_1 \le 4$, and $\beta \in [0,1]$, equality is realizable by a nonzero CPTNI branch after normalizing its Choi operator.

Proof. Partition each K_j into 2×2 blocks $(A_j, B_j; C_j, D_j)$ indexed by the second bit. Orthogonal projection onto $V \otimes I_2$ replaces its diagonal blocks by $(A_j + D_j)/2$ and removes the others. Therefore

$$4(1-w) = \sum_j \left[\tfrac{1}{2}\|A_j - D_j\|_F^2 + \|B_j\|_F^2 + \|C_j\|_F^2\right].$$

Concatenate the first columns of these blocks over j. Their squared norms obey

$$\|A_0\|^2 = g_0(1-\beta), \quad \|C_0\|^2 = g_0\beta, \quad \|B_0\|^2 = \alpha, \quad \|D_0\|^2 = g_1 - \alpha,$$

with $0 \le \alpha \le g_1$. The reverse triangle inequality gives

$$4(1-w) \ge g_0\beta + \alpha + \tfrac{1}{2}\left(\sqrt{g_0(1-\beta)} - \sqrt{g_1 - \alpha}\right)^2.$$

For fixed g_0, g_1, β, the right side is nondecreasing in α; its interior derivative is $\frac{1}{2} + \frac{\sqrt{g_0(1-\beta)}}{2\sqrt{g_1-\alpha}} \ge 0$. Setting $\alpha = 0$ proves (1), with endpoint cases obtained by continuity.

For attainment, put $u = (4 - g_0 - g_1)/2 \geq 0$ and take normalized Kraus operators

$$K_0 = \mathrm{diag}\left(\sqrt{g_0(1-\beta)}, \sqrt{g_1}, \sqrt{u}, \sqrt{u}\right), \qquad K_1 = \sqrt{g_0\beta}\,|01\rangle\langle 00|.$$

They satisfy $\sum K_j^\dagger K_j = \mathrm{diag}(g_0, g_1, u, u)$, have normalized Choi trace one, and saturate every inequality. Multiplying both Kraus operators by any sufficiently small nonzero common scalar makes their sum-of-effects at most I_4, without changing C, w, g_0, g_1 or conditional β. This gives the promised CPTNI branch. □

CPTP case. If $G = I_4$, then $g_0 = g_1 = 1$ and (1) reduces to the original sharp inequality $4(1-w) \geq \beta + \frac{1}{2}(1 - \sqrt{1-\beta})^2$. More generally, equal relative successes $g_0 = g_1 = 1$ suffice for this formula; full trace preservation is not logically required for this particular inequality.

Inverting a bound. For a separately justified lower population $w \geq w_0$, define $L = 2(g_0 + g_1) - 8(1 - w_0)$. Whenever the data are compatible and $g_0 > 0$, equation (1) yields

$$\sqrt{g_0(1-\beta)} + \sqrt{g_1} \geq \sqrt{\max\{0, L\}}, \quad \beta \leq 1 - \left[\frac{\max\{0, \sqrt{\max\{0, L\}} - \sqrt{g_1}\}}{\sqrt{g_0}}\right]^2. \tag{2}$$

When $g_0 = g_1 = 1$ and the supplied exact model populations give $w_0 = 0.99$, (2) gives $\beta \leq 0.039797974644666\ldots$. This is a conditional **population** bound, not a detector measurement. Independently measured success rates and a common applicable branch are prerequisites to interpreting g_0, g_1 physically.

3. Theorem: sharp Choi-purity interval for one prepared input

Fix an input $|\psi\rangle = |00\rangle$ with nonzero branch success. Define

$$t = \mathrm{Tr}\left[C\,(I_4 \otimes |\psi^*\rangle\langle\psi^*|)\right] = \frac{\langle\psi|H|\psi\rangle}{\mathrm{Tr}\,H}, \qquad 0 < t \leq 1.$$

Here t is the *fraction of the trace-normalized Choi state in the selected input-reference block*, not the raw physical success probability. Let $\sigma = \mathcal{B}(|\psi\rangle\langle\psi|)/\,\mathrm{Tr}\,\mathcal{B}(|\psi\rangle\langle\psi|)$ be the full 4×4 normalized output, $P = \mathrm{Tr}\,\sigma^2$, and $\lambda = \lambda_{\max}(\sigma)$.

Theorem 2. Among all normalized Choi states of nonzero CPTNI operations with these exact t and σ, the complete attainable purity interval is

$$\boxed{t^2 P + \frac{(1-t)^2}{12} \;\leq\; \mathrm{Tr}\,C^2 \;\leq\; t^2 P + (1-t)^2 + 2t(1-t)\lambda.} \tag{3}$$

Proof. Let $P_0 = I_4 \otimes |\psi^*\rangle\langle\psi^*|$ have rank four. In the decomposition $P_0 \oplus (I - P_0)$, write C in blocks $A = t\sigma$, B, and D, with $D \geq 0$ supported on a twelve-dimensional space and $\mathrm{Tr}\,D = 1 - t$. Then

$$\mathrm{Tr}\,C^2 = t^2 P + \mathrm{Tr}\,D^2 + 2\|B\|_2^2.$$

The lower bound follows from $\mathrm{Tr}\,D^2 \geq (1-t)^2/12$ and $\|B\|_2^2 \geq 0$, with equality for $B = 0, D = (1-t)I_{12}/12$. For the upper bound, $\mathrm{Tr}\,D^2 \leq (1-t)^2$. Block positivity gives $B = A^{1/2} K D^{1/2}$ for a contraction K, so $\|B\|_2^2 \leq \|A\|_{\mathrm{op}}\,\mathrm{Tr}\,D = t\lambda(1-t)$.

To attain the upper bound, take a top eigenvector $|u_1\rangle$ of σ and any normalized vector $|v\rangle$ in the complementary twelve-dimensional input-reference block. Set

$$|\chi\rangle = \sqrt{t\lambda}\,|u_1\rangle \otimes |\psi^*\rangle + \sqrt{1-t}\,|v\rangle, \quad C_{\max} = |\chi\rangle\langle\chi| + t\sum_{j=2}^{4} \lambda_j |u_j, \psi^*\rangle\langle u_j, \psi^*|.$$

This positive unit-trace state fixes the desired input block and attains the upper formula. Every positive normalized bipartite Choi operator defines a CP map; multiplying that map by a sufficiently small positive factor makes it trace-nonincreasing while leaving its normalized Choi state and conditional output unchanged. Finally, convexly mixing the two attaining normalized Choi states fixes t, σ and continuously covers all intermediate purities. □

Trace-preserving specialization. CPTP implies $t = 1/4$. Formula (3) becomes the previously supplied

$$\boxed{(P + 3/4)/16 \;\le \operatorname{Tr} C^2 \le (9 + 6\lambda + P)/16.}$$

The supplied model's conditional β bound and first-output spectrum yield the earlier 0.807826398288 outward ceiling when the additional readout premises imply the stated bound on β. Equation (3) is sharp when only t, σ are fixed, and its CPTNI constructions need not be TP for $t = 1/4$; the separate TP attainability construction must be invoked to claim TP sharpness.

Corollary: partial local output information

Let $a \in [1/2, 1]$ be the larger eigenvalue of the first output marginal of σ, and $b = \min\{\beta, 1-\beta\} \in [0, 1/2]$ for a calibrated second-output rank-one effect. Define

$$\begin{aligned} x_* &= \operatorname{clip}\left(\frac{a+b}{2} - \frac{1}{4},\, \max(0, a+b-1),\, \min(a,b)\right), \\ P_{\min}(a,b) &= (a-x_*)^2 + x_*^2 + (1-a-b+x_*)^2 + (b-x_*)^2, \\ m &= \min(1, a+b). \end{aligned}$$

Among all normalized CPTNI Choi states with the given t, a, b, the sharp interval is

$$\boxed{t^2 P_{\min}(a,b) + \frac{(1-t)^2}{12} \;\le \operatorname{Tr} C^2 \le\; t^2[m^2 + (1-m)^2] + (1-t)^2 + 2t(1-t)m.} \tag{4}$$

The endpoints follow by minimizing joint-output purity under the measured local marginals and, separately, attaining $\lambda_{\max}(\sigma) = m$, $\operatorname{Tr}\sigma^2 = m^2 + (1-m)^2$. Explicit two-qubit outputs and the block constructions of Theorem 2 attain both, and continuity fills the interval. At $t = 1/4$ this recovers the numerical and algebraic bounds reported for the CPTP local-output model.

Corollary: pure-branch exclusion does not require trace preservation

A **rank-one Choi state** for any nonzero CP operation has a single effective Kraus operator K. For a pure input with nonzero branch success, its normalized output is a rank-one projector on $K|\psi\rangle$. Hence the two output-qubit marginals must have the same unordered eigenvalues. Consequently the local readings permit a rank-one Choi state only if

$$\boxed{1 - a \le \beta \le a.}$$

Conversely, whenever this inequality holds, a pure joint output with the specified local readings exists by Schmidt decomposition and a rotation of the second Schmidt basis. The single Kraus operator $K = |\varphi\rangle\langle\psi|$ is trace-nonincreasing and yields that output on $|\psi\rangle$. Therefore **this is necessary and sufficient for a rank-one CPTNI explanation when only these local output data are fixed**. The same condition was previously established for a pure unitary-channel

explanation in the CPTP class, but the relevant explanation after postselection need not be unitary. Other joint/Choi observations can rule out rank one even when these local readings alone do not.

A mixed conditional joint output from a known pure input excludes rank-one normalized Choi states of a *single, applicable* CP branch, regardless of global trace preservation. If t is separately known and strictly positive, (3) also supplies a quantitative Choi-purity ceiling. Without a positive lower bound on t, a mixed output alone need not give a uniform numerical gap below purity one.

Paired preparations and conditional output fidelity

Exact compatibility of two first-qubit marginals for one outcome-selected operation

Model and criterion

Consider a single nonzero completely positive, trace-nonincreasing operation on two qubits. Its success effect is H, its mean success on the maximally mixed input is $p_\star=\mathrm{Tr}(H)/4$, and $g_i=p_i/p_\star$ for success on inputs $|00\rangle$ and $|01\rangle$ ($i=0,1$). Assume both $p_i>0$. The trace-one Choi state C has reference marginal $H^{\mathsf{T}}/\mathrm{Tr}(H)$. Its population w in the operator subspace $\{L\otimes I_2\}$ is bounded below by w_0. Denote the normalized first-output-qubit states by ρ_0,ρ_1. Write $F=\|\sqrt{\rho_0}\sqrt{\rho_1}\|_1$ for root fidelity.

Fix admissible g_0,g_1 ($g_0+g_1\le 4$) and impose only these relative successes, the two first marginals and $w\ge w_0$. Such a branch exists if and only if

$$g_0+g_1-2\sqrt{g_0g_1}\,F(\rho_0,\rho_1)\le 8(1-w_0).$$

Necessity

Scale the Kraus matrices by $1/\sqrt{p_\star}$ and write their second-qubit blocks as A_j,B_j,C_j,D_j in row order. Projection onto $\{L\otimes I_2\}$ gives $4(1-w)=\Sigma_j[\tfrac12\|A_j-D_j\|^2+\|B_j\|^2+\|C_j\|^2]$ with Frobenius norms. In a common dilation let ψ_0,ψ_1 be the normalized success vectors; flip the second output bit of ψ_1. The squared norm of $\sqrt{g_0}\,\psi_0-\sqrt{g_1}\,\psi_1$ after this alignment is

$$\Sigma_j\,[\|(A_j-D_j)|0\rangle\|^2+\|(C_j-B_j)|0\rangle\|^2]\le 8(1-w).$$

The real part of their overlap is at most $F(\rho_0,\rho_1)$, by the purification characterization of fidelity. Expanding the squared norm proves necessity. The same overlap lower bound applies to any common environmental measurement. The output bit flip is an algebraic comparison of the vectors and leaves the first marginal intact.

Sufficiency

Choose purifications η_0,η_1 of the prescribed first marginals with real overlap F. For the first two input columns use $\sqrt{g_0}\,\eta_0$ with second output 0 and $\sqrt{g_1}\,\eta_1$ with second output 1. For the other columns repeat one normalized first-qubit/environment vector in a disjoint environment subspace, with second outputs 0 and 1 and common norm $\sqrt{[(4-g_0-g_1)/2]}$. The four columns are orthogonal. The projection deficit is exactly the left side divided by eight. Scale all columns together to make the operation trace-nonincreasing. If $p_\star$ is fixed, use its scale whenever all four raw success probabilities lie in $[0,1]$.

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A test using all attempted preparations

Keep the success frequency and both conditional outcomes for each input

Observable probability inequality

Let r_{iy} = Pr(success and binary first-qubit result y | input i), y=0,1, with the same binary effect in both arms. Record unsuccessful attempts as well. Classical root fidelity of the two normalized output distributions is at least $F(\rho_0,\rho_1)$. Multiplying the criterion on page 1 by p★ gives the necessary test

$$\Sigma_y \left(\sqrt{r_{0y}} - \sqrt{r_{1y}}\right)^2 \le 8\, p\star\, (1-w_0).$$

Each rate uses all attempted preparations in its arm as its denominator. For a rank-one projective readout with only these two arm distributions specified, commuting first-qubit states diagonal in its basis attain the classical fidelity. For a general known effect, the displayed test is necessary. Extra measured state or detector constraints can further reduce compatibility.

Sharp difference for the supplied first marginal

At $g_0=g_1=1$, $w_0=0.99$ gives $F \ge c=0.96$. Put $s=\sqrt{(1-c^2)}=0.28$. For a projector whose arm-0 probability is $x<c^2$, the arm-1 probability is at most $G(x)=(c\sqrt{x}+s\sqrt{(1-x)})^2$, by the angle between the binary square-root probability vectors. If the arm-0 larger eigenvalue is $a\approx 0.529472934504$, then $x \in [1-a,a]$ for a rank-one projector.

The maximum of $G(x)-x$ on $[0,c^2]$ is at $x=(1-s)/2$; the difference decreases beyond that point (differentiate the concave function). Here $(1-s)/2=0.36 < 1-a$. Exchanging outcomes bounds the opposite sign. A maximizing trace-distance measurement can be a rank-one projector for two qubit states, so every common binary effect obeys

$$\left|\Pr_1(\text{click} \mid \text{success}) - \Pr_0(\text{click} \mid \text{success})\right| \le 0.272954.$$

The underlying maximum is 0.27295396145309825…. It is attained in the reduced class by diagonal states with event probabilities $x=1-a$ and $G(x)$, using page 1. Attainment under all supplied joint-projector constraints is open.

Equal raw successes can leave the test uninformative

Take orthogonal first marginals, $g_0=g_1=0.04$ and $p\star=0.5$. The page 1 construction gives other relative successes 1.96, and $w=0.99$. The tested inputs both succeed with probability 0.02; the other basis inputs succeed with probability 0.98. A common projector distinguishes the first marginals perfectly. This is a mathematical counterexample to using equality of the two success rates alone. It does not impose the reference value of a.

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Neighboring output and Choi purity

A single population deficit constrains both second-qubit flip probabilities

Equal relative successes at the model point

Keep $g_0=g_1=1$, $w\ge 0.99$ and the supplied value of a. The following bounds allow a common nonzero branch; trace preservation is sufficient. Write Ω_i for its full conditional output and ρ_i for its first marginal. Let β,α be the conditional probabilities of a second-bit flip on inputs 0,1 respectively. The projection identity on page 1 yields $\beta+\alpha \le 4(1-w) \le 0.04$. A bound from the first marginal and the measured second-bit population gives $\lambda\max(\Omega_0) \le a+\beta$ and $\lambda\max(\Omega_1) \le \lambda\max(\rho_1)+\alpha$.

For clarity, a positive two-by-two output block matrix with diagonal blocks A,D has largest eigenvalue $\le \|A\|+\|D\|$ by block positivity and Cauchy–Schwarz. The first diagonal block is bounded by the first marginal; the other by the flip probability. Exchange the two blocks on input 1. Page 2 gives $\lambda\max(\rho_1) \le G(a) < 0.793184183697$ by testing the top projector of ρ_1; G is increasing up to c^2.

Upper limit for the unmeasured joint output

Apply the binary probability bound G to the environment projector for its largest arm-1 eigenvalue. On arm 0 this projector has probability $\le a+\beta$; the environment and full joint output have the same nonzero spectra on each successful pure input. Here $a+\beta < c^2$. Consequently,

$$\lambda\max(\Omega_1) \le \min\{\, G(a+\beta),\, G(a)+0.04-\beta \,\} < 0.810954608659.$$

Split at $b=0.022229575038$. For $\beta\le b$ use $G(a+b)$; for $\beta\ge b$ use $G(a)+0.04-b$. Both are below the displayed decimal by outward interval arithmetic. The submitted single-arm comparison gave 0.824617080524.

Upper limit for normalized Choi purity

The environment marginal purifying C is the sum of the four basis-input environment states weighted by $g_i/4$; a zero-success input contributes zero. The two weights considered here are 1/4 each and the remaining weights sum to 1/2. For every environment ray, the first two probabilities sum to at most $a+\beta+G(a)+\alpha$. Thus $\lambda\max(C) \le t=[2+a+G(a)+0.04]/4$. Since $t\ge ½$, the spectrum $(t,1-t,0,\ldots)$ majorizes the spectrum of C, yielding

$$\mathrm{Tr}(C^2) \le t^2+(1-t)^2 < 0.732104302723.$$

Both upper bounds use the shared flip budget without optimizing every compatibility constraint. Their sharp values remain open. The two-input distance lower bound 0.173002789690 in the submitted report concerns unitary Choi states in the specified channel model.

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